

28. 16-Bit Timer Pulse Unit (TPUa)

This MCU has on-chip 16-bit timer pulse units (TPU) comprising six-channel 16-bit timers.

In this section, “PCLK” is used to refer to PCLKB.

28.1 Overview

Specifications of the TPU are listed in Table 28.1. Functions of TPU are listed in Table 28.2.

Figure 28.1 shows a block diagram of TPU.

Table 28.1 Specifications of TPU

Item	Description
Pulse input/output	Maximum 16
Count clocks	Seven or eight types are provided for each channel.
Settable operations	• Waveform output at compare match • Input capture function (noise filters can be set) • Counter clear operation • Simultaneous writing to multiple timer counters (TCNT) • Simultaneous clearing by compare match and input capture • Synchronous input/output for registers by counter synchronous operation • Maximum of 15-phase PWM output by combination with synchronous operation • Cascaded operation
TPU0 and TPU3	Buffer operation can be set.
TPU1, TPU2, TPU4, and TPU5	Phase counting mode can be set.
Interrupt sources	26 sources
Buffer operation	Automatic transfer of register data
Generation of trigger	Programmable pulse generator (PPG) output trigger can be generated. Conversion start trigger for the A/D converter can be generated.
Event linking (output)	Six types of event signal can be output to the ELC. • Compare match A (TPU0 to TPU3) • Compare match B (TPU0 to TPU3) • Compare match C (TPU0, TPU3) • Compare match D (TPU0, TPU3) • Overflow (TPU0 to TPU3) • Underflow (TPU1, TPU2)
Event linking (input)	Any of the three operations in response to event input is possible. • Starting counts (TPU0 to TPU3) • Restarting counts (TPU0 to TPU3) • Input capture operation (TPU0 to TPU3)
Low power consumption function	Module stop state can be set.

Table 28.2 TPU Functions (1/2)

Item		TPU0	TPU1	TPU2	TPU3	TPU4	TPU5
Count clocks		PCLK/1 PCLK/4 PCLK/16 PCLK/64 TCLKA TCLKB TCLKC TCLKD	PCLK/1 PCLK/4 PCLK/16 PCLK/64 PCLK/256 TCLKA TCLKB	PCLK/1 PCLK/4 PCLK/16 PCLK/64 PCLK/1024 TCLKA TCLKB TCLKC	PCLK/1 PCLK/4 PCLK/16 PCLK/64 PCLK/256 PCLK/1024 PCLK/4096 TCLKA	PCLK/1 PCLK/4 PCLK/16 PCLK/64 PCLK/1024 TCLKA TCLKC	PCLK/1 PCLK/4 PCLK/16 PCLK/64 PCLK/256 TCLKA TCLKC TCLKD
External clocks for phase counting mode		Not possible	TCLKA TCLKB	TCLKC TCLKD	Not possible	TCLKC TCLKD	TCLKA TCLKB
Timer general registers		TGRA TGRB TGRC*1 TGRD*1	TGRA TGRB	TGRA TGRB	TGRA TGRB TGRC*1 TGRD*1	TGRA TGRB	TGRA TGRB
I/O pins		TIOCA0 TIOCB0 TIOCC0 TIOCD0	TIOCA1 TIOCB1	TIOCA2 TIOCB2	TIOCA3 TIOCB3 TIOCC3 TIOCD3	TIOCA4 TIOCB4	TIOCA5 TIOCB5
Counter clear function (y = A to D)		TGRy compare match or input capture	TGRy compare match or input capture	TGRy compare match or input capture	TGRy compare match or input capture	TGRy compare match or input capture	TGRy compare match or input capture
Compare match output	Low output	Possible	Possible	Possible	Possible	Possible	Possible
	High output	Possible	Possible	Possible	Possible	Possible	Possible
	Toggle output	Possible	Possible	Possible	Possible	Possible	Possible
Input capture function		Possible	Possible	Possible	Possible	Possible	Possible
Synchronous operation		Possible	Possible	Possible	Possible	Possible	Possible
PWM mode		Possible	Possible	Possible	Possible	Possible	Possible
Phase counting mode		Not possible	Possible	Possible	Not possible	Possible	Possible
Buffer operation		Possible	Not possible	Not possible	Possible	Not possible	Not possible
DTC activation (y = A to D)		TGRy compare match or input capture	TGRy compare match or input capture	TGRy compare match or input capture	TGRy compare match or input capture	TGRy compare match or input capture	TGRy compare match or input capture
DMAC activation		TGRA compare match or input capture	TGRA compare match or input capture	TGRA compare match or input capture	TGRA compare match or input capture	TGRA compare match or input capture	TGRA compare match or input capture
A/D conversion start trigger		TGRA compare match or input capture	TGRA compare match or input capture	TGRA compare match or input capture	TGRA compare match or input capture	TGRA compare match or input capture	Not possible
PPG trigger		TGRA/TGRB compare match or input capture	TGRA/TGRB compare match or input capture	TGRA/TGRB compare match or input capture	TGRA/TGRB compare match or input capture	Not possible	Not possible
Interrupt sources		5 sources • Compare match or input capture 0A • Compare match or input capture 0B • Compare match or input capture 0C • Compare match or input capture 0D • Overflow	4 sources • Compare match or input capture 1A • Compare match or input capture 1B • Overflow • Underflow	4 sources • Compare match or input capture 2A • Compare match or input capture 2B • Overflow • Underflow	5 sources • Compare match or input capture 3A • Compare match or input capture 3B • Compare match or input capture 3C • Compare match or input capture 3D • Overflow	4 sources • Compare match or input capture 4A • Compare match or input capture 4B • Overflow • Underflow	4 sources • Compare match or input capture 5A • Compare match or input capture 5B • Overflow • Underflow

Table 28.2 TPU Functions (2/2)

Item	TPU0	TPU1	TPU2	TPU3	TPU4	TPU5
Event linking (output)	5 sources • Compare match 0A • Compare match 0B • Compare match 0C • Compare match 0D • Overflow	4 sources • Compare match 1A • Compare match 1B • Overflow • Underflow	4 sources • Compare match 2A • Compare match 2B • Overflow • Underflow	5 sources • Compare match 3A • Compare match 3B • Compare match 3C • Compare match 3D • Overflow	Not possible	Not possible
Event linking (input)	• Starting counts • Restarting counts • Input capture operation (data is captured in TGRA)	• Starting counts • Restarting counts • Input capture operation (data is captured in TGRA)	• Starting counts • Restarting counts • Input capture operation (data is captured in TGRA)	• Starting counts • Restarting counts • Input capture operation (data is captured in TGRA)	Not possible	Not possible
Module stop setting*2	MSTPCRA.MSTPA13 bit					

Note 1. TGRC and TGRD can be set as a buffer register.

Note 2. For details, see section 11, Low Power Consumption.

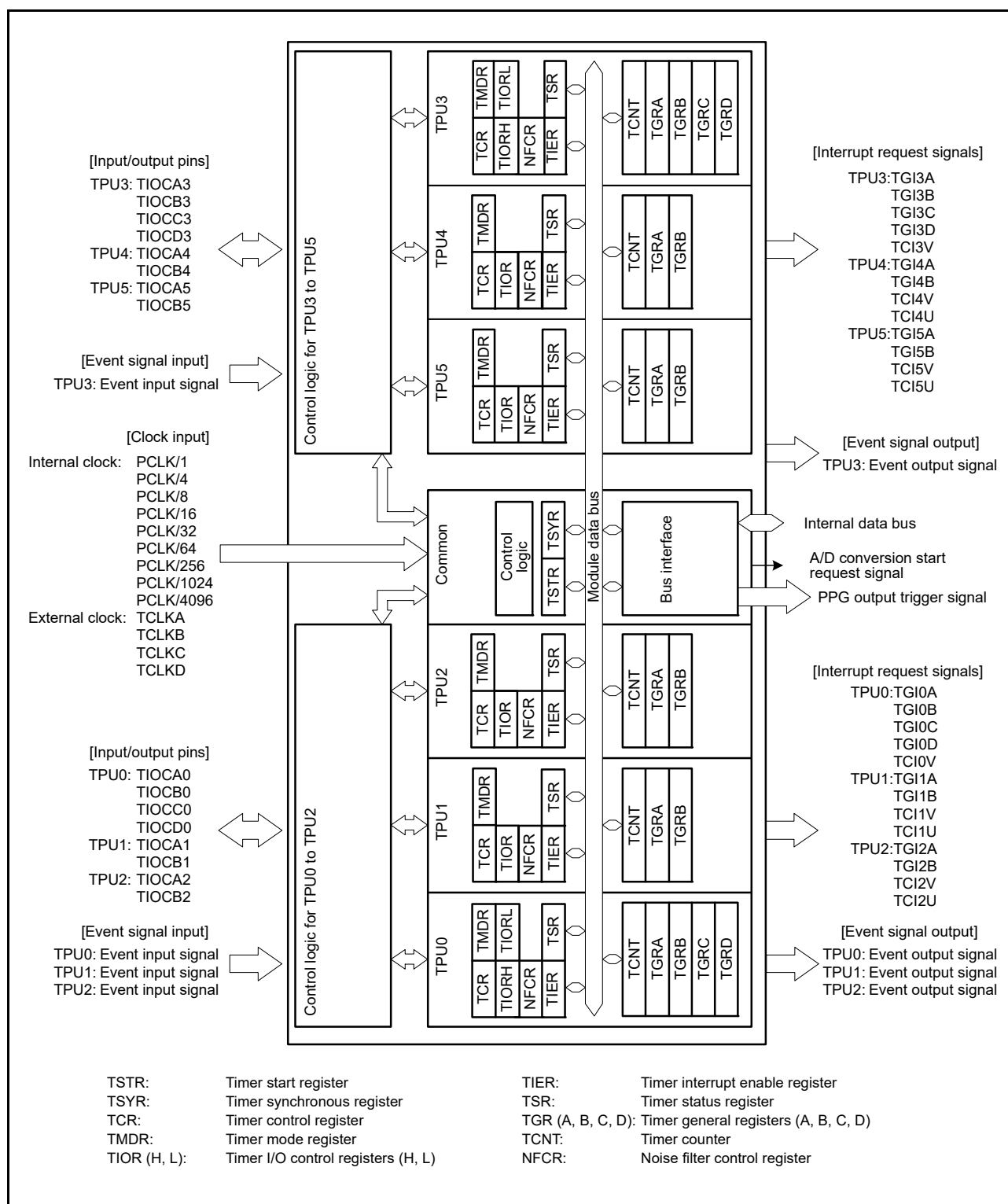


Figure 28.1 Block Diagram of TPU

Table 28.3 lists the input/output pins of the TPU.

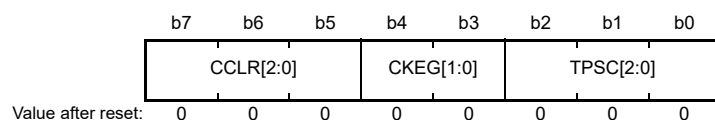
Table 28.3 Pin Configuration of TPU

Channel	Pin Name	I/O	Description
Common	TCLKA	Input	External clock A input pin (TPU1 and TPU5 phase counting mode A phase input)
	TCLKB	Input	External clock B input pin (TPU1 and TPU5 phase counting mode B phase input)
	TCLKC	Input	External clock C input pin (TPU2 and TPU4 phase counting mode A phase input)
	TCLKD	Input	External clock D input pin (TPU2 and TPU4 phase counting mode B phase input)
TPU0	TIOCA0	I/O	TPU0.TGRA input capture input/output compare output/PWM output pin
	TIOCB0	I/O	TPU0.TGRB input capture input/output compare output/PWM output pin
	TIOCC0	I/O	TPU0.TGRC input capture input/output compare output/PWM output pin
	TIOCD0	I/O	TPU0.TGRD input capture input/output compare output/PWM output pin
TPU1	TIOCA1	I/O	TPU1.TGRA input capture input/output compare output/PWM output pin
	TIOCB1	I/O	TPU1.TGRB input capture input/output compare output/PWM output pin
TPU2	TIOCA2	I/O	TPU2.TGRA input capture input/output compare output/PWM output pin
	TIOCB2	I/O	TPU2.TGRB input capture input/output compare output/PWM output pin
TPU3	TIOCA3	I/O	TPU3.TGRA input capture input/output compare output/PWM output pin
	TIOCB3	I/O	TPU3.TGRB input capture input/output compare output/PWM output pin
	TIOCC3	I/O	TPU3.TGRC input capture input/output compare output/PWM output pin
	TIOCD3	I/O	TPU3.TGRD input capture input/output compare output/PWM output pin
TPU4	TIOCA4	I/O	TPU4.TGRA input capture input/output compare output/PWM output pin
	TIOCB4	I/O	TPU4.TGRB input capture input/output compare output/PWM output pin
TPU5	TIOCA5	I/O	TPU5.TGRA input capture input/output compare output/PWM output pin
	TIOCB5	I/O	TPU5.TGRB input capture input/output compare output/PWM output pin

28.2 Register Descriptions

28.2.1 Timer Control Register (TCR)

Address(es): TPU0.TCR 0008 8110h, TPU1.TCR 0008 8120h, TPU2.TCR 0008 8130h, TPU3.TCR 0008 8140h, TPU4.TCR 0008 8150h, TPU5.TCR 0008 8160h



Bit	Symbol	Bit Name	Description	R/W
b2 to b0	TPSC[2:0]	Timer Prescaler Select	See Table 28.4 to Table 28.9.	R/W
b4, b3	CKEG[1:0]	Input Clock Edge Select	See Table 28.10.	R/W
b7 to b5	CCLR[2:0]	Counter Clear Source Select*1	See Table 28.11 and Table 28.12.	R/W

Note 1. Bit 7 is reserved in TPU1, TPU2, TPU4, and TPU5. These bits are read as 0. The write value should be 0.

TPUm.TCR settings should be made while TPUm.TCNT operation is stopped.

TPSC[2:0] Bits (Timer Prescaler Select)

These bits select the TCNT clock. The clock source can be selected independently for each channel.

To select the external clock as the clock source, set the bit in the port direction register (PDR) for the corresponding pin to 0 (input port), and set the bit in the port mode register (PMR) to 1 (uses the pin as an I/O port for peripheral functions). For details, see section 22, I/O Ports.

CKEG[1:0] Bits (Input Clock Edge Select)

These bits select the input clock edge.

When the internal clock is counted using both edges, the input clock period is halved (e.g. Both edges of PCLK/4 = PCLK/2 rising edge).

Internal clock edge selection is valid when the input clock is PCLK/4 or slower. This setting is ignored if the input clock is PCLK/1, or when overflow/underflow of another channel is selected.

Table 28.4 Bits TPSC[2:0] (TPU0)

Channel	Bits TPSC[2:0]			Description
	b2	b1	b0	
TPU0	0	0	0	Internal clock: counts on PCLK/1
	0	0	1	Internal clock: counts on PCLK/4
	0	1	0	Internal clock: counts on PCLK/16
	0	1	1	Internal clock: counts on PCLK/64
	1	0	0	External clock: counts on TCLKA pin input
	1	0	1	External clock: counts on TCLKB pin input
	1	1	0	External clock: counts on TCLKC pin input
	1	1	1	External clock: counts on TCLKD pin input

Table 28.5 Bits TPSC[2:0] (TPU1)

Channel	Bits TPSC[2:0]			Description
	b2	b1	b0	
TPU1	0	0	0	Internal clock: counts on PCLK/1
	0	0	1	Internal clock: counts on PCLK/4
	0	1	0	Internal clock: counts on PCLK/16
	0	1	1	Internal clock: counts on PCLK/64
	1	0	0	External clock: counts on TCLKA pin input
	1	0	1	External clock: counts on TCLKB pin input
	1	1	0	Internal clock: counts on PCLK/256
	1	1	1	Counts on TPU2.TCNT overflow/underflow

Note: This setting is invalid when TPU1 is in phase counting mode.

Table 28.6 Bits TPSC[2:0] (TPU2)

Channel	Bits TPSC[2:0]			Description
	b2	b1	b0	
TPU2	0	0	0	Internal clock: counts on PCLK/1
	0	0	1	Internal clock: counts on PCLK/4
	0	1	0	Internal clock: counts on PCLK/16
	0	1	1	Internal clock: counts on PCLK/64
	1	0	0	External clock: counts on TCLKA pin input
	1	0	1	External clock: counts on TCLKB pin input
	1	1	0	External clock: counts on TCLKC pin input
	1	1	1	Internal clock: counts on PCLK/1024

Note: This setting is invalid when TPU2 is in phase counting mode.

Table 28.7 Bits TPSC[2:0] (TPU3)

Channel	Bits TPSC[2:0]			Description
	b2	b1	b0	
TPU3	0	0	0	Internal clock: counts on PCLK/1
	0	0	1	Internal clock: counts on PCLK/4
	0	1	0	Internal clock: counts on PCLK/16
	0	1	1	Internal clock: counts on PCLK/64
	1	0	0	External clock: counts on TCLKA pin input
	1	0	1	Internal clock: counts on PCLK/1024
	1	1	0	Internal clock: counts on PCLK/256
	1	1	1	Internal clock: counts on PCLK/4096

Table 28.8 Bits TPSC[2:0] (TPU4)

Channel	Bits TPSC[2:0]			Description
	b2	b1	b0	
TPU4	0	0	0	Internal clock: counts on PCLK/1
	0	0	1	Internal clock: counts on PCLK/4
	0	1	0	Internal clock: counts on PCLK/16
	0	1	1	Internal clock: counts on PCLK/64
	1	0	0	External clock: counts on TCLKA pin input
	1	0	1	External clock: counts on TCLKC pin input
	1	1	0	Internal clock: counts on PCLK/1024
	1	1	1	Counts on TPU5.TCNT overflow/underflow

Note: This setting is invalid when TPU4 is in phase counting mode.

Table 28.9 Bits TPSC[2:0] (TPU5)

Channel	Bits TPSC[2:0]			Description
	b2	b1	b0	
TPU5	0	0	0	Internal clock: counts on PCLK/1
	0	0	1	Internal clock: counts on PCLK/4
	0	1	0	Internal clock: counts on PCLK/16
	0	1	1	Internal clock: counts on PCLK/64
	1	0	0	External clock: counts on TCLKA pin input
	1	0	1	External clock: counts on TCLKC pin input
	1	1	0	Internal clock: counts on PCLK/256
	1	1	1	External clock: counts on TCLKD pin input

Note: This setting is invalid when TPU5 is in phase counting mode.

Table 28.10 Bits CKEG[1:0]

Bits CKEG[1:0]		Input Clock	
b4	b3	Internal Clock	External clock
0	0	Counted at falling edge	Counted at rising edge
0	1	Counted at rising edge	Counted at falling edge
1	0	Counted at both edges	Counted at both edges
1	1	Counted at both edges	Counted at both edges

Table 28.11 Bits CCLR[2:0] (TPU0, TPU3)

Channel	Bits CCLR[2:0]			Description
	b7	b6	b5	
TPU0, TPU3	0	0	0	TCNT clearing disabled
	0	0	1	TCNT cleared by TGRA compare match/input capture
	0	1	0	TCNT cleared by TGRB compare match/input capture
	0	1	1	TCNT cleared by counter clearing for another channel performing synchronous clearing/ synchronous operation*2
	1	0	0	TCNT clearing disabled
	1	0	1	TCNT cleared by TGRC compare match/input capture*1
	1	1	0	TCNT cleared by TGRD compare match/input capture*1
	1	1	1	TCNT cleared by counter clearing for another channel performing synchronous clearing/ synchronous operation*2

Note 1. When TGRC or TGRD is used as a buffer register, TCNT is not cleared because the buffer register setting has priority, and compare match/input capture does not occur.

Note 2. Synchronous operation is selected by setting the TPUA.TSYR.SYNCj bit (j = 0, 3) to 1.

Table 28.12 Bits CCLR[2:0] (TPU1, TPU2, TPU4, TPU5)

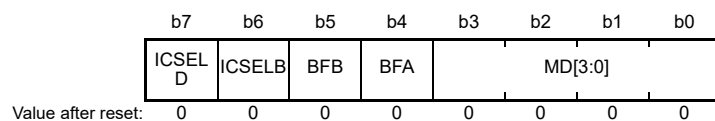
Channel	Bits CCLR[2:0]			Description
	b7*1	b6	b5	
TPU1, TPU2, TPU4, TPU5	0	0	0	TCNT clearing disabled
	0	0	1	TCNT cleared by TGRA compare match/input capture
	0	1	0	TCNT cleared by TGRB compare match/input capture
	0	1	1	TCNT cleared by counter clearing for another channel performing synchronous clearing/ synchronous operation*2
	1	0	0	Setting prohibited
	1	0	1	Setting prohibited
	1	1	0	Setting prohibited
	1	1	1	Setting prohibited

Note 1. This bit is reserved in TPU1, TPU2, TPU4, and TPU5. This bit is read as 0. The write value should be 0.

Note 2. Synchronous operation is selected by setting the TPUA.TSYR.SYNCj bit (j = 1, 2, 4, 5) to 1.

28.2.2 Timer Mode Register (TMDR)

Address(es): TPU0.TMDR 0008 8111h, TPU1.TMDR 0008 8121h, TPU2.TMDR 0008 8131h, TPU3.TMDR 0008 8141h, TPU4.TMDR 0008 8151h, TPU5.TMDR 0008 8161h



Bit	Symbol	Bit Name	Description	R/W
b3 to b0	MD[3:0]	Mode Select	b3 b0 0 0 0 0: Normal operation 0 0 0 1: Setting prohibited 0 0 1 0: PWM mode 1 0 0 1 1: PWM mode 2 0 1 0 0: Phase counting mode 1*1 0 1 0 1: Phase counting mode 2*1 0 1 1 0: Phase counting mode 3*1 0 1 1 1: Phase counting mode 4*1 Settings other than above are prohibited.	R/W
b4	BFA	Buffer Operation A*2	0: TPUm.TGRA operates normally 1: TPUm.TGRA and TPUm.TGRC used together for buffer operation (m = 0, 3)	R/W
b5	BFB	Buffer Operation B*2	0: TPUm.TGRB operates normally 1: TPUm.TGRB and TPUm.TGRD used together for buffer operation (m = 0, 3)	R/W
b6	ICSELB	TGRB Input Capture Input Select	0: Input capture input source is TIOCBn pin 1: Input capture input source is TIOCA n pin (n = 0 to 5)	R/W
b7	ICSELD	TGRD Input Capture Input Select*2	0: Input capture input source is TIOCDn pin 1: Input capture input source is TIOCCn pin (n = 0, 3)	R/W

Note 1. Phase counting mode cannot be set for TPU0 and TPU3. A 0 should be written to bit 2 for them.

Note 2. These bits are reserved in TPU1, TPU2, TPU4, and TPU5. These bits are read as 0. The write value should be 0.

TPUm.TMDR settings should be made while TPUm.TCNT operation is stopped.

BFA Bit (Buffer Operation A)

Specifies whether TPUm.TGRA (m = 0, 3) is to normally operate, or TPUm.TGRA and TPUm.TGRC (m = 0, 3) are to be used together for buffer operation.

When TGRC is used as a buffer register, TGRC input capture/output compare is not generated.

BFB Bit (Buffer Operation B)

Specifies whether TPUm.TGRB (m = 0, 3) is to normally operate, or TPUm.TGRB and TPUm.TGRD (m = 0, 3) are to be used together for buffer operation.

When TGRD is used as a buffer register, TGRD input capture/output compare is not generated.

ICSELB Bit (TGRB Input Capture Input Select)

Selects the input capture input for TPUm.TGRB (m = 0 to 5).

This function allows measurement of high-level width and period of the input pulse on a TIOCA n input pin.

ICSELD Bit (TGRD Input Capture Input Select)

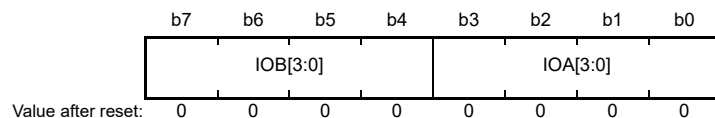
Selects the input capture input for TPUm.TGRD (m = 0, 3).

This function allows measurement of high-level width and period of the input pulse on a TIOCCn input pin.

28.2.3 Timer I/O Control Register (TIORH, TIORL, TIOR)

- TPU0.TIORH, TPU1.TIOR, TPU2.TIOR, TPU3.TIORH, TPU4.TIOR, TPU5.TIOR

Address(es): TPU0.TIORH 0008 8112h, TPU1.TIOR 0008 8122h, TPU2.TIOR 0008 8132h,
TPU3.TIORH 0008 8142h, TPU4.TIOR 0008 8152h, TPU5.TIOR 0008 8162h

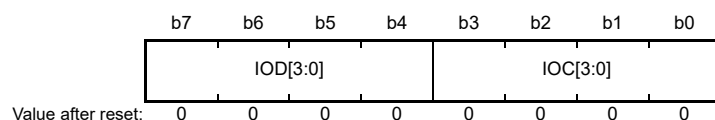


Bit	Symbol	Bit Name	Description	R/W
b3 to b0	IOA[3:0]	TGRA Control	See Table 28.13 to Table 28.18.*1	R/W
b7 to b4	IOB[3:0]	TGRB Control	See Table 28.13 to Table 28.18.*1	R/W

Note 1. If the IO[n:3:0] bit (n = A, B) values are changed to output disabled (0000b or 0100b) during low/high/toggle output on compare match, the TIOCAN/TIOCBn pin (n = 0 to 5) is placed in high impedance state.

- TPU0.TIORL, TPU3.TIORL

Address(es): TPU0.TIORL 0008 8113h, TPU3.TIORL 0008 8143h



Bit	Symbol	Bit Name	Description	R/W
b3 to b0	IOC[3:0]	TGRC Control	See Table 28.19 and Table 28.20.*1	R/W
b7 to b4	IOD[3:0]	TGRD Control	See Table 28.19 and Table 28.20.*1	R/W

Note 1. If the IO[n:3:0] bit (n = C, D) values are changed to output disabled (0000b or 0100b) during low/high/toggle output on compare match, the TIOCCn/TIOCDn pin (n = 0, 3) is placed in high impedance state.

TPU has two TIORH registers, one for TPU0 and TPU3, and two TIORL registers, one for TPU0 and TPU3, and also has four TIOR registers, one for TPU1, TPU2, TPU4, and TPU5. Thus the TPU has eight timer I/O control registers in total.

TIORH, TIORL, and TIOR control registers TGRA, TGRB, TGRC, and TGRD.

Note that TIORH, TIORL, and TIOR are affected by the TMDR setting. For details, see Table 28.13 to Table 28.20.

The initial output specified by TIORH, TIORL, and TIOR is valid when the counter is stopped (the TPUA.TSTR.CSTj bit (j = 0 to 5) is cleared to 0). In PWM mode 2, the output at the time when the TCNT is cleared to 0 is specified as the initial output.

When buffer operation has been selected for register TGRC or TGRD, the settings of the IOC[3:0] or IOD[3:0] bits become ineffective, and the TGRC or TGRD register simply operates as a buffer.

To specify the input capture pin in TIORH, TIORL, or TIOR, set the bit in the port direction register (PDR) for the corresponding pin to 0 (input port), and set the bit in the port mode register (PMR) to 1 (uses the pin as an I/O port for peripheral functions). For details, see section 22, I/O Ports.

IOA[3:0] Bits (TGRA Control)

Select the function of TPU_m.TGRA (m = 0 to 5).

IOB[3:0] Bits (TGRB Control)

Select the function of TPU_m.TGRB (m = 0 to 5).

IOC[3:0] Bits (TGRC Control)

Select the function of TPU_m.TGRC (m = 0, 3).

IOD[3:0] Bits (TGRD Control)

Select the function of TPU_m.TGRD (m = 0, 3).

Table 28.13 TPU0.TIORH

Bits IOA[3:0]				Description	
b3	b2	b1	b0	TPU0.TGRA Function	TIOCA0 Pin Function and Related Issue
0	0	0	0	Output compare register	Output disabled
0	0	0	1		Initial output is low output; low output at compare match
0	0	1	0		Initial output is low output; high output at compare match
0	0	1	1		Initial output is low output; toggle output at compare match
0	1	0	0		Output disabled
0	1	0	1		Initial output is high output; low output at compare match
0	1	1	0		Initial output is high output; high output at compare match
0	1	1	1		Initial output is high output; toggle output at compare match
1	0	0	0	Input capture register	Capture input source is TIOCA0 pin; input capture at rising edge
1	0	0	1		Capture input source is TIOCA0 pin; input capture at falling edge
1	0	1	x		Capture input source is TIOCA0 pin; input capture at both edges
1	1	x	x		Capture input source is TPU1 count clock; input capture at TPU1.TCNT count-up/count-down*1

Bits IOB[3:0]				Description	
b7	b6	b5	b4	TPU0.TGRB Function	TIOCB0 Pin Function and Related Issue
0	0	0	0	Output compare register	Output disabled
0	0	0	1		Initial output is low output; low output at compare match
0	0	1	0		Initial output is low output; high output at compare match
0	0	1	1		Initial output is low output; toggle output at compare match
0	1	0	0		Output disabled
0	1	0	1		Initial output is high output; low output at compare match
0	1	1	0		Initial output is high output; high output at compare match
0	1	1	1		Initial output is high output; toggle output at compare match
1	0	0	0	Input capture register	Capture input source is TIOCB0 or TIOCA0 pin*2; input capture at rising edge
1	0	0	1		Capture input source is TIOCB0 or TIOCA0 pin*2; input capture at falling edge
1	0	1	x		Capture input source is TIOCB0 or TIOCA0 pin*2; input capture at both edges
1	1	x	x		Capture input source is TPU1 count clock; input capture at TPU1.TCNT count-up/count-down*1

x: Don't care

Note 1. When the TPSC[2:0] bits in TPU1.TCR are set to 000b and PCLK/1 is used as the TPU1.TCNT count clock, this setting is invalid and input capture is not generated.

Note 2. Selected by the ICSELB bit in TPU0.TMDR.

Table 28.14 TPU1.TIOR

Bits IOA[3:0]				Description	
b3	b2	b1	b0	TPU1.TGRA Function	TIOCA1 Pin Function and Related Issue
0	0	0	0	Output compare register	Output disabled
0	0	0	1		Initial output is low output; low output at compare match
0	0	1	0		Initial output is low output; high output at compare match
0	0	1	1		Initial output is low output; toggle output at compare match
0	1	0	0		Output disabled
0	1	0	1		Initial output is high output; low output at compare match
0	1	1	0		Initial output is high output; high output at compare match
0	1	1	1		Initial output is high output; toggle output at compare match
1	0	0	0	Input capture register	Capture input source is TIOCA1 pin; input capture at rising edge
1	0	0	1		Capture input source is TIOCA1 pin; input capture at falling edge
1	0	1	x		Capture input source is TIOCA1 pin; input capture at both edges
1	1	x	x		Capture input source is TPU0.TGRA compare match/input capture; input capture at generation of TPU0.TGRA compare match/input capture

Bits IOB[3:0]				Description	
b7	b6	b5	b4	TPU1.TGRB Function	TIOCB1 Pin Function and Related Issue
0	0	0	0	Output compare register	Output disabled
0	0	0	1		Initial output is low output; low output at compare match
0	0	1	0		Initial output is low output; high output at compare match
0	0	1	1		Initial output is low output; toggle output at compare match
0	1	0	0		Output disabled
0	1	0	1		Initial output is high output; low output at compare match
0	1	1	0		Initial output is high output; high output at compare match
0	1	1	1		Initial output is high output; toggle output at compare match
1	0	0	0	Input capture register	Capture input source is TIOCB1 or TIOCA1 pin ^{*1} ; input capture at rising edge
1	0	0	1		Capture input source is TIOCB1 or TIOCA1 pin ^{*1} ; input capture at falling edge
1	0	1	x		Capture input source is TIOCB1 or TIOCA1 pin ^{*1} ; input capture at both edges
1	1	x	x		Capture input source is TPU0.TGRC compare match/input capture; input capture at generation of TPU0.TGRC compare match/input capture

x: Don't care

Note 1. Selected by the ICSELB bit in TPU1.TMDR.

Table 28.15 TPU2.TIOR

Bits IOA[3:0]				Description	
b3	b2	b1	b0	TPU2.TGRA Function	TIOCA2 Pin Function and Related Issue
0	0	0	0	Output compare register	Output disabled
0	0	0	1		Initial output is low output; low output at compare match
0	0	1	0		Initial output is low output; high output at compare match
0	0	1	1		Initial output is low output; toggle output at compare match
0	1	0	0		Output disabled
0	1	0	1		Initial output is high output; low output at compare match
0	1	1	0		Initial output is high output; high output at compare match
0	1	1	1		Initial output is high output; toggle output at compare match
1	x	0	0	Input capture register	Capture input source is TIOCA2 pin; input capture at rising edge
1	x	0	1		Capture input source is TIOCA2 pin; input capture at falling edge
1	x	1	x		Capture input source is TIOCA2 pin; input capture at both edges

Bits IOB[3:0]				Description	
b7	b6	b5	b4	TPU2.TGRB Function	TIOCB2 Pin (Function and Related Issue)
0	0	0	0	Output compare register	Output disabled
0	0	0	1		Initial output is low output; low output at compare match
0	0	1	0		Initial output is low output; high output at compare match
0	0	1	1		Initial output is low output; toggle output at compare match
0	1	0	0		Output disabled
0	1	0	1		Initial output is high output; low output at compare match
0	1	1	0		Initial output is high output; high output at compare match
0	1	1	1		Initial output is high output; toggle output at compare match
1	x	0	0	Input capture register	Capture input source is TIOCB2 or TIOCA2 pin ^{*1} ; input capture at rising edge
1	x	0	1		Capture input source is TIOCB2 or TIOCA2 pin ^{*1} ; input capture at falling edge
1	x	1	x		Capture input source is TIOCB2 or TIOCA2 pin ^{*1} ; input capture at both edges

x: Don't care

Note 1. Selected by the ICSELB bit in TPU2.TMDR.

Table 28.16 TPU3.TIORH

Bits IOA[3:0]				Description	
b3	b2	b1	b0	TPU3.TGRA Function	TIOCA3 Pin Function and Related Issue
0	0	0	0	Output compare register	Output disabled
0	0	0	1		Initial output is low output; low output at compare match
0	0	1	0		Initial output is low output; high output at compare match
0	0	1	1		Initial output is low output; toggle output at compare match
0	1	0	0		Output disabled
0	1	0	1		Initial output is high output; low output at compare match
0	1	1	0		Initial output is high output; high output at compare match
0	1	1	1		Initial output is high output; toggle output at compare match
1	0	0	0	Input capture register	Capture input source is TIOCA3 pin; input capture at rising edge
1	0	0	1		Capture input source is TIOCA3 pin; input capture at falling edge
1	0	1	x		Capture input source is TIOCA3 pin; input capture at both edges
1	1	x	x		Capture input source is TPU4 count clock; input capture at TPU4.TCNT count-up/count-down*1

Bits IOB[3:0]				Description	
b7	b6	b5	b4	TPU3.TGRB Function	TIOCB3 Pin Function and Related Issue
0	0	0	0	Output compare register	Output disabled
0	0	0	1		Initial output is low output; low output at compare match
0	0	1	0		Initial output is low output; high output at compare match
0	0	1	1		Initial output is low output; toggle output at compare match
0	1	0	0		Output disabled
0	1	0	1		Initial output is high output; low output at compare match
0	1	1	0		Initial output is high output; high output at compare match
0	1	1	1		Initial output is high output; toggle output at compare match
1	0	0	0	Input capture register	Capture input source is TIOCB3 or TIOCA3 pin*2; input capture at rising edge
1	0	0	1		Capture input source is TIOCB3 or TIOCA3 pin*2; input capture at falling edge
1	0	1	x		Capture input source is TIOCB3 or TIOCA3 pin*2; input capture at both edges
1	1	x	x		Capture input source is TPU4 count clock; input capture at TPU4.TCNT count-up/count-down*1

x: Don't care

Note 1. When the TPSC[2:0] bits in TPU4.TCR are set to 000b and PCLK/1 is used as the TPU4.TCNT count clock, this setting is invalid and input capture is not generated.

Note 2. Selected by the ICSELB bit in TPU3.TMDR.

Table 28.17 TPU4.TIOR

Bits IOA[3:0]				Description	
b3	b2	b1	b0	TPU4.TGRA Function	TIOCA4 Pin Function and Related Issue
0	0	0	0	Output compare register	Output disabled
0	0	0	1		Initial output is low output; low output at compare match
0	0	1	0		Initial output is low output; high output at compare match
0	0	1	1		Initial output is low output; toggle output at compare match
0	1	0	0		Output disabled
0	1	0	1		Initial output is high output; low output at compare match
0	1	1	0		Initial output is high output; high output at compare match
0	1	1	1		Initial output is high output; toggle output at compare match
1	0	0	0	Input capture register	Capture input source is TIOCA4 pin; input capture at rising edge
1	0	0	1		Capture input source is TIOCA4 pin; input capture at falling edge
1	0	1	x		Capture input source is TIOCA4 pin; input capture at both edges
1	1	x	x		Capture input source is TPU3.TGRA compare match/input capture; input capture at generation of TPU3.TGRA compare match/input capture

Bits IOB[3:0]				Description	
b7	b6	b5	b4	TPU4.TGRB Function	TIOCB4 Pin Function and Related Issue
0	0	0	0	Output compare register	Output disabled
0	0	0	1		Initial output is low output; low output at compare match
0	0	1	0		Initial output is low output; high output at compare match
0	0	1	1		Initial output is low output; toggle output at compare match
0	1	0	0		Output disabled
0	1	0	1		Initial output is high output; low output at compare match
0	1	1	0		Initial output is high output; high output at compare match
0	1	1	1		Initial output is high output; toggle output at compare match
1	0	0	0	Input capture register	Capture input source is TIOCB4 or TIOCA4 pin*1; input capture at rising edge
1	0	0	1		Capture input source is TIOCB4 or TIOCA4 pin*1; input capture at falling edge
1	0	1	x		Capture input source is TIOCB4 or TIOCA4 pin*1; input capture at both edges
1	1	x	x		Capture input source is TPU3.TGRC compare match/input capture; input capture at generation of TPU3.TGRC compare match/input capture

x: Don't care

Note 1. Selected by the ICSELB bit in TPU4.TMDR.

Table 28.18 TPU5.TIOR

Bits IOA[3:0]				Description	
b3	b2	b1	b0	TPU5.TGRA Function	TIOCA5 Pin Function and Related Issue
0	0	0	0	Output compare register	Output disabled
0	0	0	1		Initial output is low output; low output at compare match
0	0	1	0		Initial output is low output; high output at compare match
0	0	1	1		Initial output is low output; toggle output at compare match
0	1	0	0		Output disabled
0	1	0	1		Initial output is high output; low output at compare match
0	1	1	0		Initial output is high output; high output at compare match
0	1	1	1		Initial output is high output; toggle output at compare match
1	x	0	0	Input capture register	Capture input source is TIOCA5 pin; input capture at rising edge
1	x	0	1		Capture input source is TIOCA5 pin; input capture at falling edge
1	x	1	x		Capture input source is TIOCA5 pin; input capture at both edges

Bits IOB[3:0]				Description	
b7	b6	b5	b4	TPU5.TGRB Function	TIOCB5 Pin Function and Related Issue
0	0	0	0	Output compare register	Output disabled
0	0	0	1		Initial output is low output; low output at compare match
0	0	1	0		Initial output is low output; high output at compare match
0	0	1	1		Initial output is low output; toggle output at compare match
0	1	0	0		Output disabled
0	1	0	1		Initial output is high output; low output at compare match
0	1	1	0		Initial output is high output; high output at compare match
0	1	1	1		Initial output is high output; toggle output at compare match
1	x	0	0	Input capture register	Capture input source is TIOCB5 or TIOCA5 pin ^{*1} ; input capture at rising edge
1	x	0	1		Capture input source is TIOCB5 or TIOCA5 pin ^{*1} ; input capture at falling edge
1	x	1	x		Capture input source is TIOCB5 or TIOCA5 pin ^{*1} ; input capture at both edges

x: Don't care

Note 1. Selected by the ICSELB bit in TPU5.TMDR.

Table 28.19 TPU0.TIORL

Bits IOC[3:0]				Description	
b3	b2	b1	b0	TPU0.TGRC Function	TIOCC0 Pin Function and Related Issue
0	0	0	0	Output compare register*1	Output disabled
0	0	0	1		Initial output is low output; low output at compare match
0	0	1	0		Initial output is low output; high output at compare match
0	0	1	1		Initial output is low output; toggle output at compare match
0	1	0	0		Output disabled
0	1	0	1		Initial output is high output; low output at compare match
0	1	1	0		Initial output is high output; high output at compare match
0	1	1	1		Initial output is high output; toggle output at compare match
1	0	0	0	Input capture register*1	Capture input source is TIOCC0 pin; input capture at rising edge
1	0	0	1		Capture input source is TIOCC0 pin; input capture at falling edge
1	0	1	x		Capture input source is TIOCC0 pin; input capture at both edges
1	1	x	x		Capture input source is TPU1 count clock; input capture at TPU1.TCNT count-up/count-down*3

Bits IOD[3:0]				Description	
b7	b6	b5	b4	TPU0.TGRD Function	TIOCD0 Pin Function and Related Issue
0	0	0	0	Output compare register*2	Output disabled
0	0	0	1		Initial output is low output; low output at compare match
0	0	1	0		Initial output is low output; high output at compare match
0	0	1	1		Initial output is low output; toggle output at compare match
0	1	0	0		Output disabled
0	1	0	1		Initial output is high output; low output at compare match
0	1	1	0		Initial output is high output; high output at compare match
0	1	1	1		Initial output is high output; toggle output at compare match
1	0	0	0	Input capture register*2	Capture input source is TIOCD0 or TIOCC0 pin*4; input capture at rising edge
1	0	0	1		Capture input source is TIOCD0 or TIOCC0 pin*4; input capture at falling edge
1	0	1	x		Capture input source is TIOCD0 or TIOCC0 pin*4; input capture at both edges
1	1	x	x		Capture input source is TPU1 count clock; input capture at TPU1.TCNT count-up/count-down*3

x: Don't care

Note 1. When the BFA bit in TPU0.TMDR is set to 1 (TPU0.TGRA and TPU0.TGRC are used for buffer operation) and TPU0.TGRC is used as a buffer register, this setting is invalid and input capture/output compare is not generated.

Note 2. When the BFB bit in TPU0.TMDR is set to 1 (TPU0.TGRB and TPU0.TGRD are used for buffer operation) and TPU0.TGRD is used as a buffer register, this setting is invalid and input capture/output compare is not generated.

Note 3. When the TPSC[2:0] bits in TPU1.TCR are set to 000b and PCLK/1 is used as the TPU0.TCNT count clock, this setting is invalid and input capture is not generated.

Note 4. Selected by the ICSELD bit in TPU0.TMDR.

Table 28.20 TPU3.TIORL

Bits IOC[3:0]				Description	
b3	b2	b1	b0	TPU3.TGRC Function	TIOCC3 Pin Function and Related Issue
0	0	0	0	Output compare register*1	Output disabled
0	0	0	1		Initial output is low output; low output at compare match
0	0	1	0		Initial output is low output; high output at compare match
0	0	1	1		Initial output is low output; toggle output at compare match
0	1	0	0		Output disabled
0	1	0	1		Initial output is high output; low output at compare match
0	1	1	0		Initial output is high output; high output at compare match
0	1	1	1		Initial output is high output; toggle output at compare match
1	0	0	0	Input capture register*1	Capture input source is TIOCC3 pin; input capture at rising edge
1	0	0	1		Capture input source is TIOCC3 pin; input capture at falling edge
1	0	1	x		Capture input source is TIOCC3 pin; input capture at both edges
1	1	x	x		Capture input source is TPU4 count clock; input capture at TPU4.TCNT count-up/count-down*3

Bits IOD[3:0]				Description	
b7	b6	b5	b4	TPU3.TGRD Function	TIOCD3 Pin Function and Related Issue
0	0	0	0	Output compare register*2	Output disabled
0	0	0	1		Initial output is low output; low output at compare match
0	0	1	0		Initial output is low output; high output at compare match
0	0	1	1		Initial output is low output; toggle output at compare match
0	1	0	0		Output disabled
0	1	0	1		Initial output is high output; low output at compare match
0	1	1	0		Initial output is high output; high output at compare match
0	1	1	1		Initial output is high output; toggle output at compare match
1	0	0	0	Input capture register*2	Capture input source is TIOCD3 or TIOCC3 pin*4; input capture at rising edge
1	0	0	1		Capture input source is TIOCD3 or TIOCC3 pin*4; input capture at falling edge
1	0	1	x		Capture input source is TIOCD3 or TIOCC3 pin*4; input capture at both edges
1	1	X	x		Capture input source is TPU4 count clock; input capture at TPU4.TCNT count-up/count-down*3

x: Don't care

Note 1. When the BFA bit in TPU3.TMDR is set to 1 (TPU3.TGRA and TPU3.TGRC are used for buffer operation) and TPU3.TGRC is used as a buffer register, this setting is invalid and input capture/output compare is not generated.

Note 2. When the BFB bit in TPU3.TMDR is set to 1 (TPU3.TGRB and TPU3.TGRD are used for buffer operation) and TPU3.TGRD is used as a buffer register, this setting is invalid and input capture/output compare is not generated.

Note 3. When the TPSC[2:0] bits in TPU4.TCR are set to 000b and PCLK/1 is used as the TPU3.TCNT count clock, this setting is invalid and input capture is not generated.

Note 4. Selected by the ICSELD bit in TPU3.TMDR.

28.2.4 Timer Interrupt Enable Register (TIER)

Address(es): TPU0.TIER 0008 8114h, TPU1.TIER 0008 8124h, TPU2.TIER 0008 8134h, TPU3.TIER 0008 8144h, TPU4.TIER 0008 8154h, TPU5.TIER 0008 8164h

b7	b6	b5	b4	b3	b2	b1	b0
TTGE	—	TCIEU	TCIEV	TGIED	TGIEC	TGIEB	TGIEA

Value after reset: 0 1 0 0 0 0 0 0

Bit	Symbol	Bit Name	Description	R/W
b0	TGIEA	TGRA Interrupt Enable	0: Interrupt requests (TGImA) disabled 1: Interrupt requests (TGImA) enabled (m = 0 to 5)	R/W
b1	TGIEB	TGRB Interrupt Enable	0: Interrupt requests (TGImB) disabled 1: Interrupt requests (TGImB) enabled (m = 0 to 5)	R/W
b2	TGIEC	TGRC Interrupt Enable* ¹	0: Interrupt requests (TGImC) disabled 1: Interrupt requests (TGImC) enabled (m = 0, 3)	R/W
b3	TGIED	TGRD Interrupt Enable* ¹	0: Interrupt requests (TGImD) disabled 1: Interrupt requests (TGImD) enabled (m = 0, 3)	R/W
b4	TCIEV	Overflow Interrupt Enable	0: Interrupt requests (TCImV) disabled 1: Interrupt requests (TCImV) enabled (m = 0 to 5)	R/W
b5	TCIEU	Underflow Interrupt Enable* ²	0: Interrupt requests (TCImU) disabled 1: Interrupt requests (TCImU) enabled (m = 1, 2, 4, 5)	R/W
b6	—	Reserved	This bit is read as 1. The write value should be 1.	R/W
b7	TTGE	A/D Conversion Start Request Enable* ³	0: A/D conversion start request generation disabled 1: A/D conversion start request generation enabled	R/W

Note 1. These bits are reserved in TPU1, TPU2, TPU4, and TPU5. These bits are read as 0. The write value should be 0.

Note 2. This bit is reserved in TPU0 and TPU3. This bit is read as 0. The write value should be 0.

Note 3. This bit is reserved in TPU5. This bit is read as 0. The write value should be 0.

TTGE Bit (A/D Conversion Start Request Enable)

Enables/disables generation of A/D conversion start requests by TPU_m.TGRA (m = 0 to 4) input capture/compare match.

28.2.5 Timer Status Register (TSR)

Address(es): TPU0.TSR 0008 8115h, TPU1.TSR 0008 8125h, TPU2.TSR 0008 8135h,
TPU3.TSR 0008 8145h, TPU4.TSR 0008 8155h, TPU5.TSR 0008 8165h

b7	b6	b5	b4	b3	b2	b1	b0
TCFD	—	TCFU	TCFV	TGFD	TGFC	TGFB	TGFA

Value after reset: 1 1 0 0 0 0 0 0

Bit	Symbol	Bit Name	Description	R/W
b0	TGFA	Input Capture/Output Compare Flag A	0: Input capture to TPUm.TGRA or compare match with TPUm.TGRA has not occurred. 1: Input capture to TPUm.TGRA or compare match with TPUm.TGRA has occurred. (m = 0 to 5)	R/W*1
b1	TGFB	Input Capture/Output Compare Flag B	0: Input capture to TPUm.TGRB or compare match with TPUm.TGRB has not occurred. 1: Input capture to TPUm.TGRB or compare match with TPUm.TGRB has occurred. (m = 0 to 5)	R/W*1
b2	TGFC	Input Capture/Output Compare Flag C*2	0: Input capture to TPUm.TGRC or compare match with TPUm.TGRC has not occurred. 1: Input capture to TPUm.TGRC or compare match with TPUm.TGRC has occurred. (m = 0, 3)	R/W*1
b3	TGFD	Input Capture/Output Compare Flag D*2	0: Input capture to TPUm.TGRD or compare match with TPUm.TGRD has not occurred. 1: Input capture to TPUm.TGRD or compare match with TPUm.TGRD has occurred. (m = 0, 3)	R/W*1
b4	TCFV	Overflow Flag	0: TPUm.TCNT has not overflowed. 1: TPUm.TCNT has overflowed. (m = 0 to 5)	R/W*1
b5	TCFU	Underflow Flag*3	0: TPUm.TCNT has not underflowed. 1: TPUm.TCNT has underflowed. (m = 1, 2, 4, 5)	R/W*1
b6	—	Reserved	This bit is read as 1. The write value should be 1.	R/W
b7	TCFD	Counting Direction Flag*4	0: TPUm.TCNT counts down. 1: TPUm.TCNT counts up. (m = 1, 2, 4, 5)	R

Note 1. Only writing 0 to this bit is possible; this clears the flag.

Note 2. These bits are reserved in TPU1, TPU2, TPU4, and TPU5. The bits are read as 0. The write value should be 0.

Note 3. This bit is reserved in TPU0 and TPU3. The bit is read as 0. The write value should be 0.

Note 4. This bit is reserved in TPU0 and TPU3. The bit is read as 1. The write value should be 1.

TGFA Flag (Input Capture/Output Compare Flag A)

This status flag indicates that input capture to TPUm.TGRA or compare match with TPUm.TGRA (m = 0 to 5) has occurred.

[Setting conditions]

- When TPUm.TGRA holds the value for comparison in output-compare operations, TPUm.TCNT matches TPUm.TGRA.
- When TPUm.TGRA is serving as an input-capture register, the input-capture signal has caused transfer of the value in TPUm.TCNT to TPUm.TGRA.

[Clearing conditions]

- Activation of the DTC by the TGImA interrupt and clearing of the DTC.MRB.DISEL bit.
- Writing 0 to TGFA after reading its value as 1.

TGFB Flag (Input Capture/Output Compare Flag B)

This status flag indicates that input capture to TPUm.TGRB or compare match with TPUm.TGRB (m = 0 to 5) has occurred.

[Setting conditions]

- When TPUm.TGRB holds the value for comparison in output-compare operations, TPUm.TCNT matches TPUm.TGRB.
- When TPUm.TGRB is serving as an input-capture register, the input-capture signal has caused transfer of the value in TPUm.TCNT to TPUm.TGRB.

[Clearing conditions]

- Activation of the DTC by the TGImB interrupt and clearing of the DTC.MRB.DISEL bit.
- Writing 0 to TGFB after reading its value as 1.

TGFC Flag (Input Capture/Output Compare Flag C)

This status flag indicates that input capture to TPUm.TGRC or compare match with TPUm.TGRC (m = 0, 3) has occurred.

[Setting conditions]

- When TPUm.TGRC holds the value for comparison in output-compare operations, TPUm.TCNT matches TPUm.TGRC.
- When TPUm.TGRC is serving as an input-capture register, the input-capture signal has caused transfer of the value in TPUm.TCNT to TPUm.TGRC.

[Clearing conditions]

- Activation of the DTC by the TGImC interrupt and clearing of the DTC.MRB.DISEL bit.
- Writing 0 to TGFC after reading its value as 1.

TGFD Flag (Input Capture/Output Compare Flag D)

This status flag indicates that input capture to TPUm.TGRD or compare match with TPUm.TGRD (m = 0, 3) has occurred.

[Setting conditions]

- When TPUm.TGRD holds the value for comparison in output-compare operations, TPUm.TCNT matches TPUm.TGRD.
- When TPUm.TGRD is serving as an input-capture register, the input-capture signal has caused transfer of the value in TPUm.TCNT to TPUm.TGRD.

[Clearing conditions]

- Activation of the DTC by the TGImD interrupt and clearing of the DTC.MRB.DISEL bit.
- Writing 0 to TGFD after reading its value as 1.

TCFV Flag (Overflow Flag)

This status flag indicates an overflow of TPUm.TCNT (m = 0 to 5).

[Setting condition]

- Overflow of the value in TPUm.TCNT (TCNT counted from FFFFh to 0000h).

[Clearing condition]

- Writing 0 to TCFV after reading its value as 1.

TCFU Flag (Underflow Flag)

This status flag indicates an underflow of TPUm.TCNT (m = 1, 2, 4, 5).

[Setting condition]

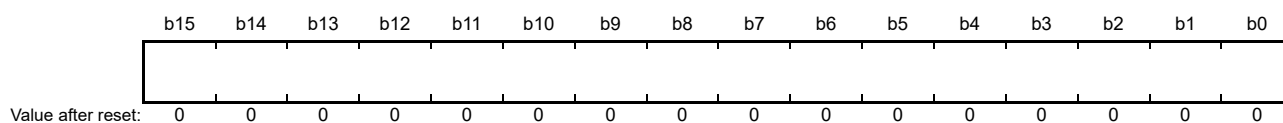
- Underflow of the value in TPUm.TCNT (TCNT counted from 0000h to FFFFh).

[Clearing condition]

- Writing 0 to TCFU after reading its value as 1.

28.2.6 Timer Counter (TCNT)

Address(es): TPU0.TCNT 0008 8116h, TPU1.TCNT 0008 8126h, TPU2.TCNT 0008 8136h,
TPU3.TCNT 0008 8146h, TPU4.TCNT 0008 8156h, TPU5.TCNT 0008 8166h



TPUm.TCNT is a readable/writable counter that counts the internal clock or external events.

28.2.7 Timer General Register A (TGRA), Timer General Register B (TGRB), Timer General Register C (TGRC), Timer General Register D (TGRD)

Address(es): TPU0.TGRA 0008 8118h, TPU0.TGRB 0008 811Ah, TPU0.TGRC 0008 811Ch, TPU0.TGRD 0008 811Eh,
TPU1.TGRA 0008 8128h, TPU1.TGRB 0008 812Ah,
TPU2.TGRA 0008 8138h, TPU2.TGRB 0008 813Ah,
TPU3.TGRA 0008 8148h, TPU3.TGRB 0008 814Ah, TPU3.TGRC 0008 814Ch, TPU3.TGRD 0008 814Eh,
TPU4.TGRA 0008 8158h, TPU4.TGRB 0008 815Ah,
TPU5.TGRA 0008 8168h, TPU5.TGRB 0008 816Ah



TPU has 16 TGR registers in total, four each for TPU0 and TPU3, and two each for TPU1, TPU2, TPU4, and TPU5. TPUm.TGRA ($m = 0$ to 5), TPUm.TGRB ($m = 0$ to 5), TPUm.TGRC ($m = 0, 3$), and TPUm.TGRD ($m = 0, 3$) are readable/writable registers with a dual function as output compare and input capture registers.

TPUm.TGRC and TPUm.TGRD can also be specified for operation as buffer registers. Register combinations during buffer operations are TPUm.TGRA—TPUm.TGRC and TPUm.TGRB—TPUm.TGRD.

28.2.8 Timer Start Register (TSTR)

Address(es): TPUA.TSTR 0008 8100h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	CST5	CST4	CST3	CST2	CST1	CST0
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	CST0	Counter Start 0	0: TCNT count operation is stopped 1: TCNT performs count operation	R/W
b1	CST1	Counter Start 1		R/W
b2	CST2	Counter Start 2		R/W
b3	CST3	Counter Start 3		R/W
b4	CST4	Counter Start 4		R/W
b5	CST5	Counter Start 5		R/W
b7, b6	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

TPUA.TSTR starts or stops TCNT operation for TPU0 to TPU5.

Before setting the operating mode in TPUm.TMDR or setting the TPUm.TCNT count clock in TPUm.TCR, stop the TPUm.TCNT operation.

CSTn Bit (Counter Start n) (n = 0 to 5)

This bit starts or stop the TCNT.

When the CSTn bit is cleared to 0 with CSTn = 1 and the corresponding TIOCyn pin (y = A to D; n = 0 to 5) specified for output, count operation stops but the output compare output level of the corresponding TIOCyn pin is retained.

If TIORH, TIORL, or TIOR is written to when the CSTn bit is 0, the pin output level will be changed to the set initial output value.

28.2.9 Timer Synchronous Register (TSYR)

Address(es): TPUA.TSYR 0008 8101h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	SYNC5	SYNC4	SYNC3	SYNC2	SYNC1	SYNC0
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	SYNC0	Timer Synchronization 0	0: TCNT operates independently (TCNT setting/clearing is unrelated to other channels)	R/W
b1	SYNC1	Timer Synchronization 1	1: TCNT performs synchronous operation*1 (TCNT synchronous setting/synchronous clearing is possible)	R/W
b2	SYNC2	Timer Synchronization 2		R/W
b3	SYNC3	Timer Synchronization 3		R/W
b4	SYNC4	Timer Synchronization 4		R/W
b5	SYNC5	Timer Synchronization 5		R/W
b7, b6	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. To set synchronous operation, the SYNCn bit (n = 0 to 5) for at least two channels must be set to 1. To set synchronous clearing, the TCNT clearing source must also be set by the TCR.CCLR[2:0] bits in addition to the SYNCn bit.

TPUA.TSYR selects independent operation or synchronous operation for TCNT of TPU0 to TPU5.

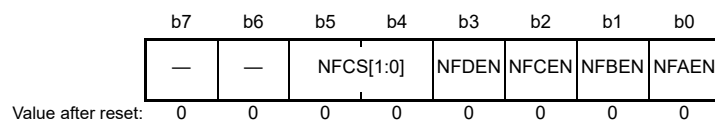
SYNCn Bit (Timer Synchronization n) (n = 0 to 5)

This bit selects whether the TCNT operation is independent of or synchronized with TCNT of other channels.

When synchronous operation is selected, synchronous setting of multiple TCNT and synchronous clearing through counter clearing on another channel are possible.

28.2.10 Noise Filter Control Register (NFCR)

Address(es): TPU0.NFCR 0008 8108h, TPU1.NFCR 0008 8109h, TPU2.NFCR 0008 810Ah,
TPU3.NFCR 0008 810Bh, TPU4.NFCR 0008 810Ch, TPU5.NFCR 0008 810Dh



Bit	Symbol	Bit Name	Description	R/W
b0	NFAEN	Noise Filter Enable A	0: The noise filter for TIOCAm is disabled. 1: The noise filter for TIOCAm is enabled. (m = 0 to 5)	R/W
b1	NFBEN	Noise Filter Enable B	0: The noise filter for TIOCBm is disabled. 1: The noise filter for TIOCBm is enabled. (m = 0 to 5)	R/W
b2	NFCEN	Noise Filter Enable C*1	0: The noise filter for TIOCCm is disabled. 1: The noise filter for TIOCCm is enabled. (m = 0, 3)	R/W
b3	NFDEN	Noise Filter Enable D*1	0: The noise filter for TIOCDm is disabled. 1: The noise filter for TIOCDm is enabled. (m = 0, 3)	R/W
b5, b4	NFCS[1:0]	Noise Filter Clock Select	b5 b4 0 0: PCLK/1 0 1: PCLK/8 1 0: PCLK/32 1 1: Clock source that drives counting	R/W
b7, b6	—	Reserved	These bits are read as 0. Writing to these bits is not possible.	R

Note 1. These bits are reserved in TPU1, TPU2, TPU4, and TPU5. The bits are read as 0. Writing to these bits is not possible.

Only set the TPUm.NFCR registers while the TPUm.TCNT is stopped.

NFAEN Bit (Noise Filter Enable A)

This bit disables or enables the noise filter for the TIOCAm pin (m = 0 to 5).

Since unexpected edges may be internally generated when the value of NFAEN is changed, select the output compare function in the timer I/O control register before changing the NFAEN value.

NFBEN Bit (Noise Filter Enable B)

This bit disables or enables the noise filter for the TIOCBm pin (m = 0 to 5).

Since unexpected edges may be internally generated when the value of NFBEN is changed, select the output compare function in the timer I/O control register before changing the NFBEN value.

NFCEN Bit (Noise Filter Enable C)

This bit disables or enables the noise filter for the TIOCCm pin (m = 0, 3).

Since unexpected edges may be internally generated when the value of NFCEN is changed, select the output compare function in the timer I/O control register before changing the NFCEN value.

NFDEN Bit (Noise Filter Enable D)

This bit disables or enables the noise filter for the TIOCDm pin ($m = 0, 3$).

Since unexpected edges may be internally generated when the value of NFDEN is changed, select the output compare function in the timer I/O control register before changing the NFDEN value.

NFCS[1:0] Bits (Noise Filter Clock Select)

These bits select the sampling clock for the noise filter.

When the count source is selected with NFCS[1:0] bits set to 11b, the clock that can be used as sampling clock are the internal clocks other than PCLK/1 specified with the TPSC[2:0] bits and the external clock. To select the PCLK/1 as both the count clock and the sampling clock, set the NFCS[1:0] bits to 00b.

The input-capture signal is sampled on rising edges of the selected clock signal. If the sampled levels match three times in a row, the given level is passed through as the input-capture signal. If the levels do not match, the existing value is retained.

After setting the NFCS[1:0] bits, wait for two selected sampling periods before setting the input capture function.

28.3 Operation

28.3.1 Basic Functions

Each channel has a TPUm.TCNT and a TPUm.TGRy register (y = A to D).

TCNT is a 16-bit up-counter, which can function as a free-running counter, periodic counter, or event counter.

TGRy can be used as an input capture register or output compare register.

(1) Counter Operation

When the CSTj bit (j = 0 to 5) in TPUA.TSTR is set to 1, the TCNT for the corresponding channel starts counting.

(a) Example of count operation setting procedure

Figure 28.2 shows an example of the count operation setting procedure.

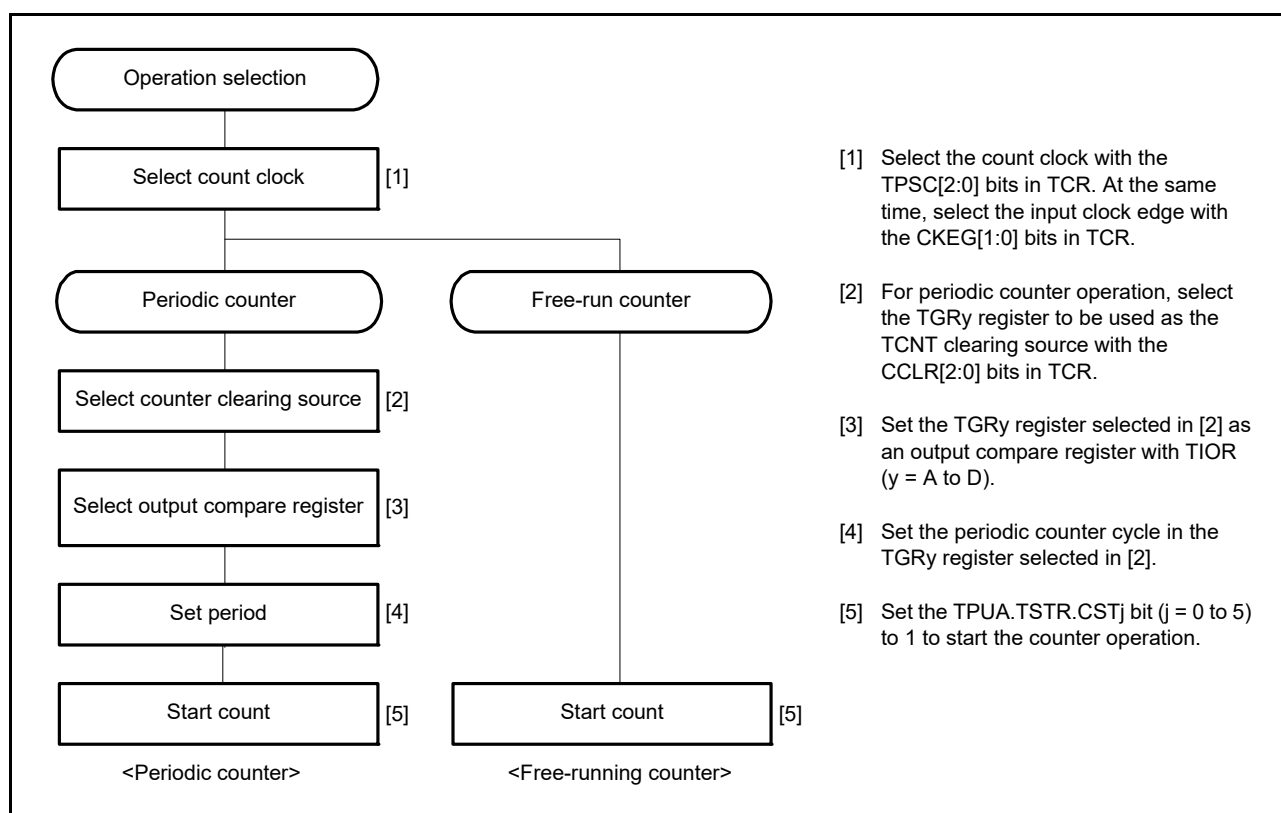


Figure 28.2 Example of Counter Operation Setting Procedure

(b) Free-running count operation and periodic count operation

Immediately after a reset, TPUM.TCNT are all set as free-running counters. When the relevant bit in TPUA.TSTR is set to 1, the corresponding TCNT starts up-count operation as a free-running counter. When TCNT overflows (changes from FFFFh to 0000h), the TPU requests an interrupt. After an overflow, TCNT restarts counting up from 0000h.

Figure 28.3 shows free-running counter operation.

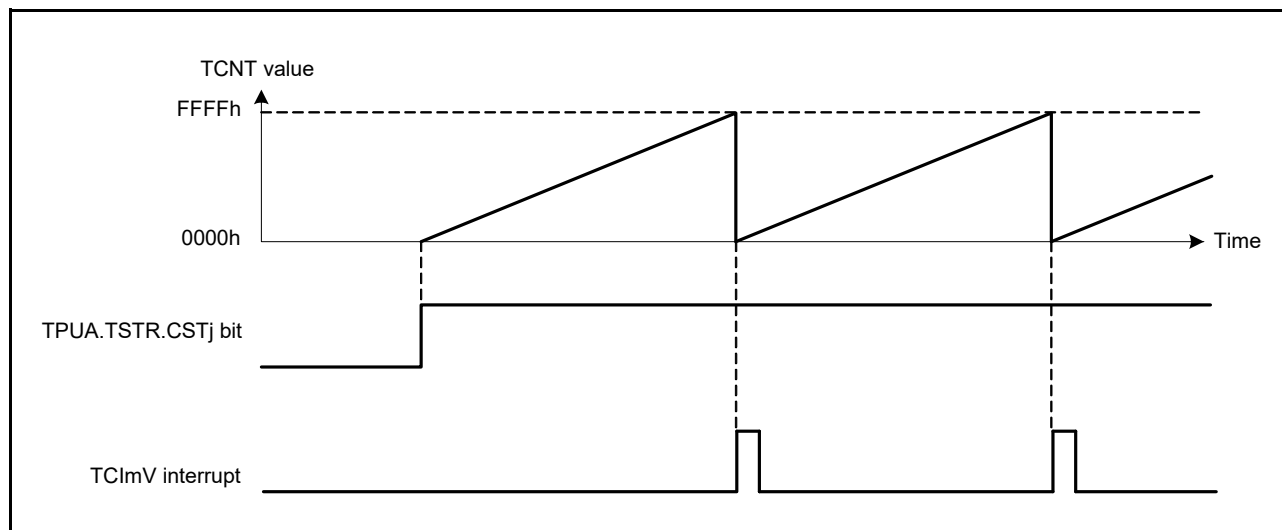


Figure 28.3 Free-Running Counter Operation

When compare match is selected as the TCNT clearing source, the TCNT for the relevant channel performs periodic count operation. The TPUM.TGRy for setting the period is set as an output compare register, and counter clearing by compare match is selected by the TPUM.TCR.CCLR[2:0] bits. After the settings have been made, TCNT starts count-up operation as a periodic counter when the corresponding bit in TPUA.TSTR is set to 1. When the count value matches the TGRy value, TCNT is cleared to 0000h.

At this time, the TPU requests an interrupt. After a compare match, TCNT restarts counting up from 0000h.

Figure 28.4 shows periodic counter operation.

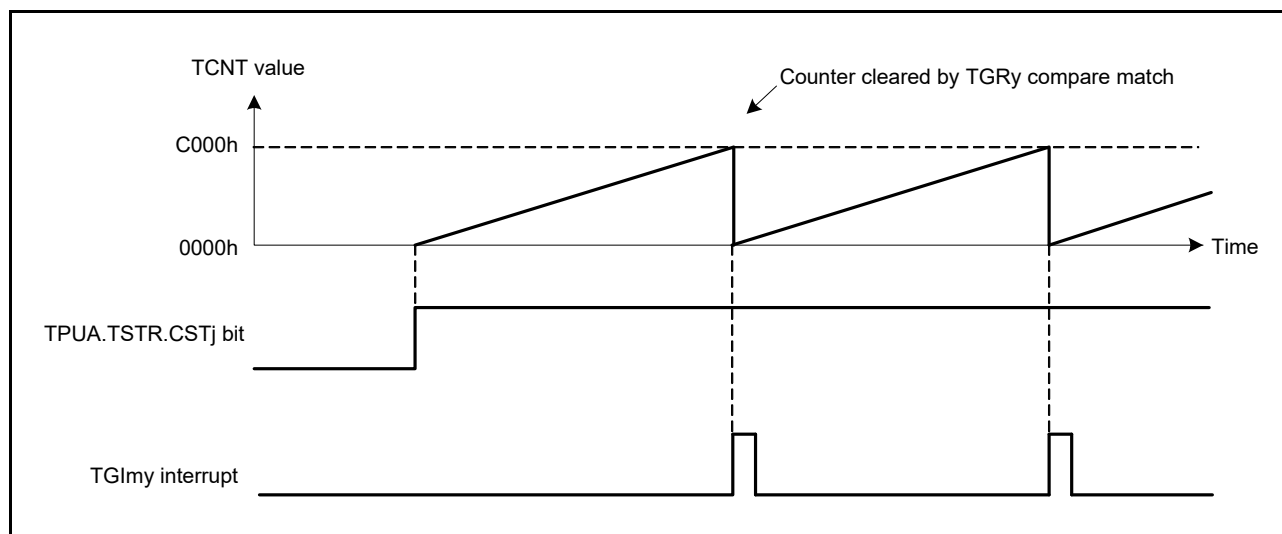


Figure 28.4 Periodic Counter Operation

(2) Waveform Output by Compare Match

The TPU can perform low, high, or toggle output from the corresponding output pin using a compare match.

(a) Example of setting procedure for waveform output by compare match

Figure 28.5 shows an example of the setting procedure for waveform output by a compare match.

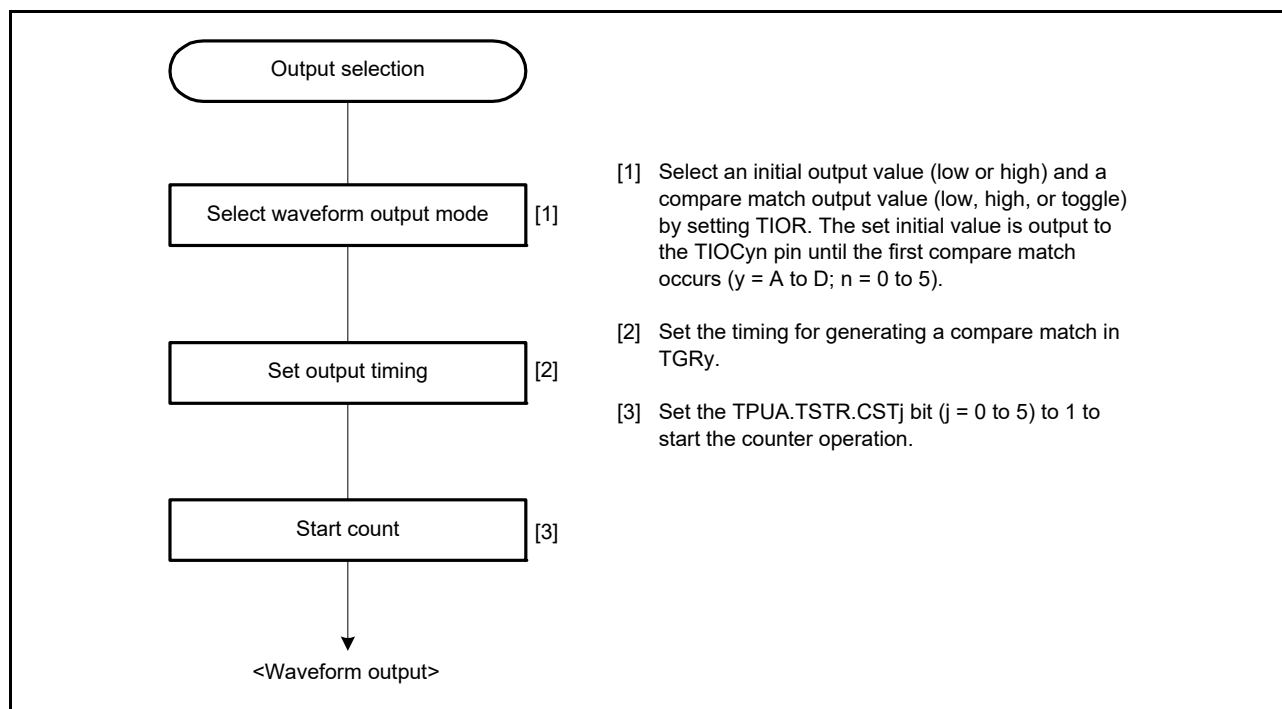


Figure 28.5 Example of Setting Procedure for Waveform Output by Compare Match

(b) Examples of waveform output operation

Figure 28.6 shows an example of low output/high output.

In this example, TPUm.TCNT has been set as a free-running counter, and settings have been made so that high is output by compare match A and low is output by compare match B. When the set level and the pin level match, the pin level does not change.

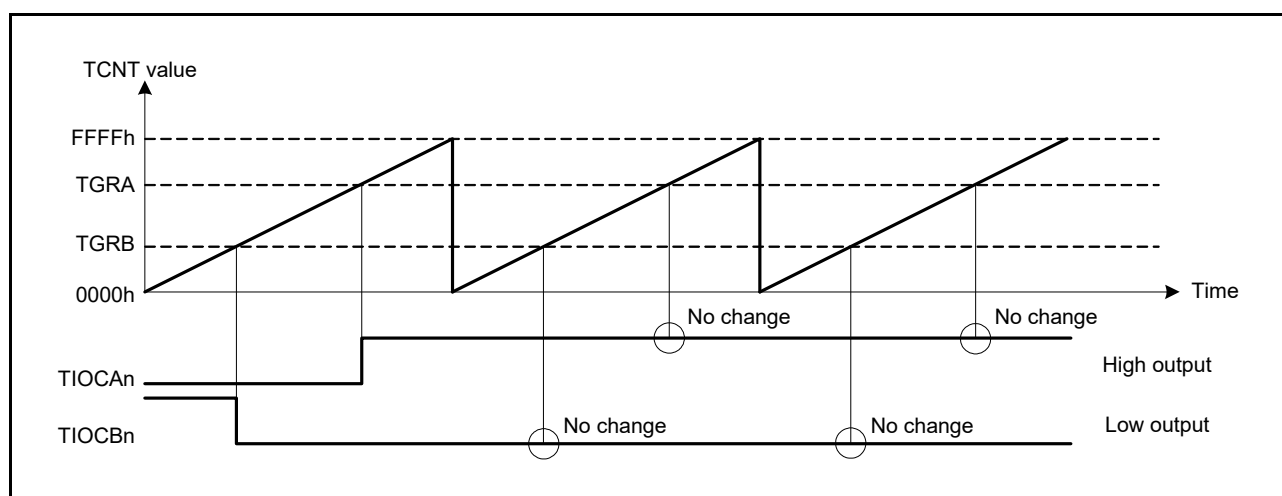


Figure 28.6 Example of Low-Output/High-Output Operation (n = 0 to 5)

Figure 28.7 shows an example of toggle output.

In this example, TPUm.TCNT has been set as a periodic counter (with counter clearing performed by compare match B), and settings have been made so that output is toggled by both compare match A and compare match B.

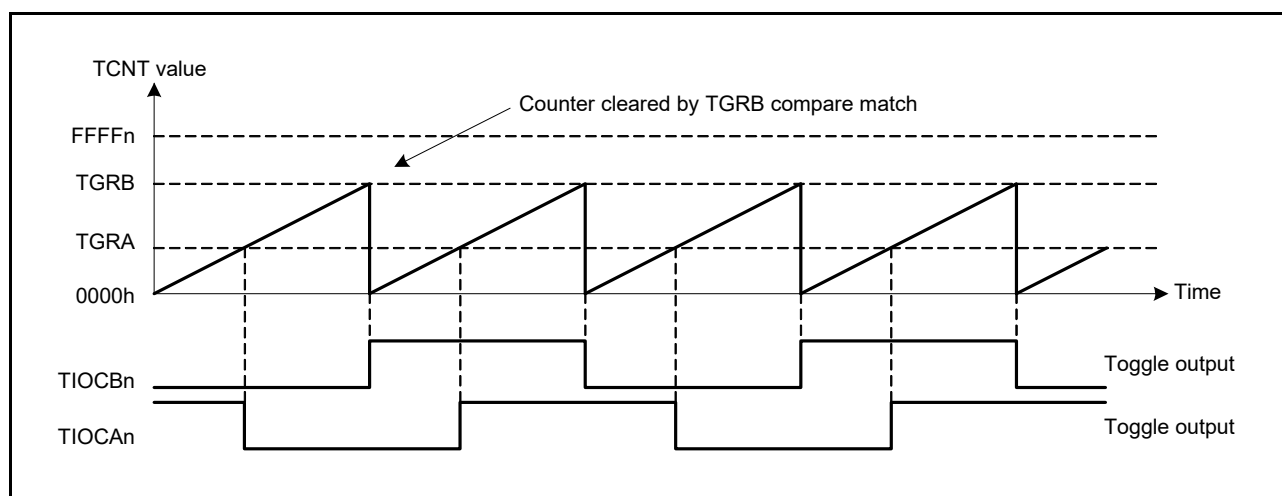


Figure 28.7 Example of Toggle Output Operation (n = 0 to 5)

(3) Input Capture Function

The TPUm.TCNT value can be transferred to TPUm.TGRy on detection of the TIOCyn pin (y = A to D; n = 0 to 5) input edge.

The rising edge, the falling edge, or both edges can be selected as the detection edge. It is also possible to specify the count clock or compare match signal of TPU0, TPU1, TPU3, and TPU4 as the input capture source. Noise filtering can be applied to the input capture input.

Note: Even if the counter is halted, an input capture is generated, and flag and interrupt signals are generated.

Note: When another channel's count clock is used as the input capture input for TPU0 and TPU3, PCLK/1 should not be selected as the count clock used for input capture input. Input capture will not be generated if PCLK/1 is selected.

(a) Example of setting procedure for input capture operation

Figure 28.8 shows an example of the setting procedure for input capture operation.

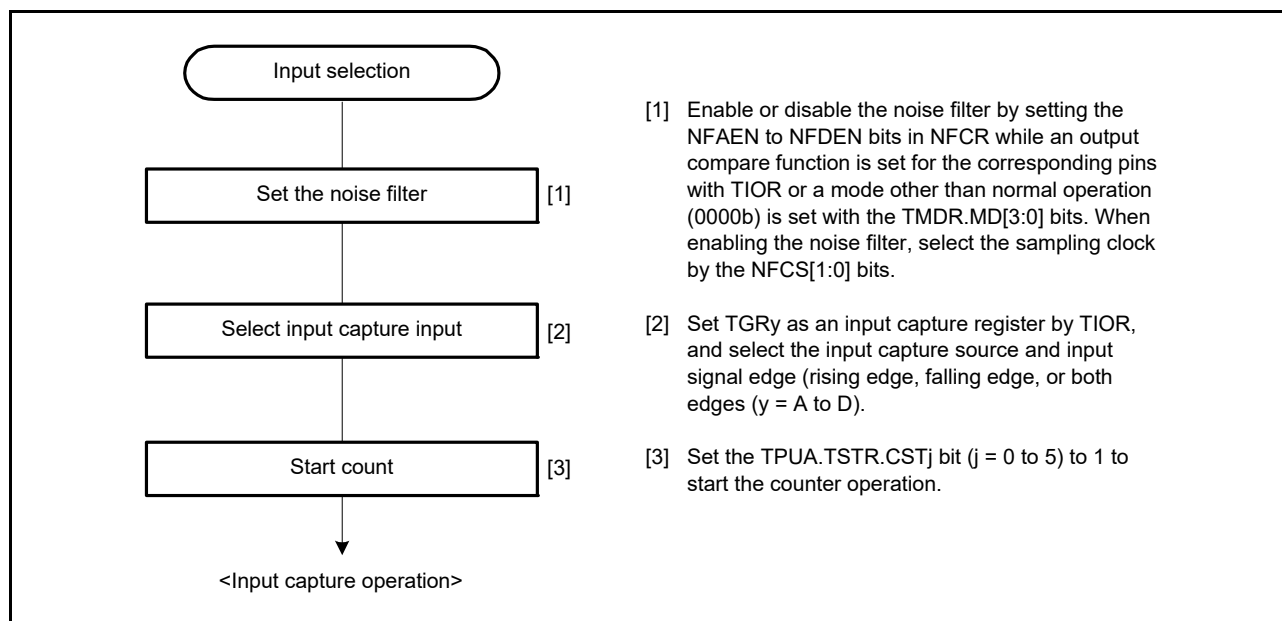


Figure 28.8 Example of Setting Procedure for Input Capture Operation

(b) Example of input capture operation

Figure 28.9 shows an example of input capture operation when the noise filter is stopped.

In this example, both rising and falling edges have been selected as the TIOCA_n pin input capture input edge, the falling edge has been selected as the TIOCB_n pin input capture input edge, and counter clearing by TPUm.TGRB input capture has been set for TPUm.TCNT.

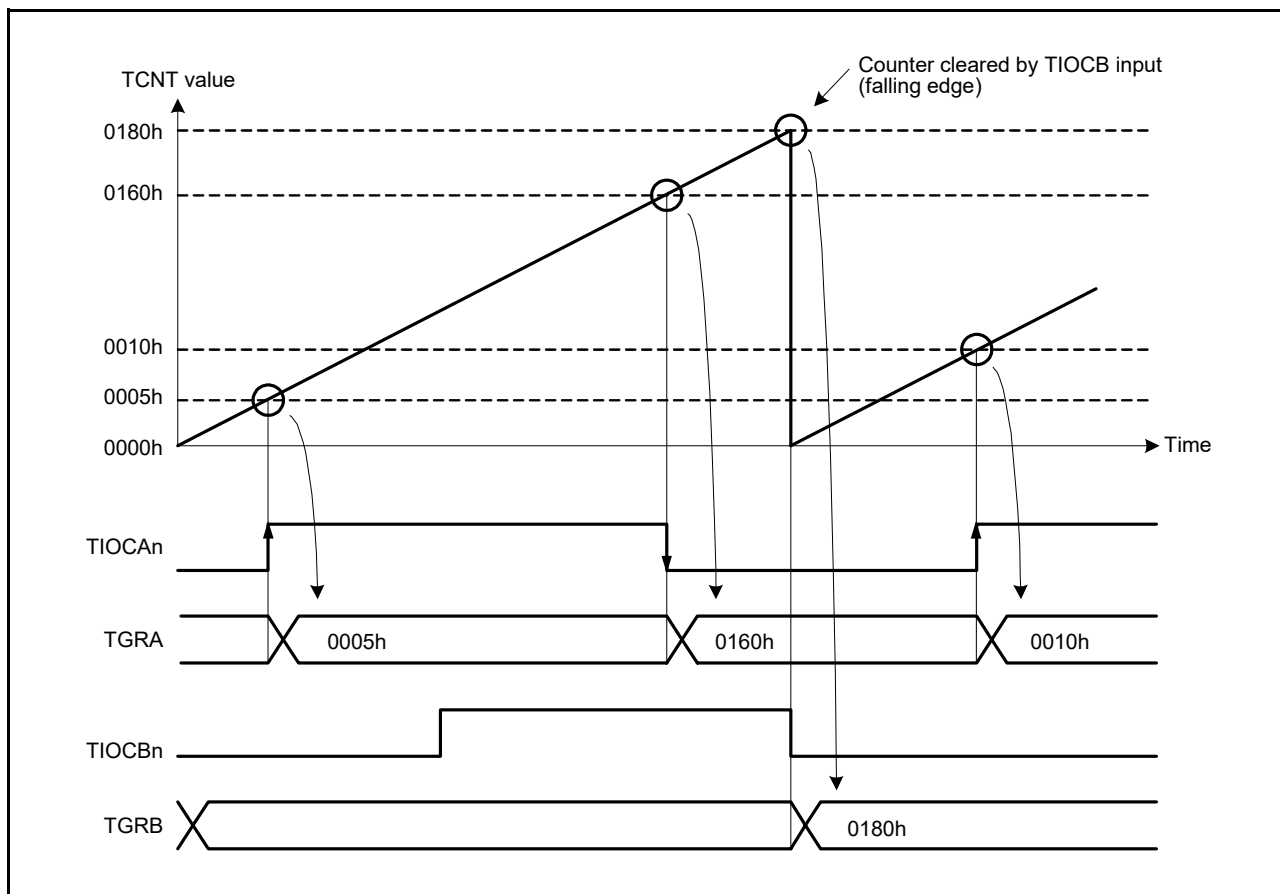


Figure 28.9 Example of Input Capture Operation (with Noise Filter Stopped) (n = 0 to 5)

When noise filtering is enabled, see Figure 28.30.

28.3.2 Synchronous Operation

In synchronous operation, the values in multiple TPUm.TCNT can be rewritten simultaneously (synchronous setting). Also, multiple TCNT can be cleared simultaneously (synchronous clearing) by making the appropriate setting in TPUm.TCR.

Synchronous operation enables TPUm.TGRy to be incremented with respect to a single time base.

TPU0 to TPU5 can all be set for synchronous operation.

(1) Example of Synchronous Operation Setting Procedure

Figure 28.10 shows an example of the synchronous operation setting procedure.

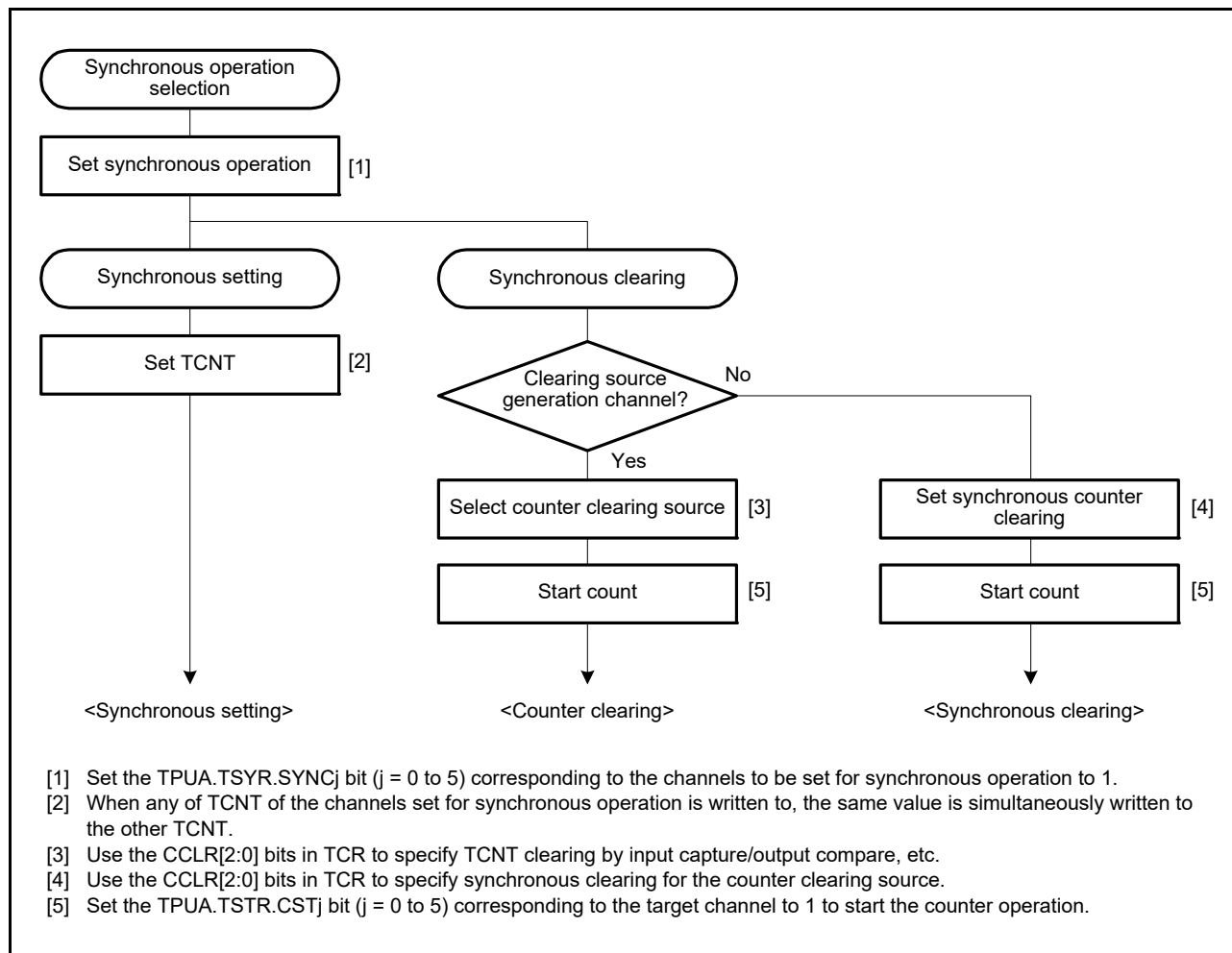


Figure 28.10 Example of Synchronous Operation Setting Procedure

(2) Example of Synchronous Operation

Figure 28.11 shows an example of synchronous operation.

In this example, synchronous operation and PWM mode 1 have been set for TPU0 to TPU2, TPU0.TGRB compare match has been set as the TPU0 counter clearing source, and synchronous clearing has been set for the TPU1 and TPU2 counter clearing source.

Three-phase PWM waveforms are output from pins TIOCA0, TIOCA1, and TIOCA2. At this time, synchronous setting and synchronous clearing by TPU0.TGRB compare match are performed for TPUm.TCNT of TPU0 to TPU2, and the data set in TPU0.TGRB is used as the PWM cycle.

For details on PWM modes, see section 28.3.5, PWM Modes.

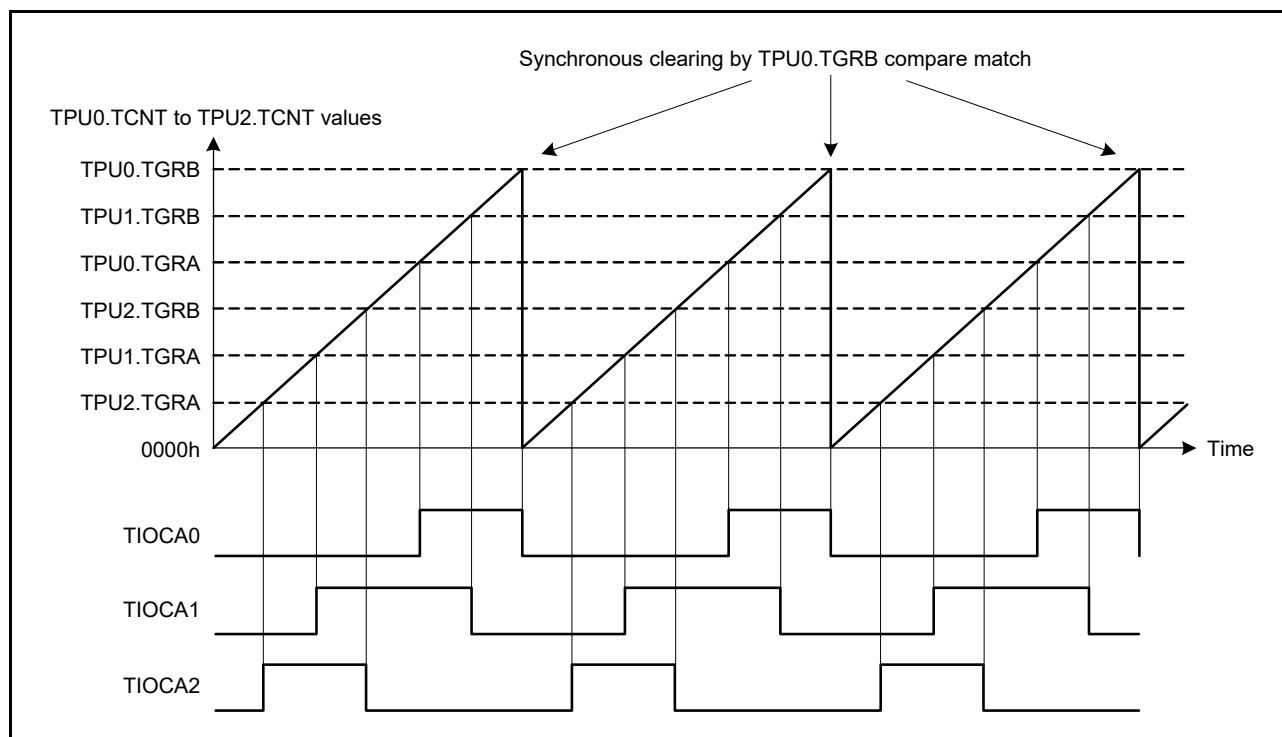


Figure 28.11 Example of Synchronous Operation

28.3.3 Buffer Operation

Buffer operation, provided for TPU0 and TPU3, enables TPUm.TGRC and TPUm.TGRD to be used as buffer registers. Buffer operation differs depending on whether TPUm.TGRy has been set as an input capture register or a compare match register.

Table 28.21 lists the register combinations used in buffer operation.

Table 28.21 Register Combinations

Channel	Timer General Register	Buffer Register
TPU0	TPU0.TGRA	TPU0.TGRC
	TPU0.TGRB	TPU0.TGRD
TPU3	TPU3.TGRA	TPU3.TGRC
	TPU3.TGRB	TPU3.TGRD

- When TPUm.TGRy is an output compare register

When a compare match occurs, the value in the buffer register for the corresponding channel is transferred to the timer general register.

This operation is shown in Figure 28.12.

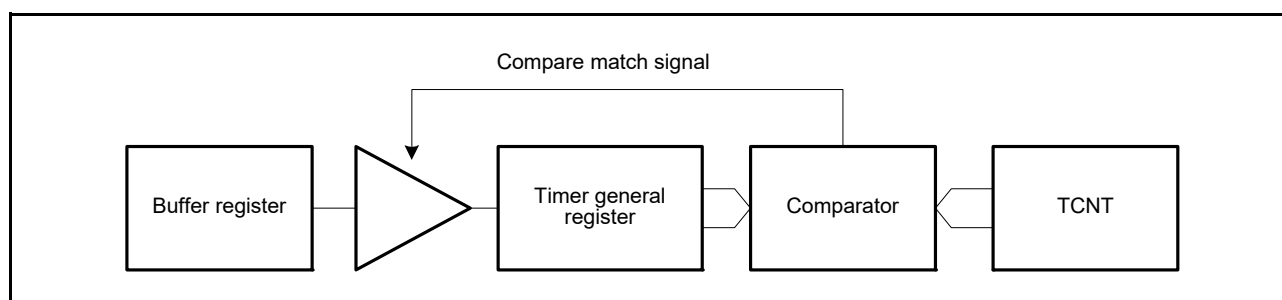


Figure 28.12 Compare Match Buffer Operation

- When TPUm.TGRy is an input capture register

When input capture occurs, the value in TPUm.TCNT is transferred to TGRy and the value previously held in TGRy is simultaneously transferred to the buffer register.

This operation is shown in Figure 28.13.

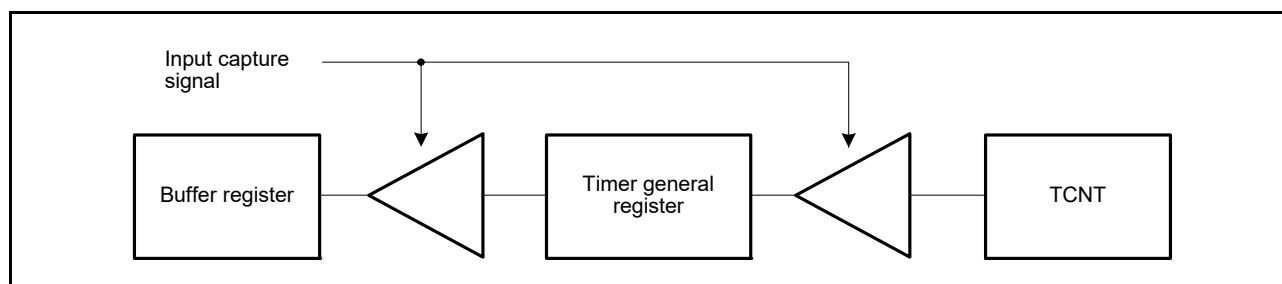


Figure 28.13 Input Capture Buffer Operation

(1) Example of Buffer Operation Setting Procedure

Figure 28.14 shows an example of the buffer operation setting procedure.

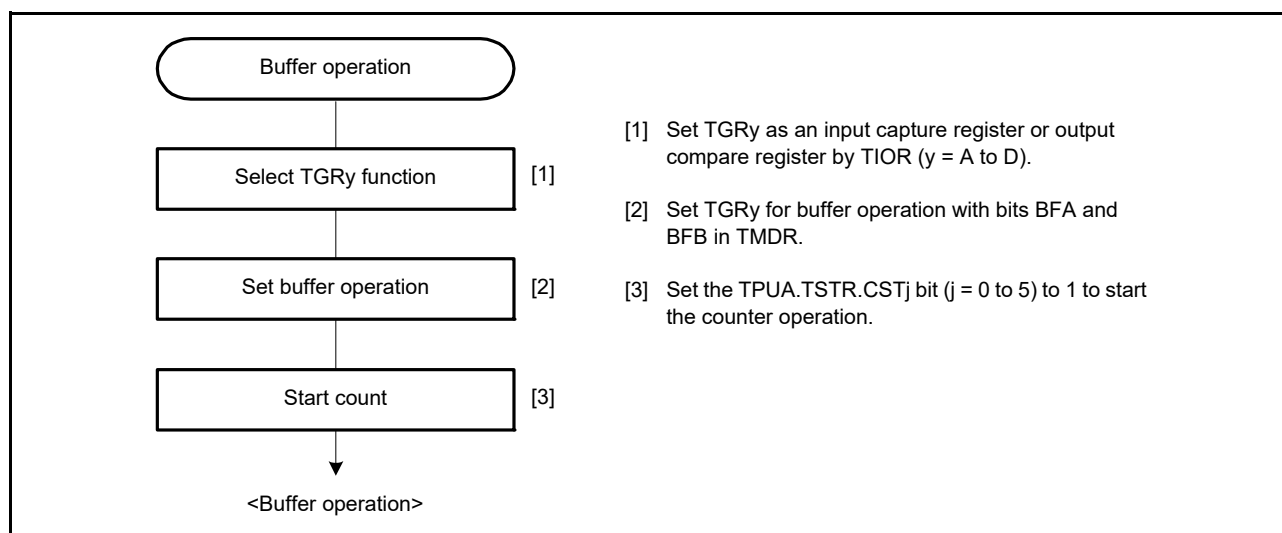


Figure 28.14 Example of Buffer Operation Setting Procedure

(2) Examples of Buffer Operation

(a) When TPUm.TGRy is an output compare register

Figure 28.15 shows an operation example in which PWM mode 1 has been set for TPU0, and buffer operation has been set for TPU0.TGRA and TPU0.TGRC. The settings used in this example are TPU0.TCNT clearing by compare match B, high output at compare match A, and low output at compare match B.

As buffer operation has been set, when compare match A occurs, the output changes and the TPU0.TGRC value is simultaneously transferred to TPU0.TGRA. This operation is repeated each time compare match A occurs.

For details on PWM modes, see section 28.3.5, PWM Modes.

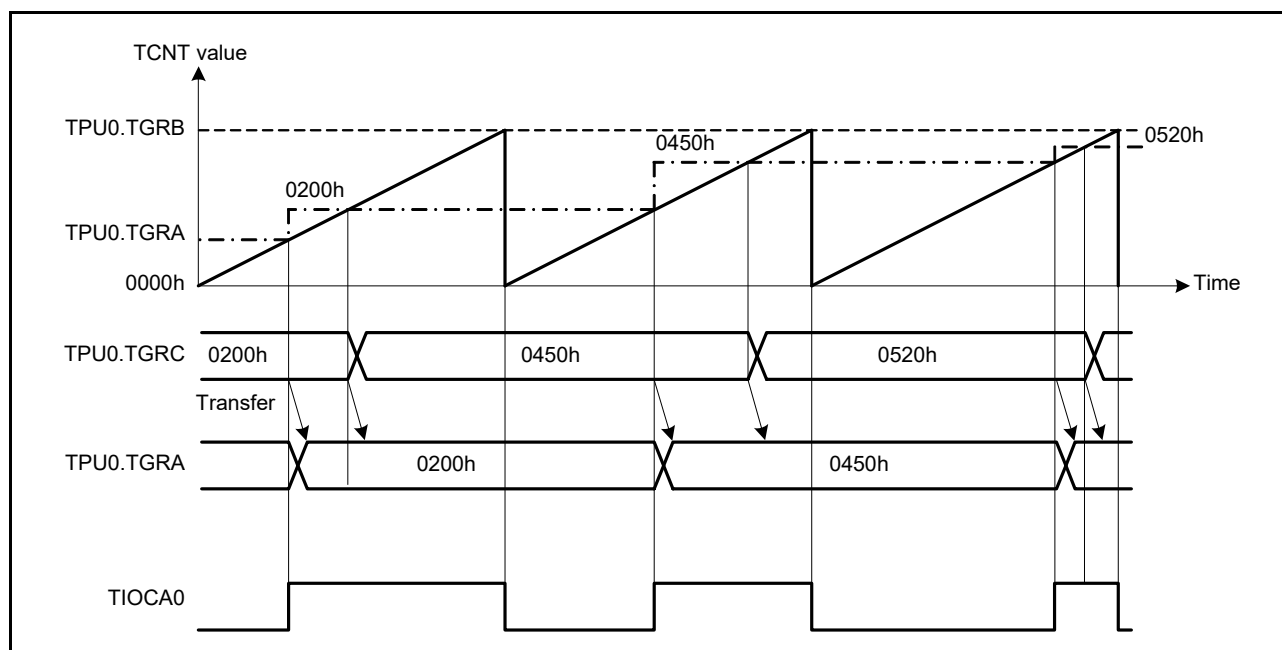


Figure 28.15 Example of Buffer Operation (1)

(b) When TPUM.TGRy is an input capture register

Figure 28.16 shows an operation example in which TPUM.TGRA has been set as an input capture register, and buffer operation has been set for the TGRA register and TPUM.TGRC.

Counter clearing by TGRA input capture has been set for TPUM.TCNT, and both rising and falling edges have been selected as the TIOCA_n pin input capture input edge.

As buffer operation has been set, when the TCNT value is stored in TGRA upon occurrence of input capture A, the value previously stored in TGRA is simultaneously transferred to TGRC.

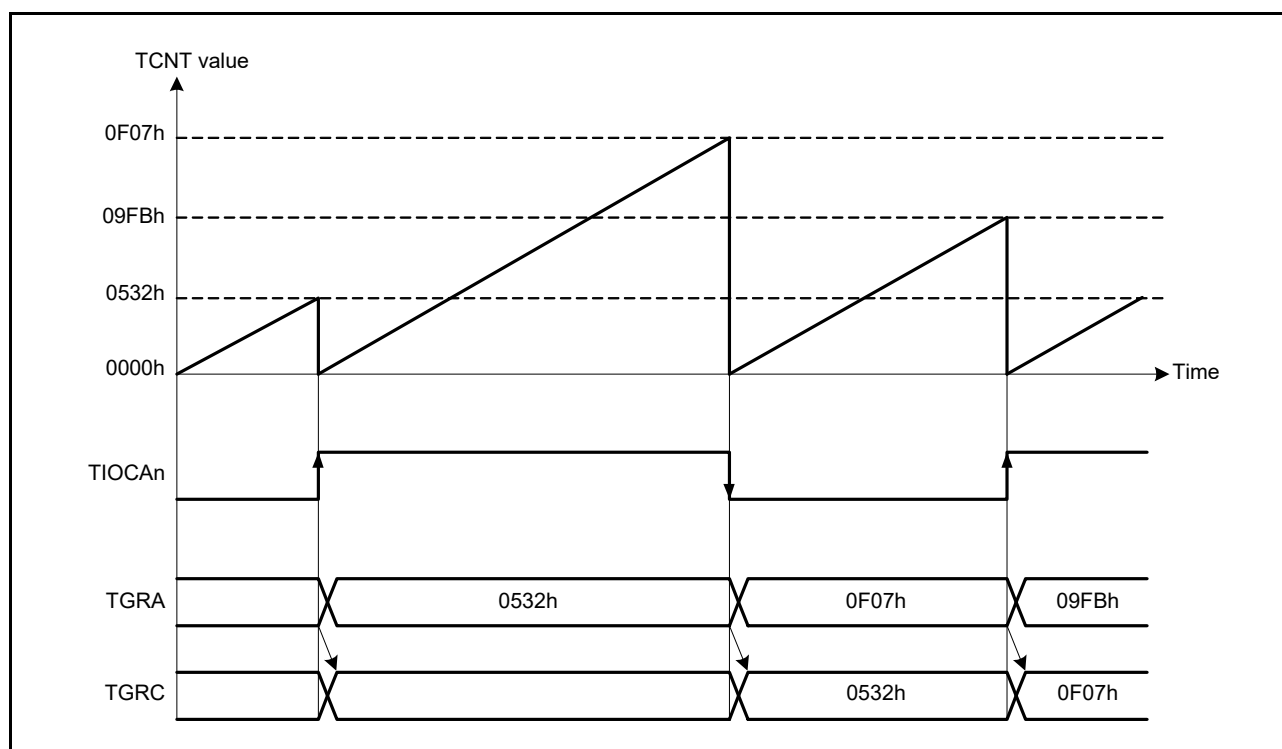


Figure 28.16 Example of Buffer Operation (2) (n = 0 to 5)

28.3.4 Cascaded Operation

In cascaded operation, two 16-bit counters for different channels are used together as a 32-bit counter.

This function works by counting the TPU1 (TPU4) count clock at overflow/underflow of TPU2.TCNT (TPU5.TCNT) as set by the TPSC[2:0] bits in TPU1.TCR (TPSC[2:0] bits in TPU4.TCR).

Underflow occurs only when the lower 16-bit TPUm.TCNT is in phase counting mode.

Table 28.22 lists the register combinations used in cascaded operation.

Note: When phase counting mode is set for TPU1 or TPU4, the count clock setting is invalid and the counter operates independently in phase counting mode.

Table 28.22 Cascaded Combinations

Combination	Upper 16 Bits	Lower 16 Bits
TPU1 and TPU2	TPU1.TCNT	TPU2.TCNT
TPU4 and TPU5	TPU4.TCNT	TPU5.TCNT

(1) Example of Cascaded Operation Setting Procedure

Figure 28.17 shows an example of the setting procedure for cascaded operation.

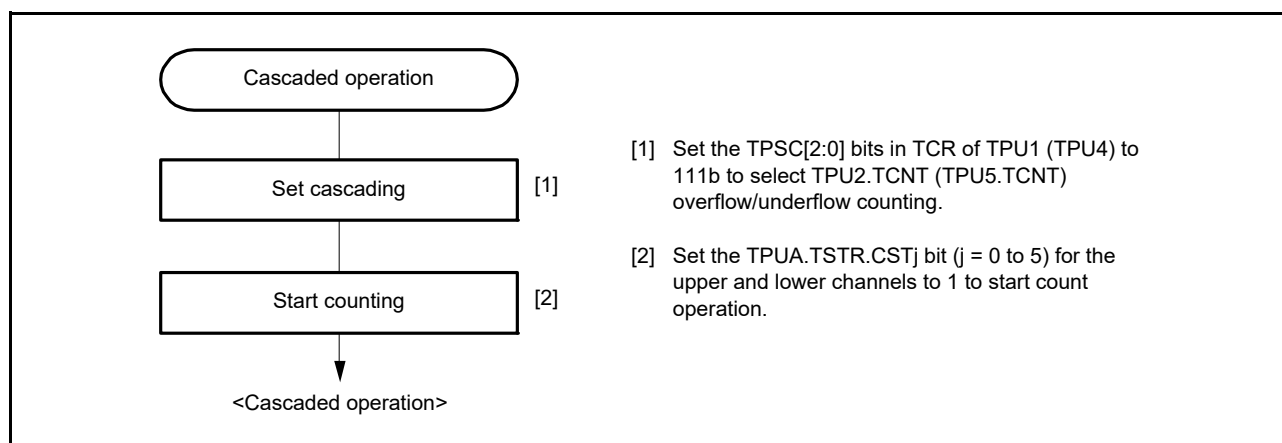


Figure 28.17 Cascaded Operation Setting Procedure

(2) Examples of Cascaded Operation

Figure 28.18 shows the operation when counting upon TPU2.TCNT overflow/underflow has been set for TPU1.TCNT, TPU1.TGRA and TPU2.TGRA have been set as input capture registers, and the rising edge of the TIOCA1 and TIOCA2 pins has been selected.

When a rising edge is input to the TIOCA1 and TIOCA2 pins simultaneously, the upper 16 bits of the 32-bit data are transferred to TPU1.TGRA, and the lower 16 bits to TPU2.TGRA. Note that a point for caution applies to simultaneous input capture in cascade operation, as described in section 28.10.11, TCNT Simultaneous Input Capture in Cascade Operation.

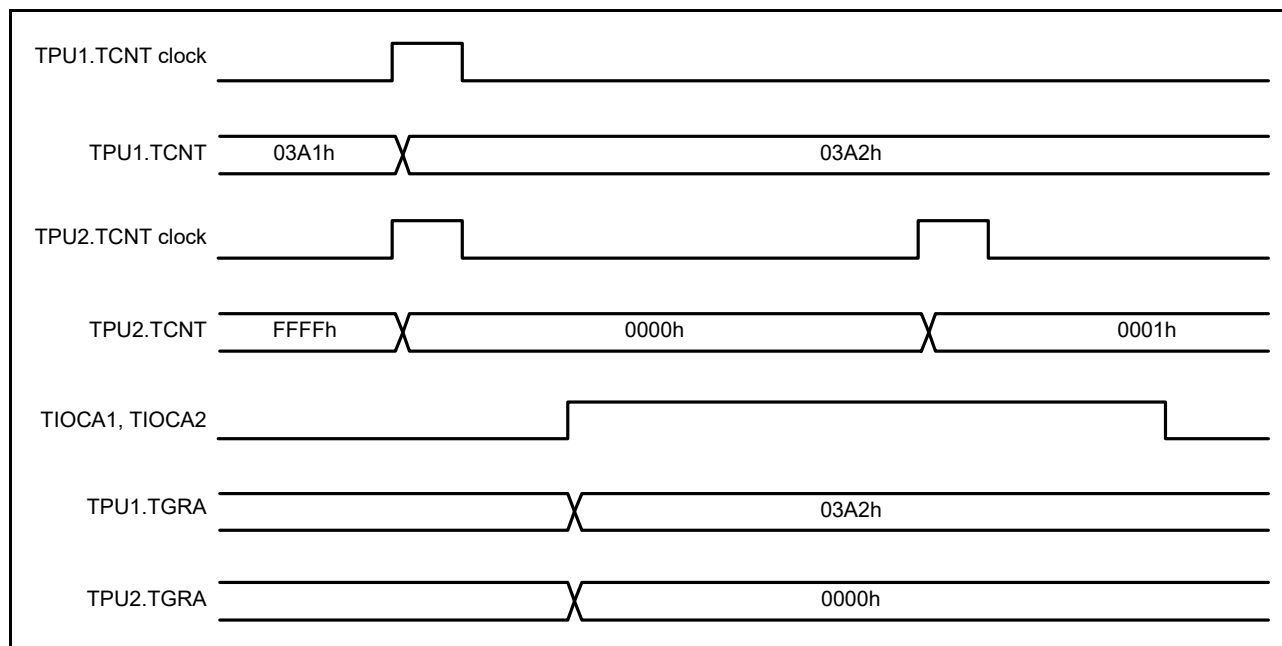


Figure 28.18 Example of Cascaded Operation (1)

Figure 28.19 shows the operation when counting upon TPU2.TCNT overflow/underflow has been set for TPU1.TCNT, and phase counting mode 1 has been specified for TPU2.

TPU1.TCNT is incremented by TPU2.TCNT overflow and decremented by TPU2.TCNT underflow.

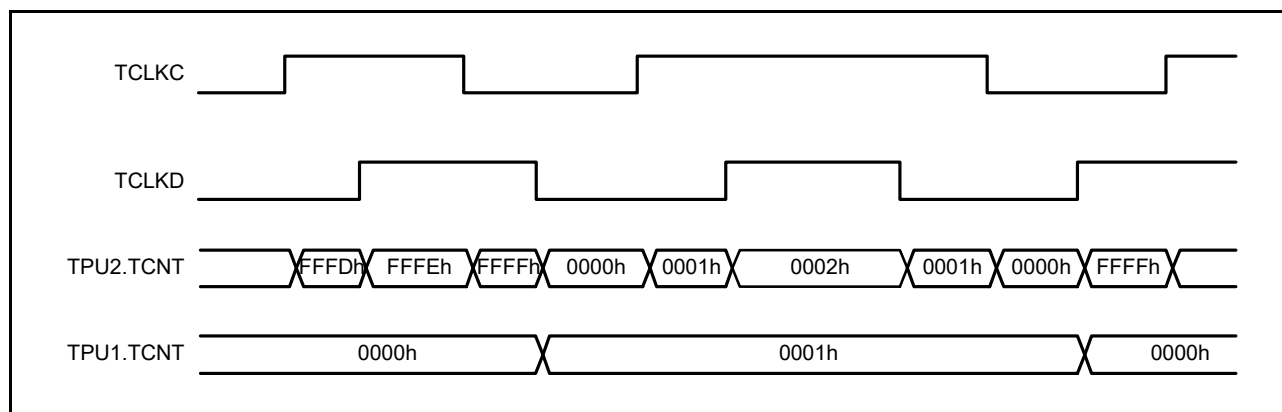


Figure 28.19 Example of Cascaded Operation (2)

28.3.5 PWM Modes

In PWM mode, PWM waveforms are output from the output pins. low, high, or toggle output can be selected as the output level in response to compare match of each TPUM.TGRy.

Settings of TGRy registers can output a PWM waveform in the range of 0% to 100% duty cycle.

Specifying TGRy compare match as the counter clearing source enables the cycle to be set in that register. All channels can be set for PWM mode independently. Synchronous operation is also possible.

There are two PWM modes, as described below.

1. PWM mode 1

PWM waveform is generated from the TIOCAN and TIOCCn pins by pairing TPUM.TGRA with TPUM.TGRB and TPUM.TGRC with TPUM.TGRD. The outputs specified by the IOA[3:0] bits in TPUM.TIOR(H) and IOC[3:0] bits in TPUM.TIORL are output from the TIOCAN and TIOCCn pins at compare matches A and C, respectively. The outputs specified by the IOB[3:0] bits in TPUM.TIOR(H) and IOD[3:0] bits in TPUM.TIORL are output from the TIOCAN and TIOCCn pins at compare matches B and D, respectively. The initial output value is the value set in TGRA or TGRC. If the set values of paired TGRy registers are identical, the output value does not change even when a compare match occurs.

In PWM mode 1, a maximum 8-phase PWM output is possible.

2. PWM mode 2

PWM waveform is generated by using one TPUM.TGRy as the cycle register and the others as duty cycle registers. The output specified in TPUM.TIORH, TPUM.TIORL, or TPUM.TIOR is performed by compare matches. Upon counter clearing by a synchronous register compare match, the output value of each pin is the initial value set in TIORH, TIORL, or TIOR. If the set values of the cycle register and duty cycle register are identical, the output value does not change even when a compare match occurs.

In PWM mode 2, a maximum 15-phase PWM waveform is possible by combined use with synchronous operation.

The correspondence between PWM output pins and registers is listed in Table 28.23.

Table 28.23 PWM Output Registers and Output Pins

Channel	Register	Output Pin	
		PWM Mode 1	PWM Mode 2
TPU0	TPU0.TGRA	TIOCA0	TIOCA0
	TPU0.TGRB		TIOCB0
	TPU0.TGRC	TIOCC0	TIOCC0
	TPU0.TGRD		TIOCD0
TPU1	TPU1.TGRA	TIOCA1	TIOCA1
	TPU1.TGRB		TIOCB1
TPU2	TPU2.TGRA	TIOCA2	TIOCA2
	TPU2.TGRB		TIOCB2
TPU3	TPU3.TGRA	TIOCA3	TIOCA3
	TPU3.TGRB		TIOCB3
	TPU3.TGRC	TIOCC3	TIOCC3
	TPU3.TGRD		TIOCD3
TPU4	TPU4.TGRA	TIOCA4	TIOCA4
	TPU4.TGRB		TIOCB4
TPU5	TPU5.TGRA	TIOCA5	TIOCA5
	TPU5.TGRB		TIOCB5

Note: In PWM mode 2, PWM waveform output is not possible for the TPUm.TGRy register in which the cycle is set.

(1) Example of PWM Mode Setting Procedure

Figure 28.20 shows an example of the PWM mode setting procedure.

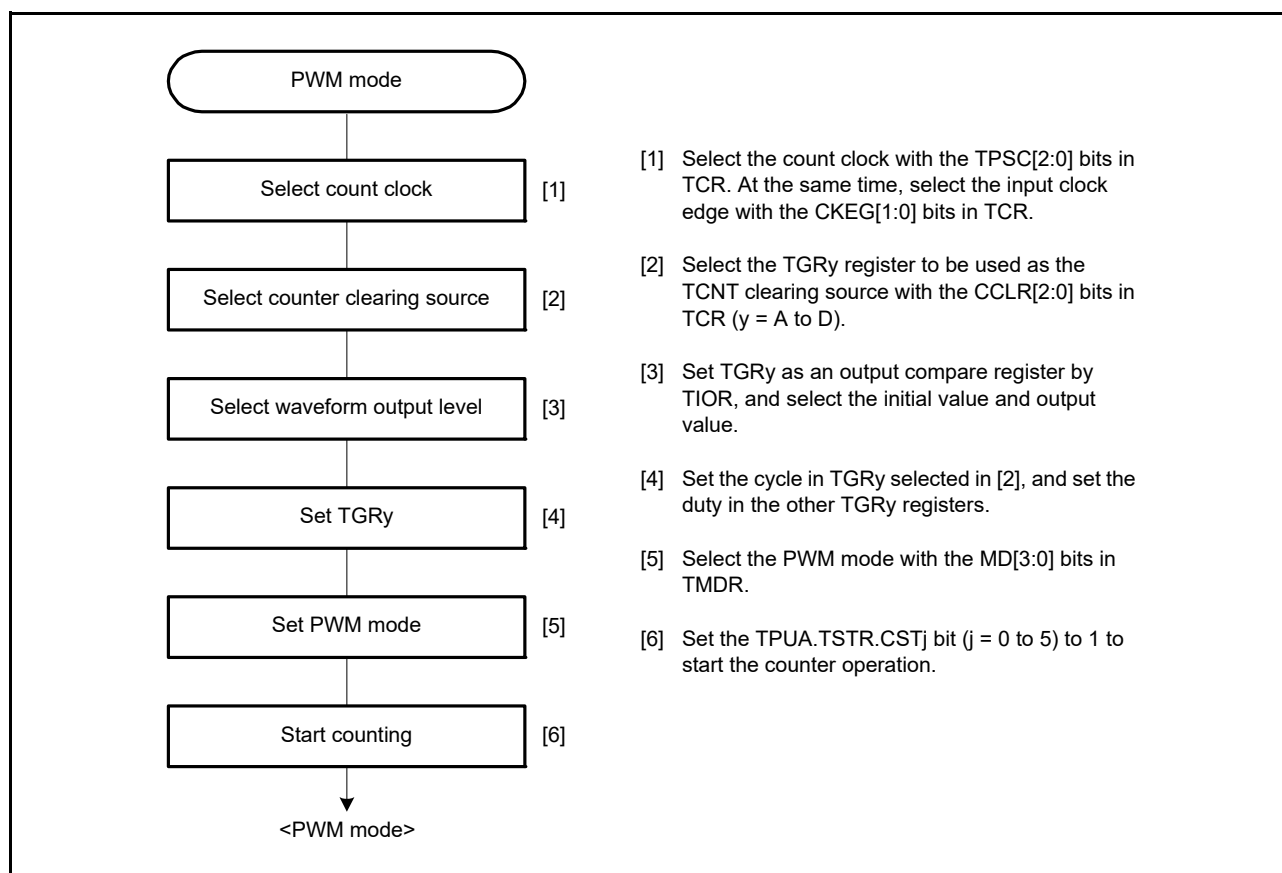


Figure 28.20 Example of PWM Mode Setting Procedure

(2) Examples of PWM Mode Operation

Figure 28.21 shows an example of PWM mode 1 operation.

In this example, TPUm.TGRA compare match is set as the TPUm.TCNT clearing source, low is set for the TGRA initial output value and output value, and high is set as the TPUm.TGRB output value.

In this case, the value set in TGRA is used as the cycle, and the value set in TGRB is used as the duty cycle.

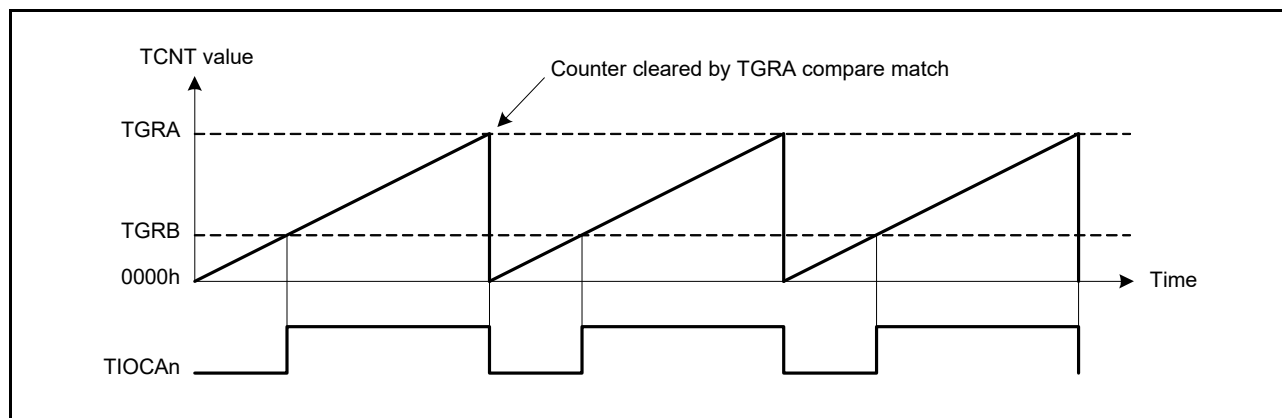


Figure 28.21 Example of PWM Mode Operation (1) (n = 0 to 5)

Figure 28.22 shows an example of PWM mode 2 operation.

In this example, synchronous operation is specified for TPU0 and TPU1, TPU1.TGRB compare match is set as the TPUm.TCNT clearing source, and low is set for the initial output value and high for the output value of the other TPUm.TGRy registers (TPU0.TGRA to TPU0.TGRD and TPU1.TGRA), to output a 5-phase PWM waveform.

In this case, the value set in TPU1.TGRB is used as the cycle, and the values set in the other TGRy registers are used as the duty cycle.

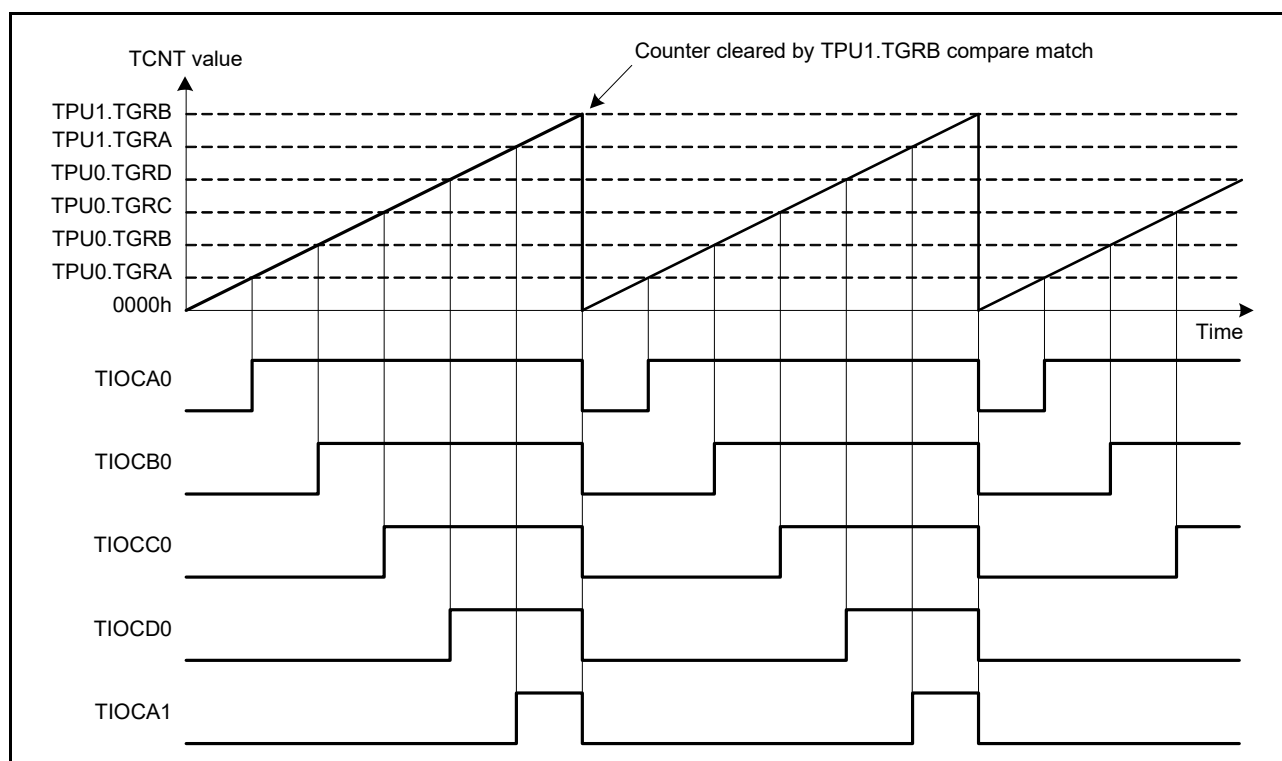


Figure 28.22 Example of PWM Mode Operation (2)

Figure 28.23 shows examples of PWM waveform output with 0% duty cycle and 100% duty cycle in PWM mode.

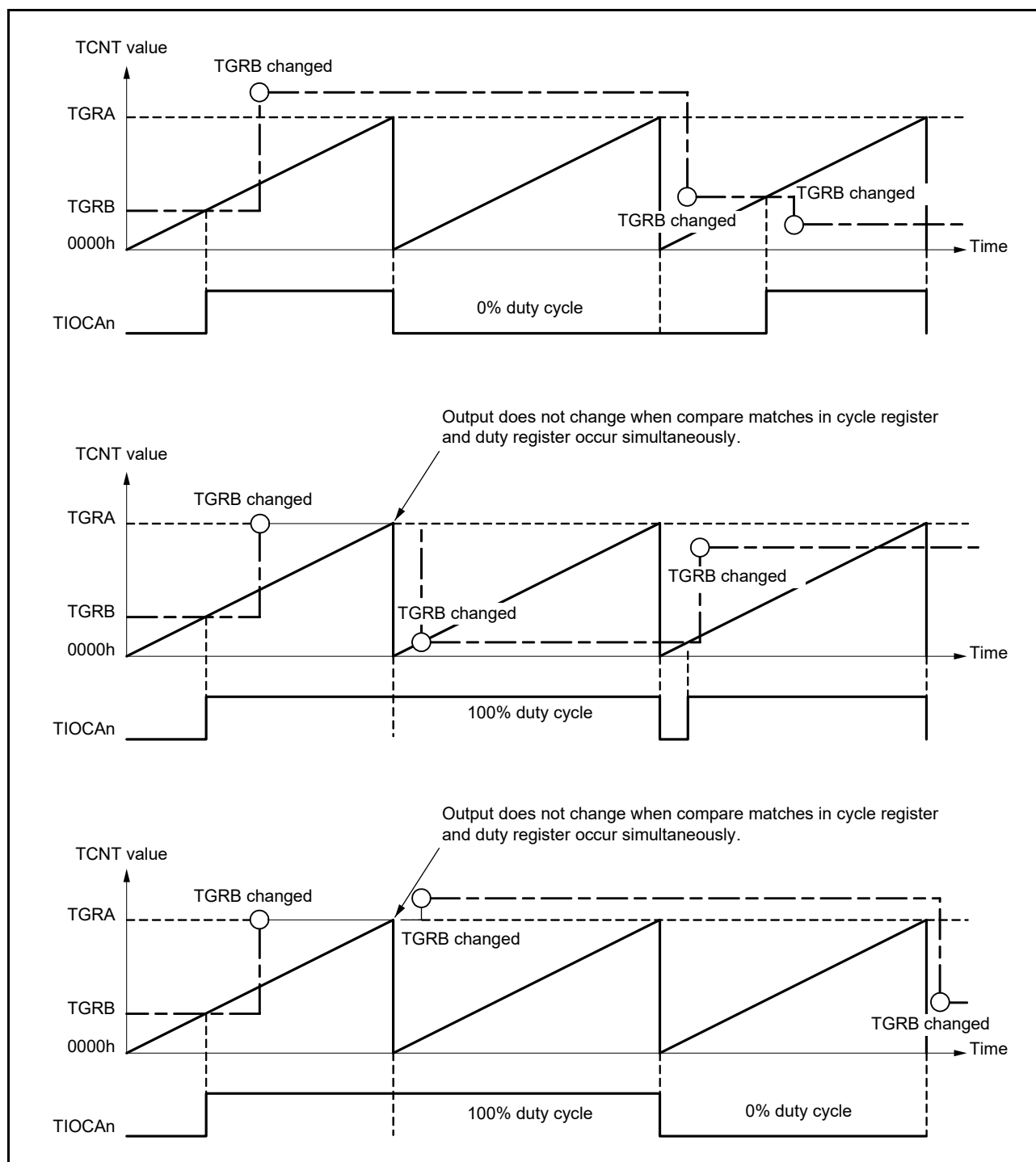


Figure 28.23 Example of PWM Mode Operation (3) (n = 0 to 5)

28.3.6 Phase Counting Mode

In phase counting mode, the phase difference between two external clock inputs is detected by the settings for channels 1, 2, 4, and 5, and TPUm.TCNT is incremented/decremented accordingly.

When phase counting mode is set, an external clock is selected as the count clock and TCNT operates as an up-/down-counter regardless of the setting of the TPSC[2:0] bits and CKEG[1:0] bits in TPUm.TCR. However, the lower 2 bits of the CCLR[2:0] bits in TPUm.TCR and the functions of TPUm.TIORH, TPUm.TIOLR, TPUm.TIOR, TPUm.TIER, and TPUm.TGRy are valid, and therefore input capture/compare match and interrupt functions are available.

When an overflow occurs while TCNT is counting up, a TCIV interrupt request is generated; when an underflow occurs while TCNT is counting down, a TCIU interrupt request is generated. The TCFD bit in TPUm.TSR is the count direction flag. Reading the TCFD flag provides an indication of whether TCNT is counting up or down.

In phase counting mode, the external clock pins TCLKA, TCLKB, TCLKC, and TCLKD can be used as 2-phase encoder pulse input.

Table 28.24 lists the correspondence between external clock pins and channels.

Table 28.24 Clock Input Pins in Phase Counting Mode

Channel	External Clock Pins	
	A-Phase	B-Phase
When TPU1 or TPU5 is set to phase counting mode	TCLKA	TCLKB
When TPU2 or TPU4 is set to phase counting mode	TCLKC	TCLKD

(1) Example of Phase Counting Mode Setting Procedure

Figure 28.24 shows an example of the phase counting mode setting procedure.

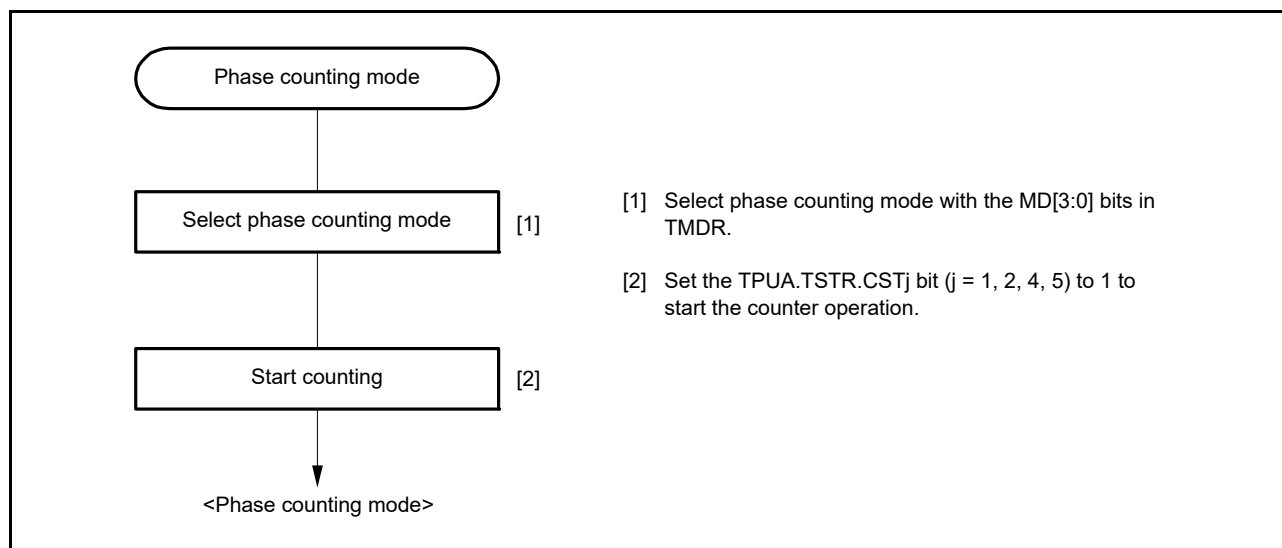


Figure 28.24 Example of Phase Counting Mode Setting Procedure

(2) Examples of Phase Counting Mode Operation

In phase counting mode, TPUm.TCNT counts up or down according to the phase difference between two external clocks. There are four modes, according to the count conditions.

(a) Phase counting mode 1

Figure 28.25 shows an example of phase counting mode 1 operation, and Table 28.25 lists the TPUm.TCNT up-/down-count conditions.

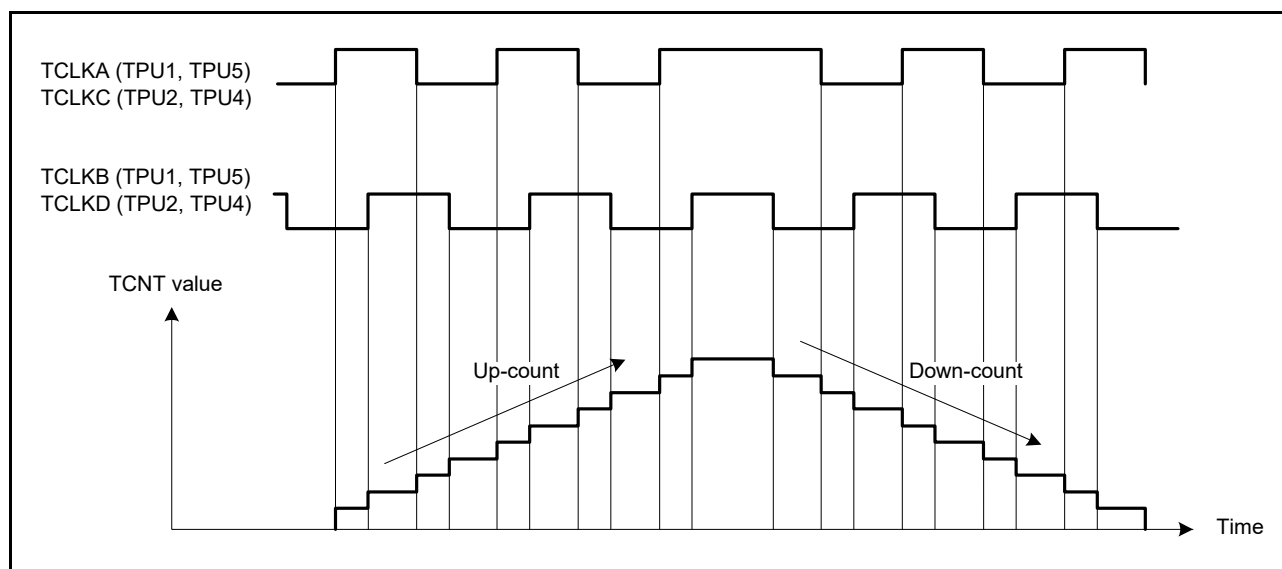


Figure 28.25 Example of Phase Counting Mode 1 Operation

Table 28.25 Up-/Down-Count Conditions in Phase Counting Mode 1

TCLKA (TPU1, TPU5) TCLKC (TPU2, TPU4)	TCLKB (TPU1, TPU5) TCLKD (TPU2, TPU4)	Operation
High		Up-count
Low		
	Low	
	High	
High		Down-count
Low		
	High	
	Low	

: Rising edge

: Falling edge

(b) Phase counting mode 2

Figure 28.26 shows an example of phase counting mode 2 operation, and Table 28.26 lists the TPUm.TCNT up-/down-count conditions.

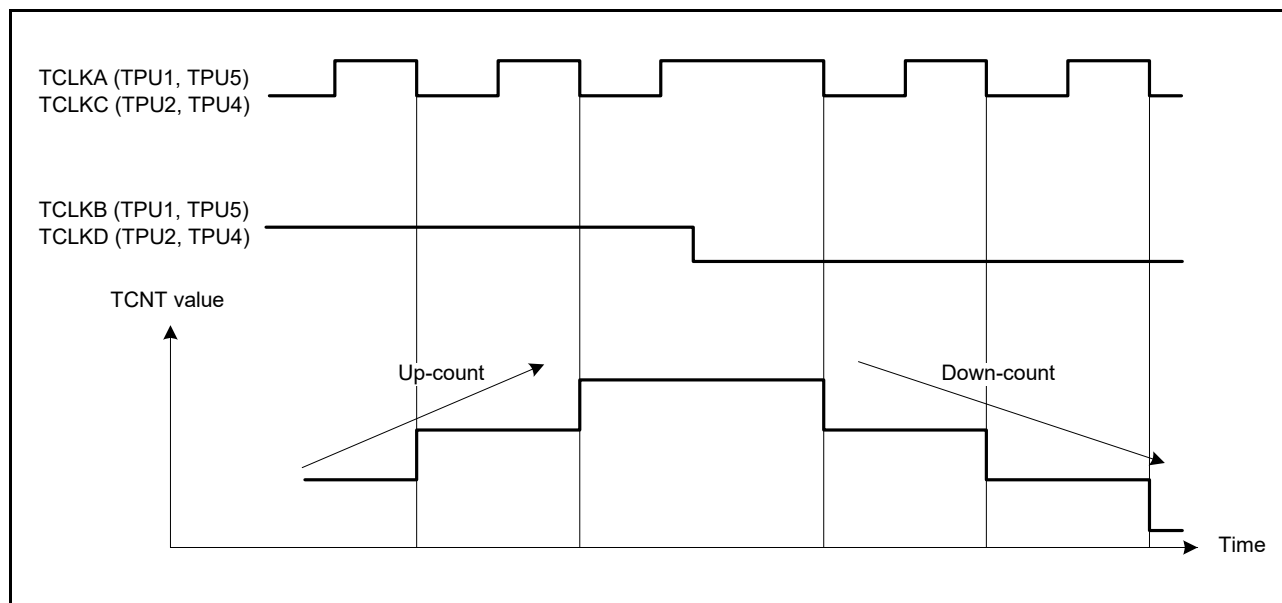


Figure 28.26 Example of Phase Counting Mode 2 Operation

Table 28.26 Up-/Down-Count Conditions in Phase Counting Mode 2

TCLKA (TPU1, TPU5) TCLKC (TPU2, TPU4)	TCLKB (TPU1, TPU5) TCLKD (TPU2, TPU4)	Operation
High		Don't care
Low		Don't care
	Low	Don't care
	High	Up-count
High		Don't care
Low		Don't care
	High	Don't care
	Low	Down-count

: Rising edge

: Falling edge

(c) Phase counting mode 3

Figure 28.27 shows an example of phase counting mode 3 operation, and Table 28.27 lists the TPU_m.TCNT up-/down-count conditions.

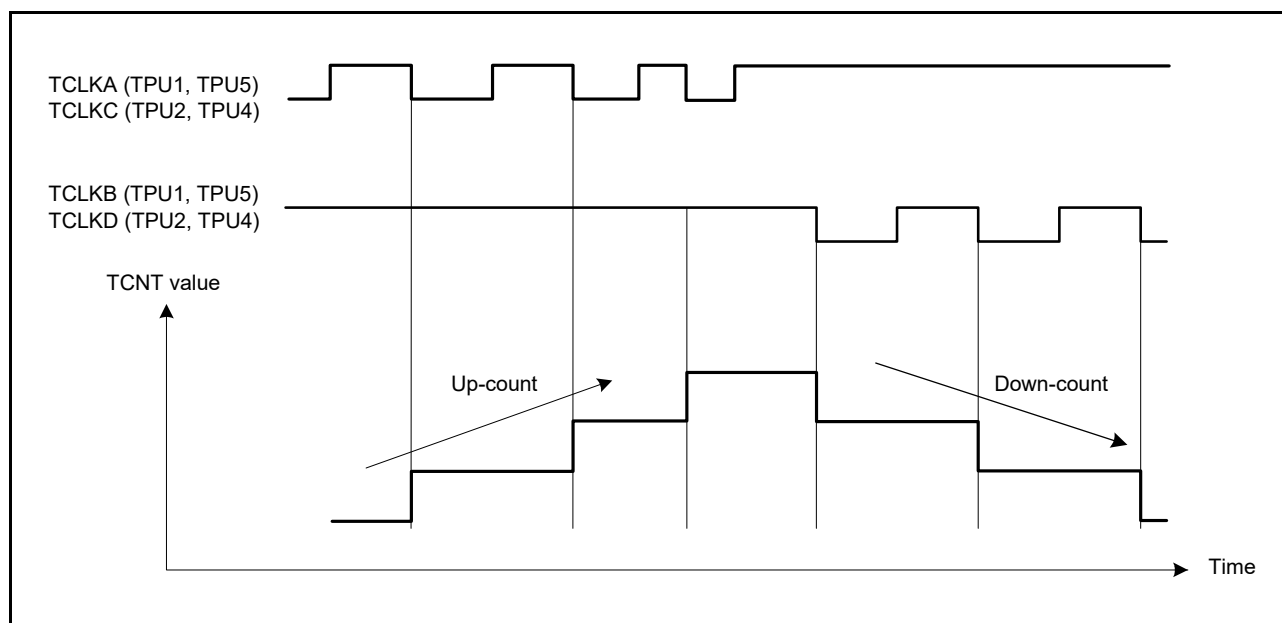


Figure 28.27 Example of Phase Counting Mode 3 Operation

Table 28.27 Up-/Down-Count Conditions in Phase Counting Mode 3

TCLKA (TPU1, TPU5) TCLKC (TPU2, TPU4)	TCLKB (TPU1, TPU5) TCLKD (TPU2, TPU4)	Operation
High		Don't care
Low		Don't care
	Low	Don't care
	High	Up-count
High		Down-count
Low		Don't care
	High	Don't care
	Low	Don't care

 : Rising edge
 : Falling edge

(d) Phase counting mode 4

Figure 28.28 shows an example of phase counting mode 4 operation, and Table 28.28 lists the TPU_m.TCNT up-/down-count conditions.

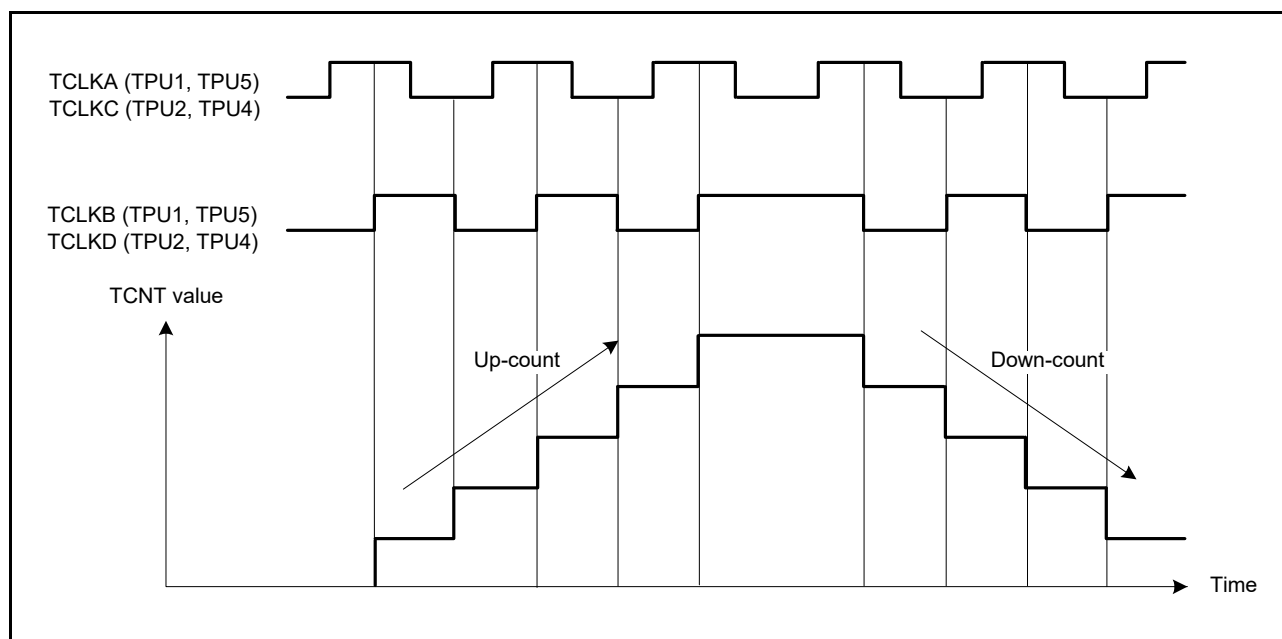


Figure 28.28 Example of Phase Counting Mode 4 Operation

Table 28.28 Up-/Down-Count Conditions in Phase Counting Mode 4

TCLKA (TPU1, TPU5) TCLKC (TPU2, TPU4)	TCLKB (TPU1, TPU5) TCLKD (TPU2, TPU4)	Operation
High		Up-count
Low		Up-count
	Low	Don't care
	High	Don't care
High		Down-count
Low		Down-count
	High	Don't care
	Low	Don't care

 : Rising edge

 : Falling edge

28.3.6.1 Phase Counting Mode Application Example

Figure 28.29 shows an example in which phase counting mode is set for TPU1, and TPU1 is coupled with TPU0 to input servo motor 2-phase encoder pulses in order to detect the position or speed.

TPU1 is set to phase counting mode 1, and the encoder pulse A-phase and B-phase are input to the TCLKA and TCLKB pins.

TPU0 operates with TPU0.TCNT clearing by TPU0.TGRC compare match; TPU0.TGRA and TPU0.TGRC are used for the compare match function and are set with the speed control cycle and position control cycle. TPU0.TGRB is used for input capture, with TPU0.TGRB and TPU0.TGRD operating in buffer mode. The TPU1 count clock is specified as the TPU0.TGRB input capture source, and the pulse width of 2-phase encoder 4-multiplication pulses is detected.

TPU1.TGRA and TPU1.TGRB for TPU1 are specified for input capture, TPU0.TGRA and TPU0.TGRC compare matches are selected as the input capture source, and the up-/down-counter values for the control cycles are stored. This procedure enables accurate position/speed detection to be achieved.

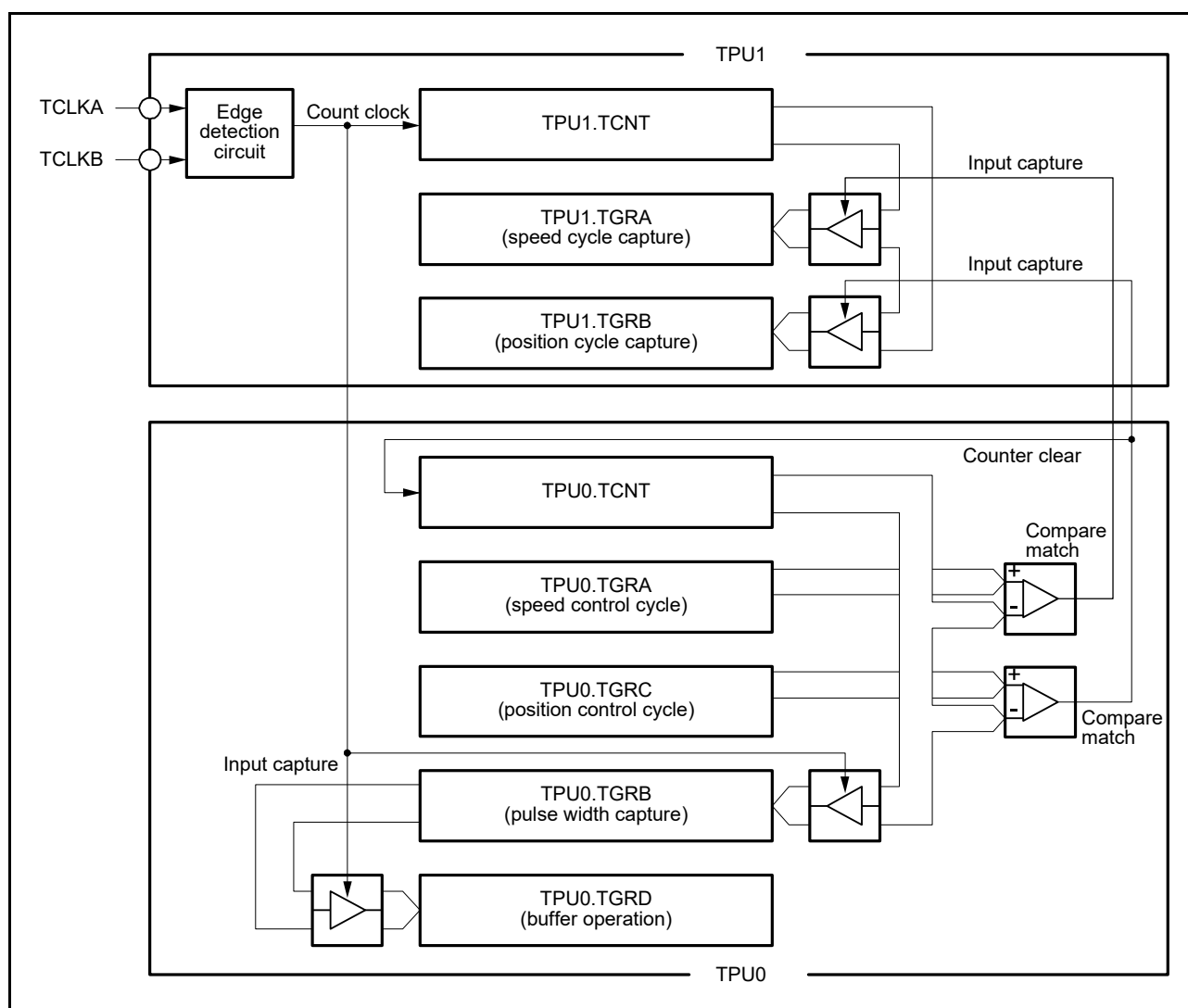


Figure 28.29 Phase Counting Mode Application Example

28.3.7 Noise Filters

Each pin for use in input capture by TPU is equipped with a noise filter. The noise filter samples the level on the pin three times at the selected sampling interval, conveys the level to the internal circuits if the samples match, and continues to convey that level until the other level is sampled from the pins three times in a row. The noise filter function can be enabled or disabled for each pin. Furthermore, sampling clock settings can be made for each channel.

Figure 28.30 is a timing chart for the noise filter.

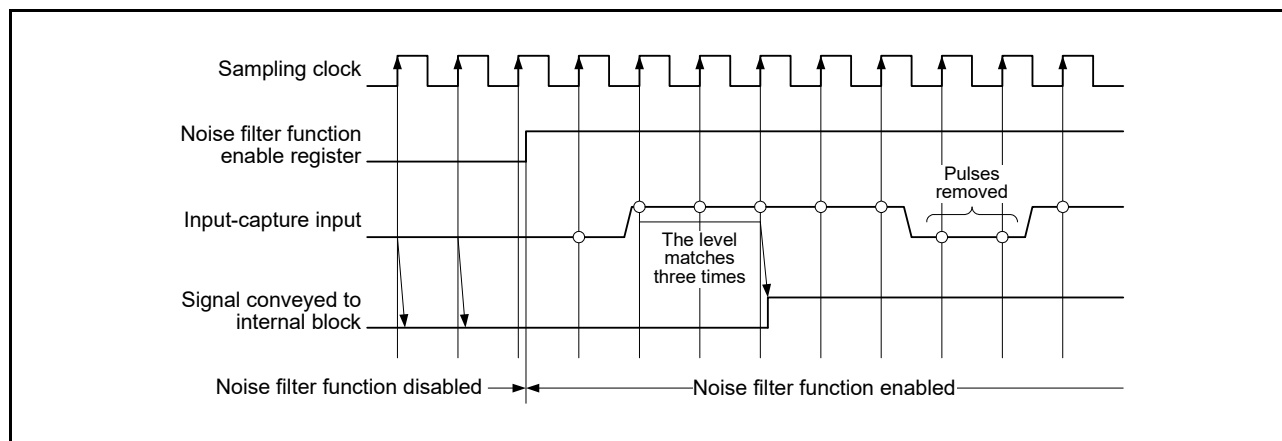


Figure 28.30 Timing Chart for the Noise Filter

If noise filtering is set, input capture operation is performed on the edges of noise-filtered signal after a minimum delay of $(\text{sampling interval} \times 2 + \text{PCLK})$ due to noise filtering for the input capture input.

28.4 Interrupt Sources

There are three kinds of TPU interrupt sources: TPUm.TGRy input capture/compare match, TPUm.TCNT overflow, and TPUm.TCNT underflow.

Relative channel priority levels can be changed by the interrupt controller, but the priority within a channel is fixed. For details, see [section 15, Interrupt Controller \(ICUD\)](#).

Table 28.29 lists the TPU interrupt sources.

Table 28.29 TPU Interrupt Sources

Channel	Name	Interrupt Source	DTC Activation	DMAC Activation
TPU0	TGI0A	TPU0.TGRA input capture/compare match	Possible	Possible
	TGI0B	TPU0.TGRB input capture/compare match	Possible	Not possible
	TGI0C	TPU0.TGRC input capture/compare match	Possible	Not possible
	TGI0D	TPU0.TGRD input capture/compare match	Possible	Not possible
	TCI0V	TPU0.TCNT overflow	Not possible	Not possible
TPU1	TGI1A	TPU1.TGRA input capture/compare match	Possible	Possible
	TGI1B	TPU1.TGRB input capture/compare match	Possible	Not possible
	TCI1V	TPU1.TCNT overflow	Not possible	Not possible
	TCI1U	TPU1.TCNT underflow	Not possible	Not possible
TPU2	TGI2A	TPU2.TGRA input capture/compare match	Possible	Possible
	TGI2B	TPU2.TGRB input capture/compare match	Possible	Not possible
	TCI2V	TPU2.TCNT overflow	Not possible	Not possible
	TCI2U	TPU2.TCNT underflow	Not possible	Not possible
TPU3	TGI3A	TPU3.TGRA input capture/compare match	Possible	Possible
	TGI3B	TPU3.TGRB input capture/compare match	Possible	Not possible
	TGI3C	TPU3.TGRC input capture/compare match	Possible	Not possible
	TGI3D	TPU3.TGRD input capture/compare match	Possible	Not possible
	TCI3V	TPU3.TCNT overflow	Not possible	Not possible
TPU4	TGI4A	TPU4.TGRA input capture/compare match	Possible	Possible
	TGI4B	TPU4.TGRB input capture/compare match	Possible	Not possible
	TCI4V	TPU4.TCNT overflow	Not possible	Not possible
	TCI4U	TPU4.TCNT underflow	Not possible	Not possible
TPU5	TGI5A	TPU5.TGRA input capture/compare match	Possible	Possible
	TGI5B	TPU5.TGRB input capture/compare match	Possible	Not possible
	TCI5V	TPU5.TCNT overflow	Not possible	Not possible
	TCI5U	TPU5.TCNT underflow	Not possible	Not possible

Note: This table lists the initial state immediately after a reset. The relative channel priority levels can be changed by the interrupt controller.

(1) Input Capture/Compare Match Interrupt

An interrupt is requested when the TGIEy bit (y = A, B, C, D) in TPUm.TIER is set to 1 by the occurrence of a TPUm.TGRy input capture/compare match on a channel. The TPU has 16 input capture/compare match interrupts, four each for TPU0 and TPU3, and two each for TPU1, TPU2, TPU4, and TPU5.

(2) Overflow Interrupt

An interrupt is requested when the TCIEV bit in TPUm.TIER is set to 1 by the occurrence of a TPUm.TCNT overflow on a channel. The TPU has six overflow interrupts, one for each channel.

(3) Underflow Interrupt

An interrupt is requested when the TCIEU bit in TPUm.TIER is set to 1 by the occurrence of a TPUm.TCNT underflow on a channel. The TPU has four underflow interrupts, one each for TPU1, TPU2, TPU4, and TPU5.

28.5 DTC Activation

The DTC can be activated by the TPUm.TGRy input capture/compare match interrupt of each channel. For details, see section 20, Data Transfer Controller (DTCb).

A total of 16 input capture/compare match interrupts can be used as DTC activation sources, four each for TPU0 and TPU3, and two each for TPU1, TPU2, TPU4, and TPU5.

28.6 DMAC Activation

The DMAC can be activated by the TPUm.TGRA input capture/compare match interrupt of each channel. For details, see section 18, DMA Controller (DMACa).

A total of six TPUm.TGRA input capture/compare match interrupts can be used as DMAC activation sources, one for each channel.

28.7 A/D Converter Activation

The TPU can activate the A/D converter by the TPUm.TGRA input capture/compare match for each channel.

When the TTGE bit in TPUm.TIER is set to 1, the TPU requests the A/D converter to start A/D conversion by the occurrence of a TPUm.TGRA input capture/compare match on a particular channel.

28.8 PPG Trigger

Input capture to or compare match with TGRA and TGRB in TPU0 to TPU3 can be made to act as a PPG1 waveform trigger. For details, see section 29, Programmable Pulse Generator (PPG).

28.9 Operation Timing

28.9.1 Input/Output Timing

(1) TPUm.TCNT Count Timing

Figure 28.31 shows TPUm.TCNT count timing in internal clock operation, and Figure 28.32 shows TCNT count timing in external clock operation.

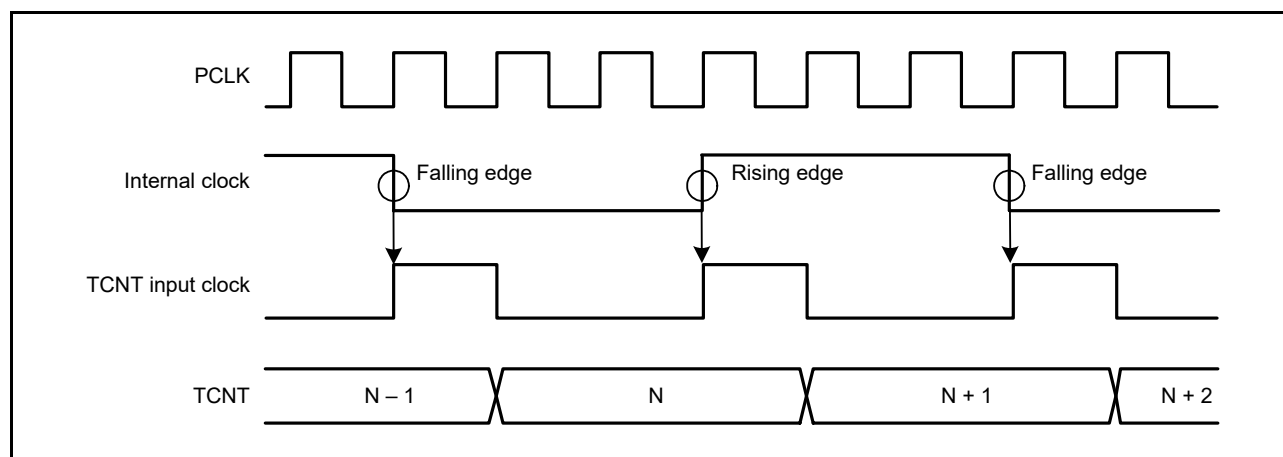


Figure 28.31 Count Timing in Internal Clock Operation

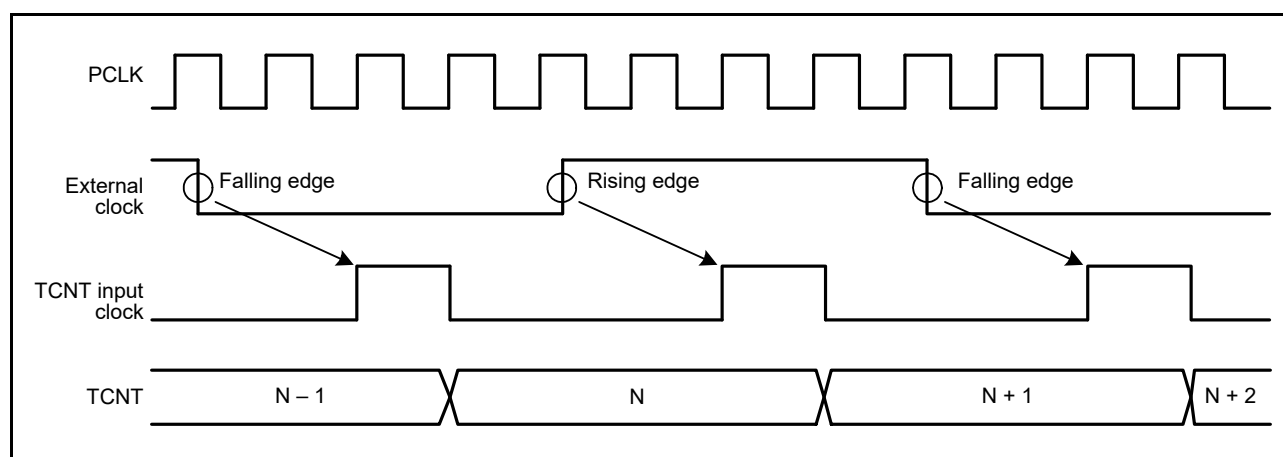


Figure 28.32 Count Timing in External Clock Operation

(2) Output Compare Output Timing

A compare match signal is generated in the final state in which $TPUm.TCNT$ and $TPUm.TGRy$ match (the point at which the count value matched by $TCNT$ is updated). When a compare match signal is generated, the output value set in $TPUm.TIORH$, $TPUm.TIORL$, or $TPUm.TIOR$ is output to the output compare output pin $TIOCyn$ ($y = A$ to D ; $n = 0$ to 5). After a match between $TCNT$ and $TGRy$, the compare match signal is not generated until the $TCNT$ input clock is generated.

Figure 28.33 shows output compare output timing.

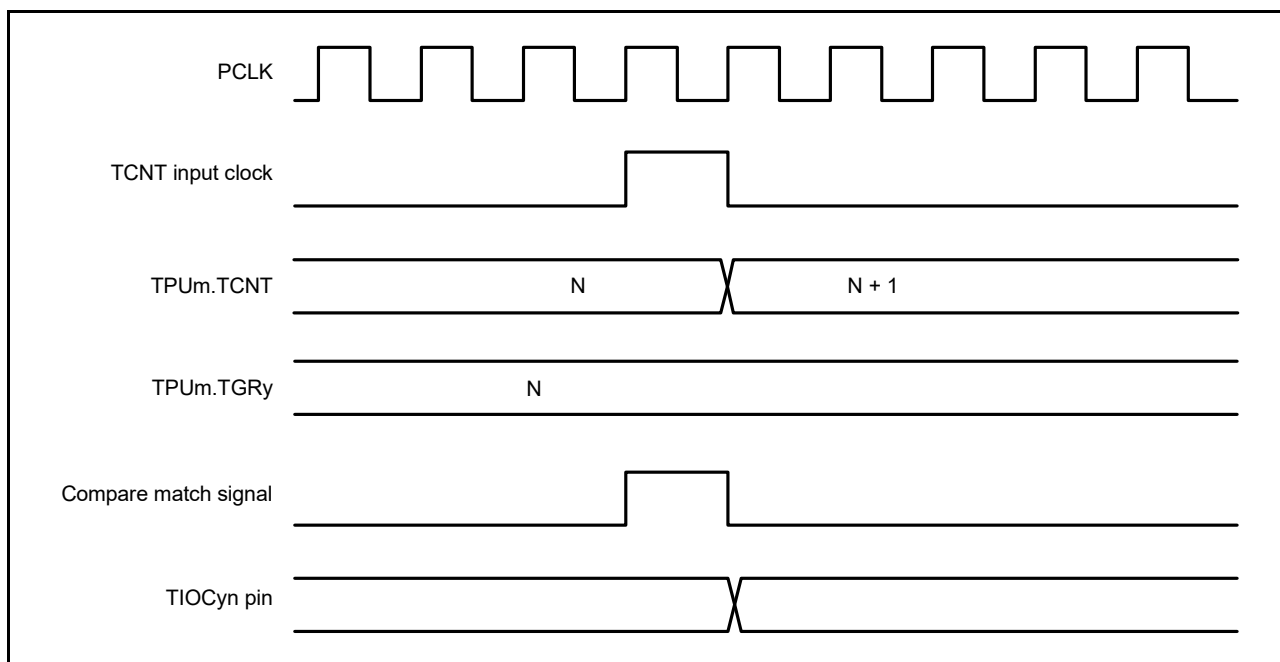


Figure 28.33 Output Compare Output Timing

(3) Input Capture Signal Timing

Figure 28.34 shows input capture signal timing.

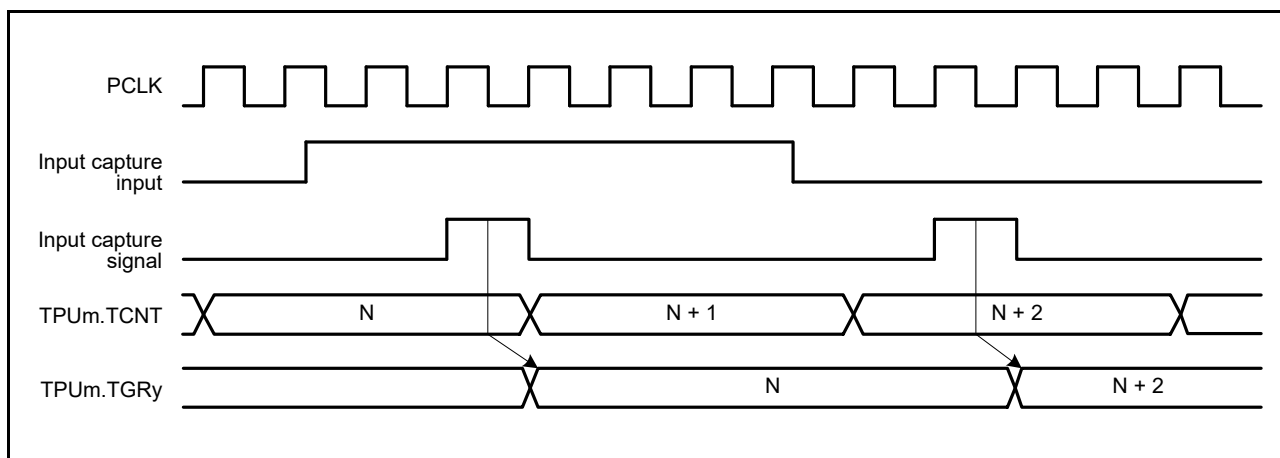


Figure 28.34 Input Capture Signal Timing

(4) Timing for Counter Clearing by Compare Match/Input Capture

Figure 28.35 shows the timing when counter clearing by compare match occurrence is specified, and Figure 28.36 shows the timing when counter clearing by input capture occurrence is specified.

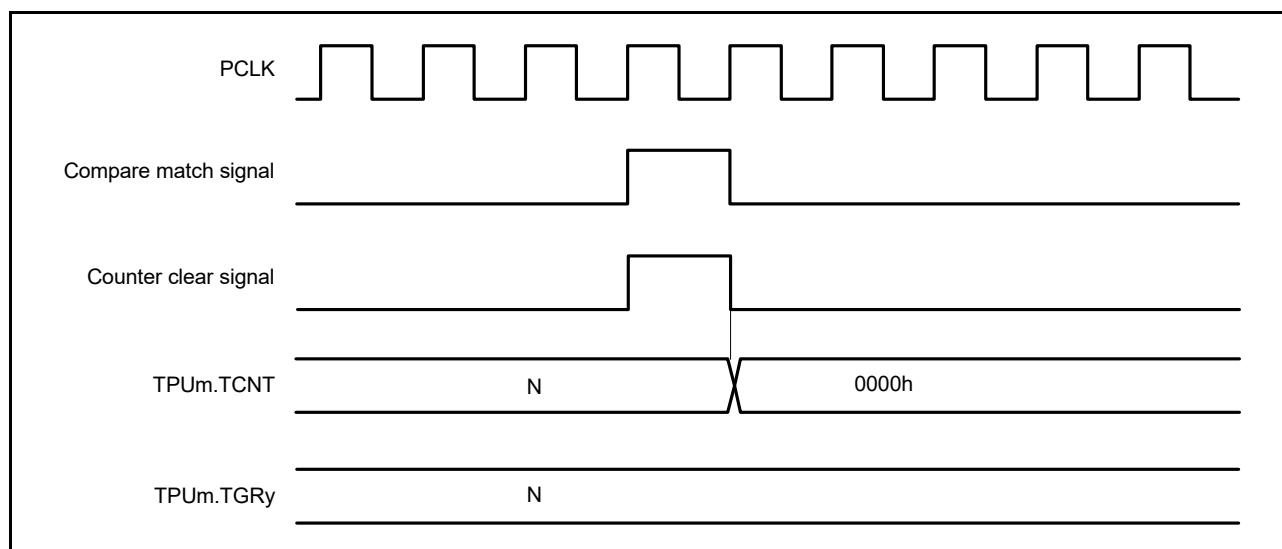


Figure 28.35 Counter Clear Timing (Compare Match)

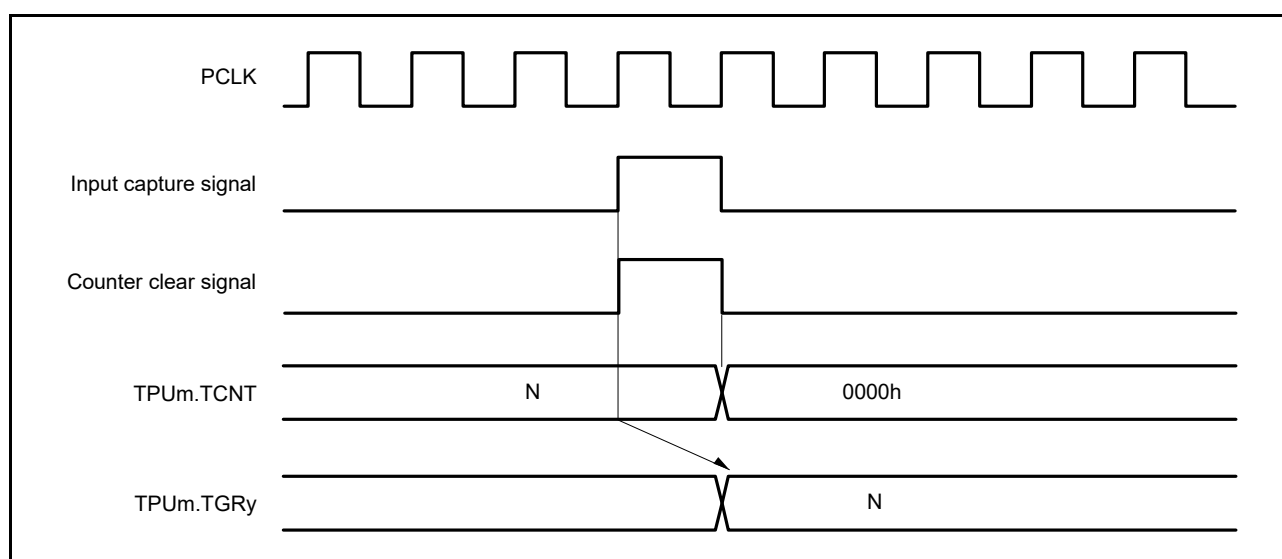


Figure 28.36 Counter Clear Timing (Input Capture)

(5) Buffer Operation Timing

Figure 28.37 and Figure 28.38 show the timings in buffer operation.

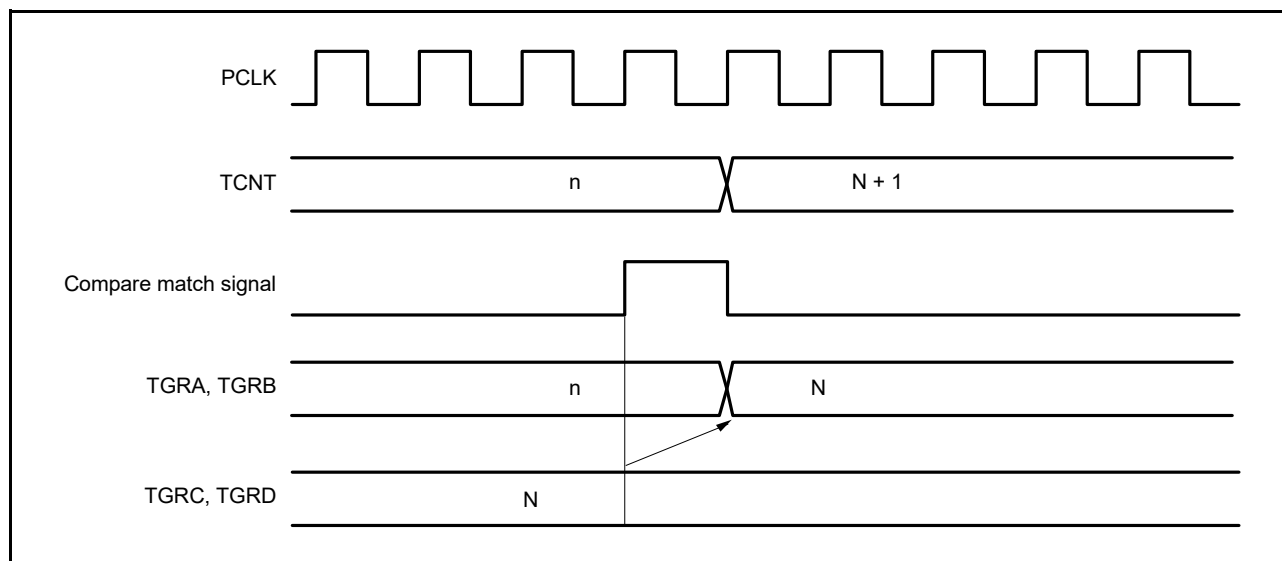


Figure 28.37 Buffer Operation Timing (Compare Match)

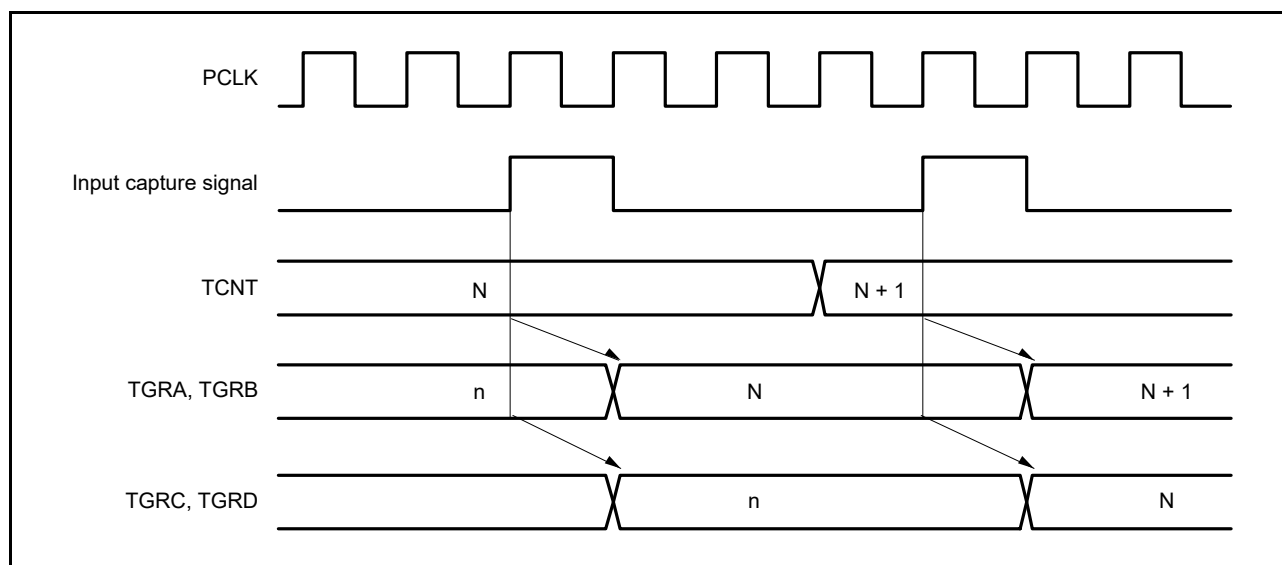


Figure 28.38 Buffer Operation Timing (Input Capture)

28.9.2 Interrupt Signal Timing

(1) Timing of Interrupt Signal Setting on Compare Match

Figure 28.39 shows the timing for setting the interrupt signal by compare match occurrence.

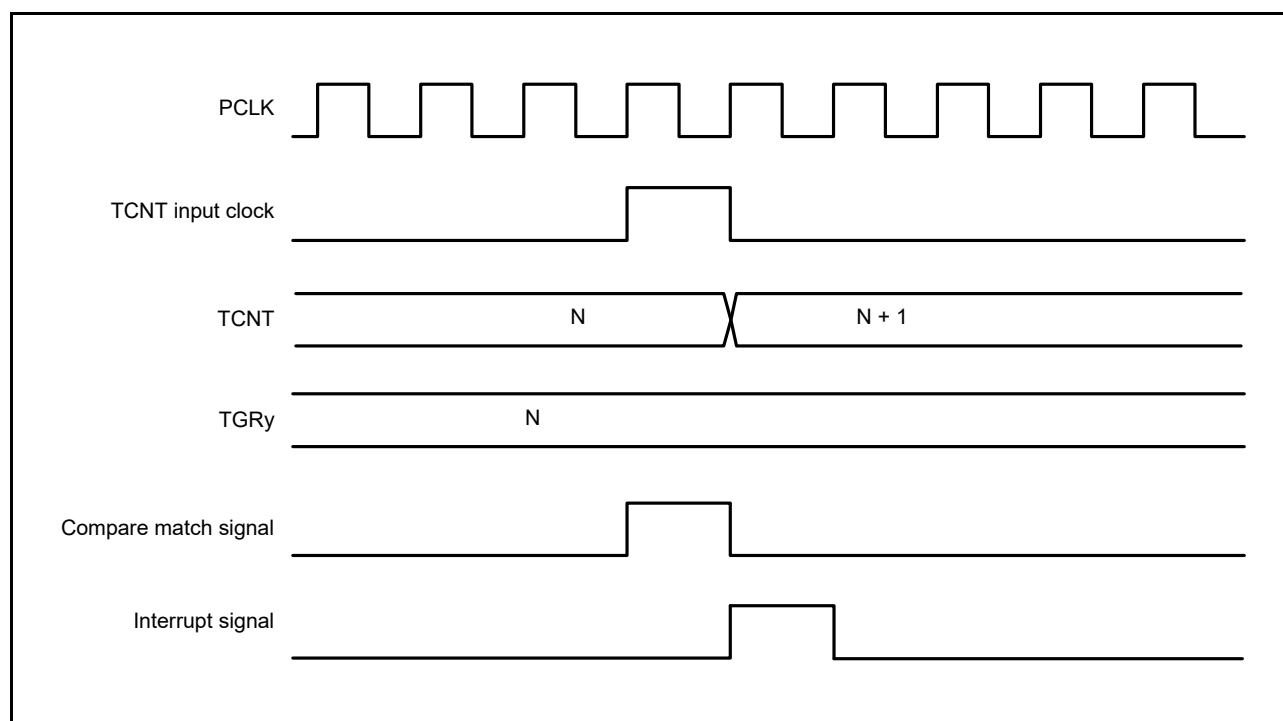


Figure 28.39 TGImy Interrupt Timing (Compare Match)

(2) Timing of Interrupt Signal Setting on Input Capture

Figure 28.40 shows the timing for setting the interrupt signal by input capture occurrence.

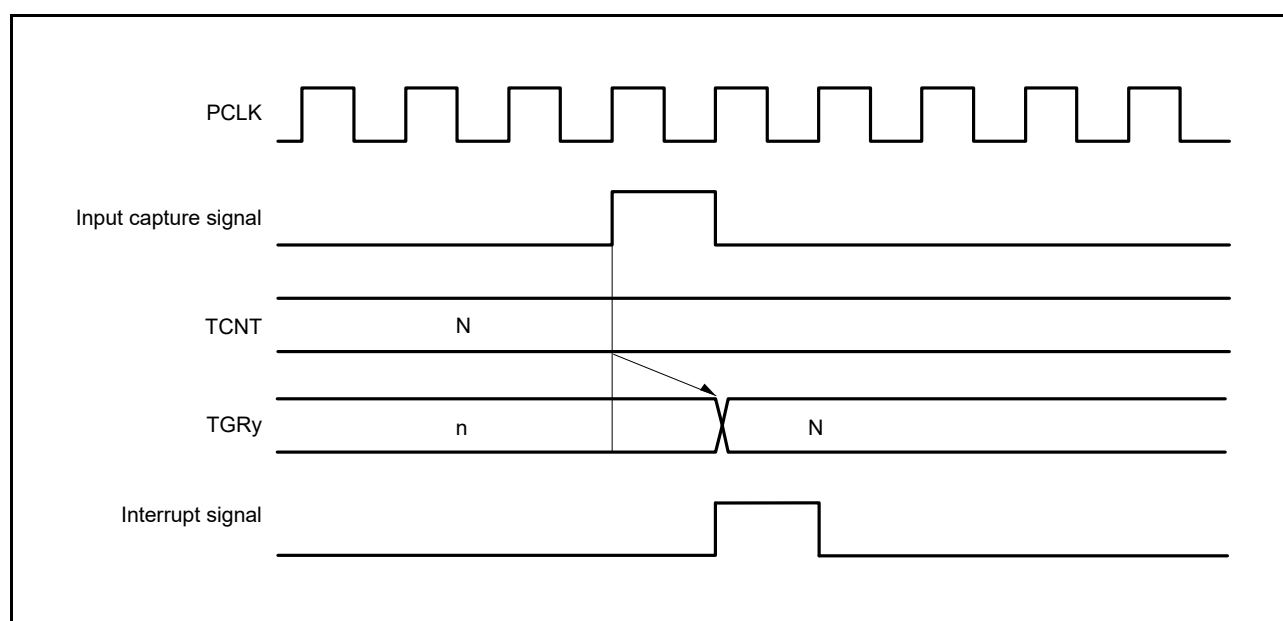


Figure 28.40 TGImy Interrupt Timing (Input Capture)

(3) Timing of TCImV/TCImU Interrupt Signal Setting

Figure 28.41 shows the timing for generating the TCImV interrupt signal by overflow occurrence.

Figure 28.42 shows the timing for generating the TCImU interrupt signal by underflow occurrence.

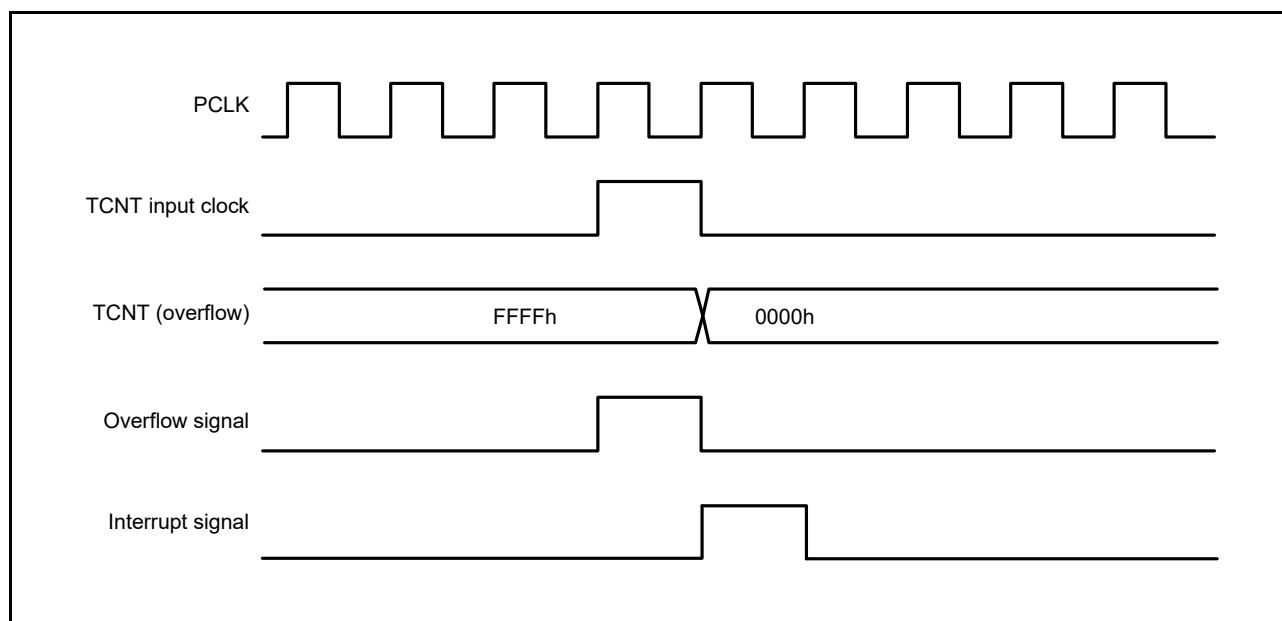


Figure 28.41 TCImV Interrupt Setting Timing

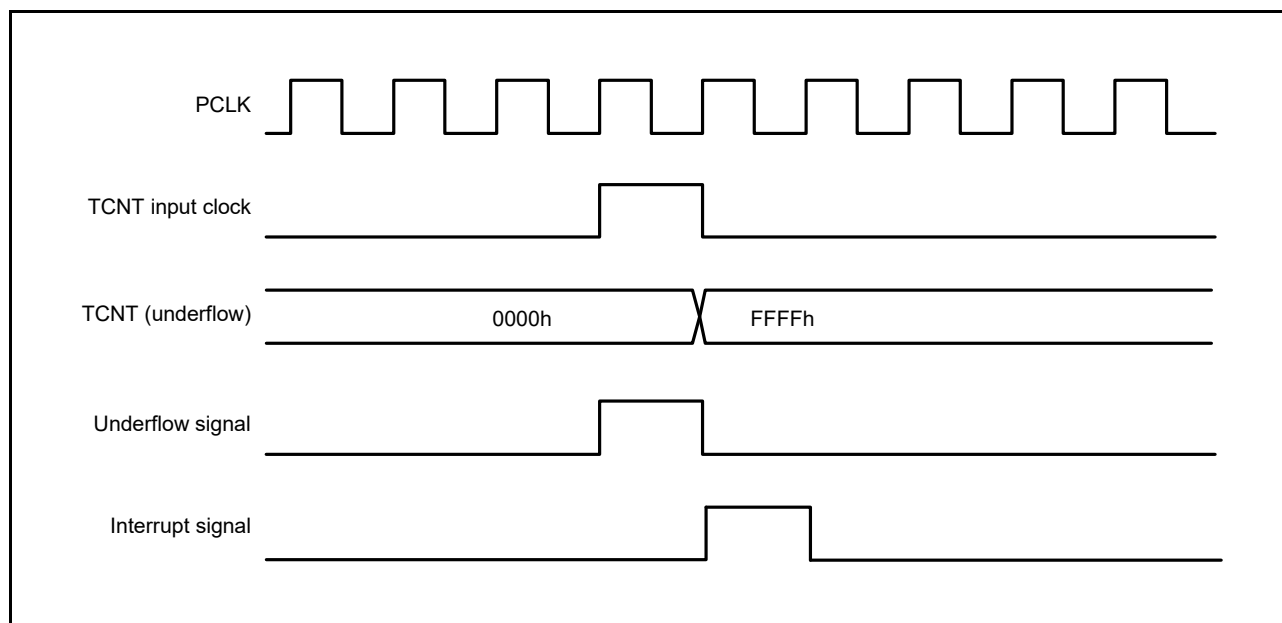


Figure 28.42 TCImU Interrupt Setting Timing

28.10 Usage Notes

28.10.1 Module Stop Function Setting

Operation of the TPU can be disabled or enabled using the module stop control register. The TPU does not operate with the initial setting. Register access is enabled by releasing the module stop state. For details, see section 11, Low Power Consumption.

28.10.2 Input Clock Restrictions

The input clock pulse width must be at least 1.5 PCLK cycles in the case of single-edge detection, and at least 2.5 PCLK cycles in the case of both-edge detection. The TPU will not operate properly with a narrower pulse width.

In phase counting mode, the phase difference and overlap between the two input clocks must be at least 1.5 PCLK cycles, and the pulse width must be at least 2.5 PCLK cycles. Figure 28.43 shows the input clock conditions in phase counting mode.

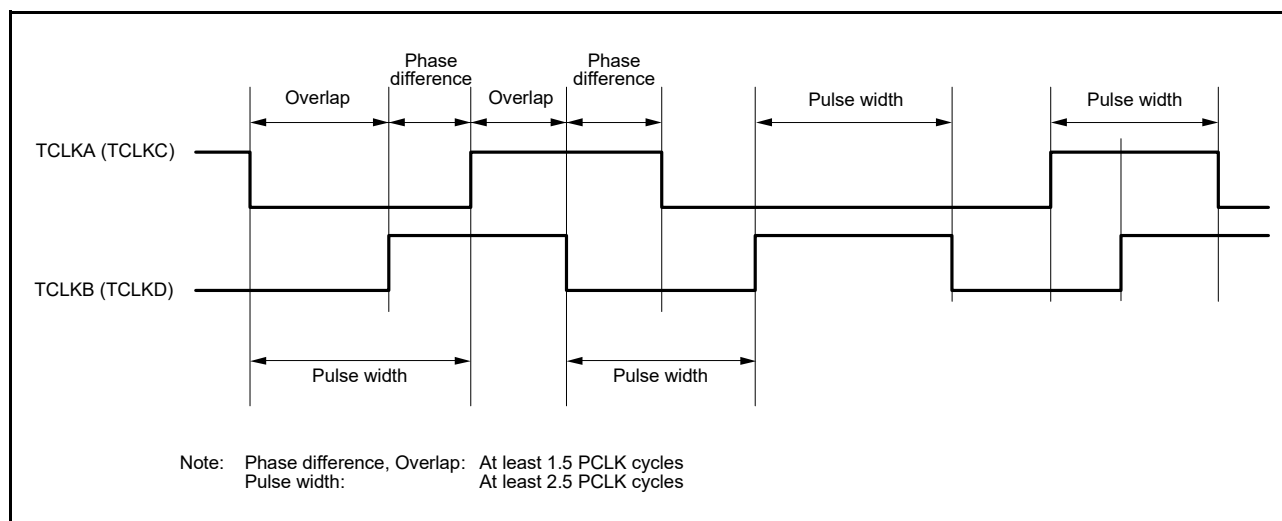


Figure 28.43 Phase Difference, Overlap, and Pulse Width in Phase Counting Mode

28.10.3 Notes on Cycle Setting

When counter clearing by compare match is set, TPUm.TCNT is cleared in the final state in which it matches the TPUm.TGRy value (the point at which the count value matched by TCNT is updated). Consequently, the actual counter frequency is given by the following formula:

$$f = \frac{f_{\text{TCNT_CLK}}}{(N + 1)}$$

f: Counter frequency
 $f_{\text{TCNT_CLK}}$: Count clock frequency
 N: TGRy set value

28.10.4 Conflict between TPUm.TCNT Write and Clear Operations

If the counter clearing signal is generated in a TCNT write cycle, TCNT clearing takes precedence and the TCNT write is not performed. Figure 28.44 shows the timing in this case.

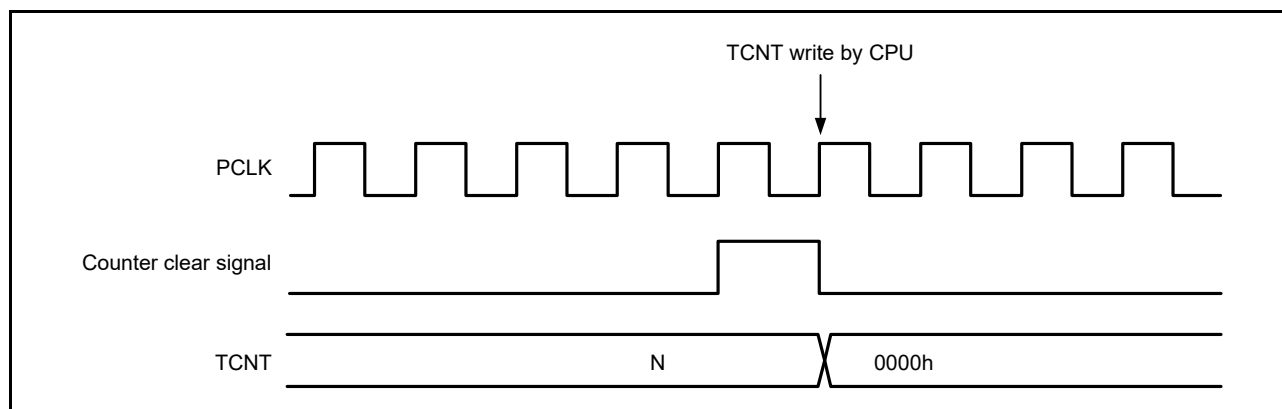


Figure 28.44 Conflict between TPUm.TCNT Write and Clear Operations

28.10.5 Conflict between TPUm.TCNT Write and Increment Operations

If incrementing occurs in a TCNT write cycle, the TCNT write takes precedence and TCNT is not incremented. Figure 28.45 shows the timing in this case.

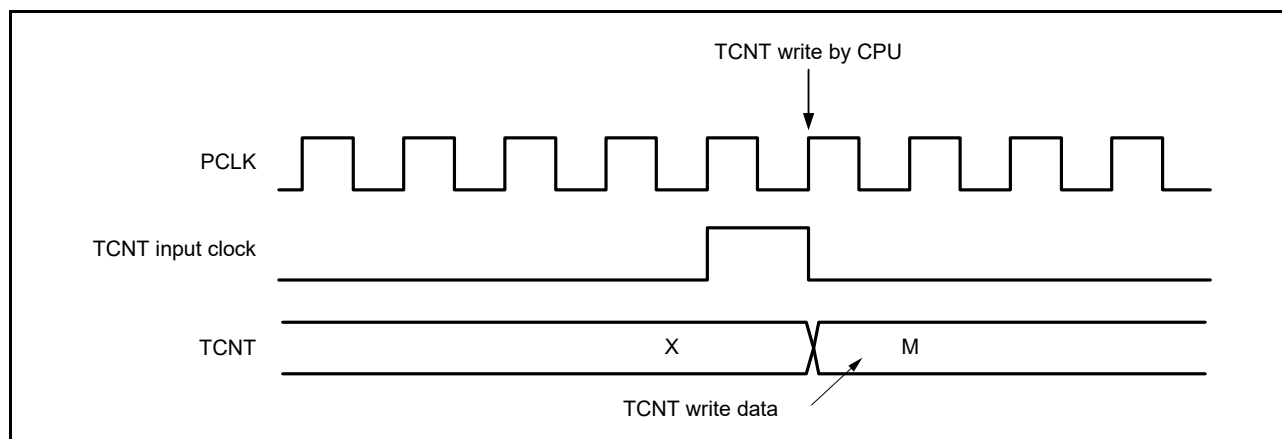


Figure 28.45 Conflict between TPUm.TCNT Write and Increment Operations

28.10.6 Conflict between TPUM.TGRy Write and Compare Match

If a compare match occurs in a TGRy write cycle, the TGRy write takes precedence and the compare match signal is disabled. A compare match also does not occur when the same value as before is written.

Figure 28.46 shows the timing in this case.

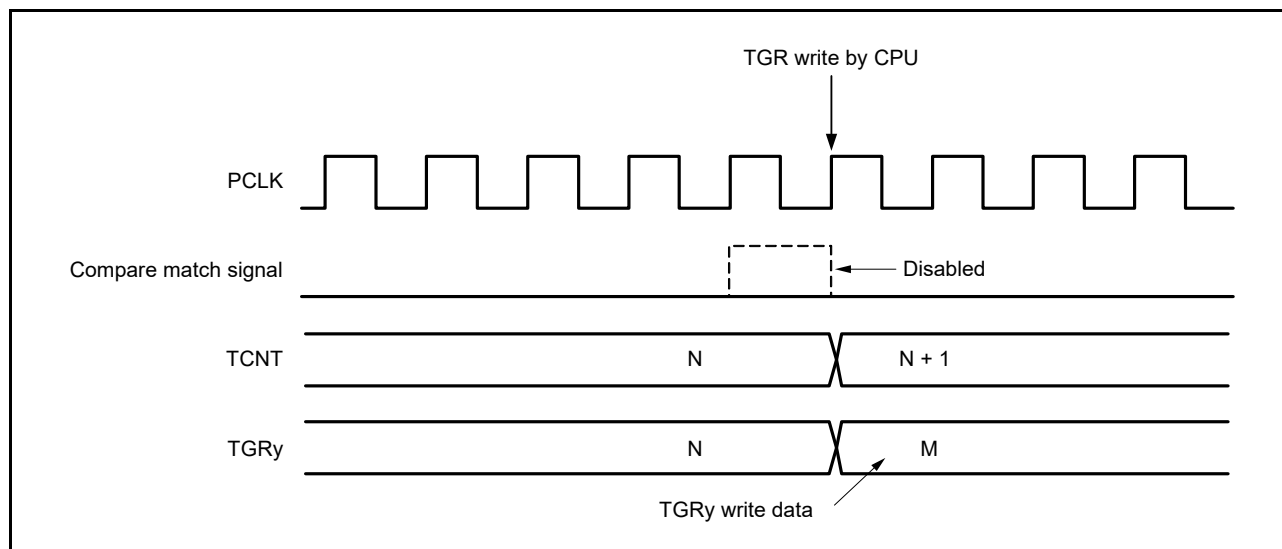


Figure 28.46 Conflict between TPUM.TGRy Write and Compare Match

28.10.7 Conflict between Buffer Register Write and Compare Match

If a compare match occurs in a TPUM.TGRy write cycle, the data transferred to TGRy by the buffer operation will be the data before writing.

Figure 28.47 shows the timing in this case.

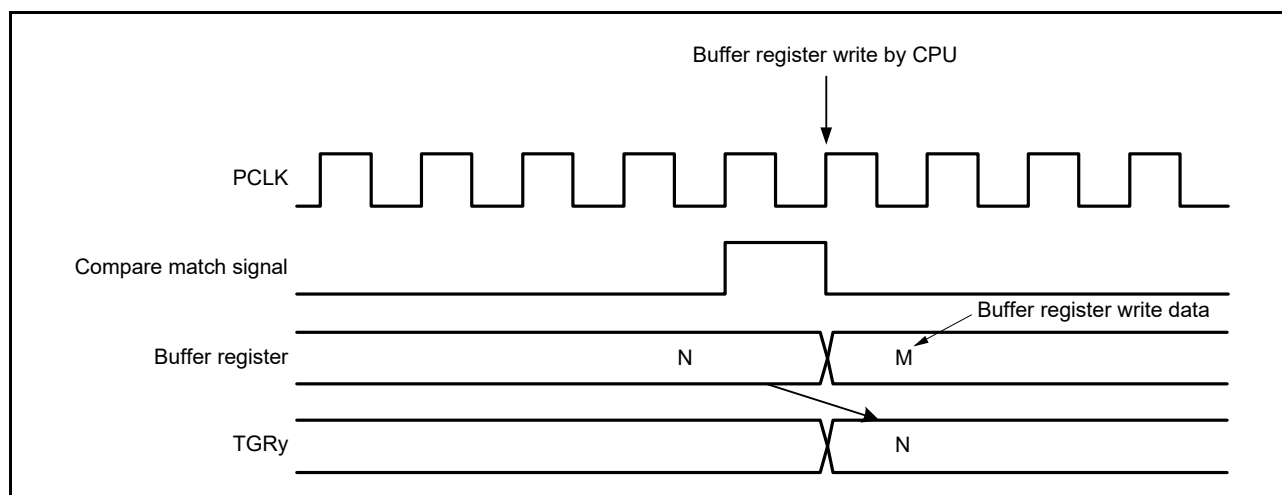


Figure 28.47 Conflict between Buffer Register Write and Compare Match

28.10.8 Conflict between TPUm.TGRy Read and Input Capture

If the input capture signal is generated in a TGRy read cycle, the data that is read will be the data before input capture transfer.

Figure 28.48 shows the timing in this case.

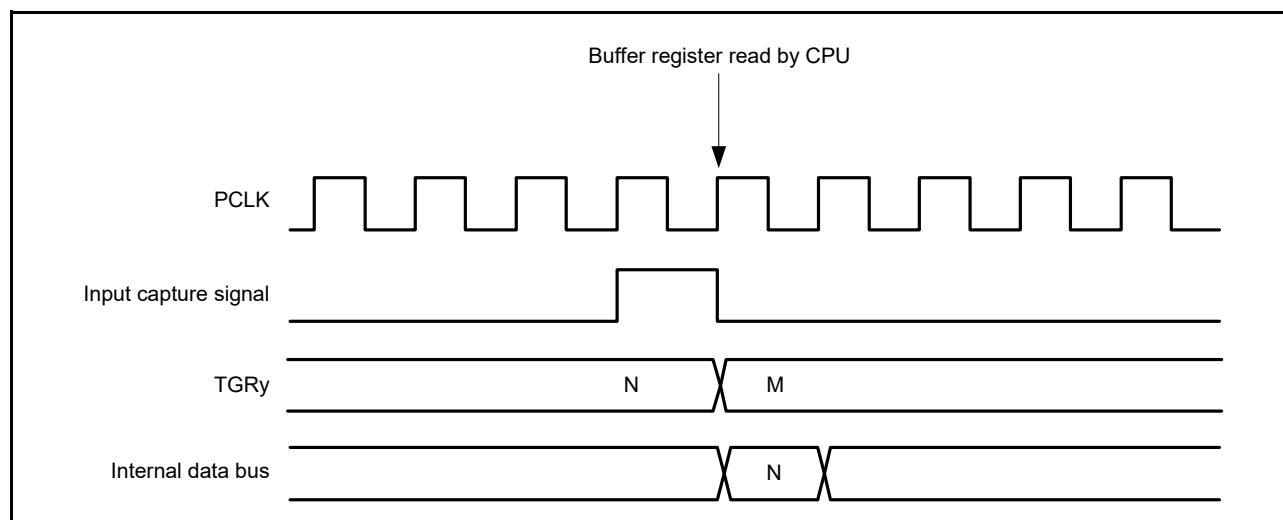


Figure 28.48 Conflict between TPUm.TGRy Read and Input Capture

28.10.9 Conflict between TPUm.TGRy Write and Input Capture

If the input capture signal is generated in a TGRy write cycle, the input capture operation takes precedence and the write to TGRy is not performed. Figure 28.49 shows the timing in this case.

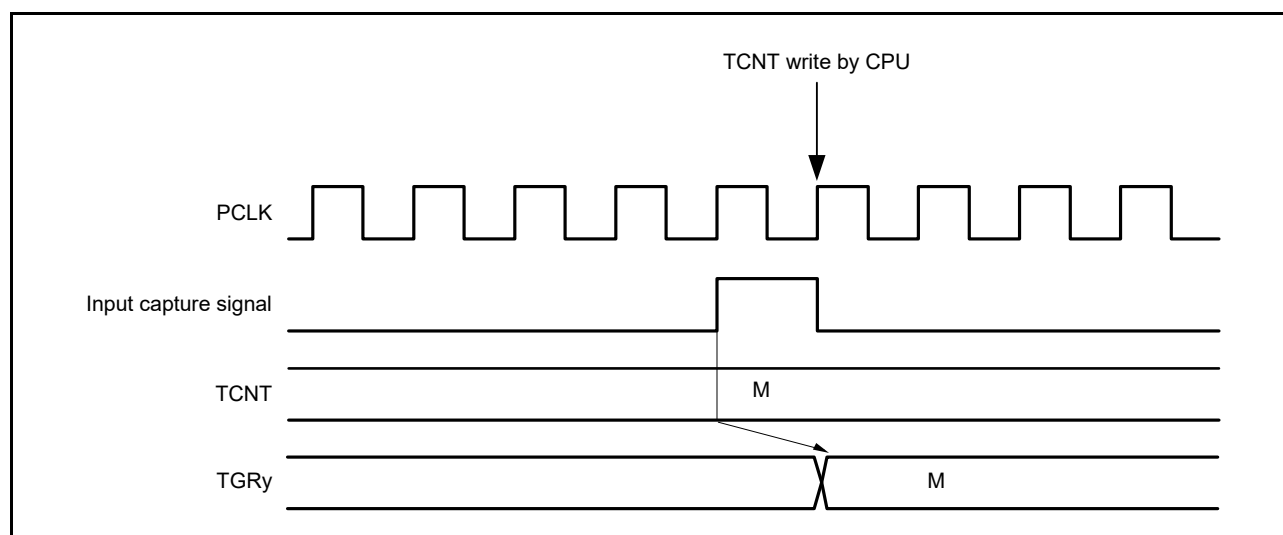


Figure 28.49 Conflict between TPUm.TGRy Write and Input Capture

28.10.10 Conflict between Buffer Register Write and Input Capture

If the input capture signal is generated in a buffer register write cycle, the buffer operation takes precedence and the write to the buffer register is not performed. Figure 28.50 shows the timing in this case.

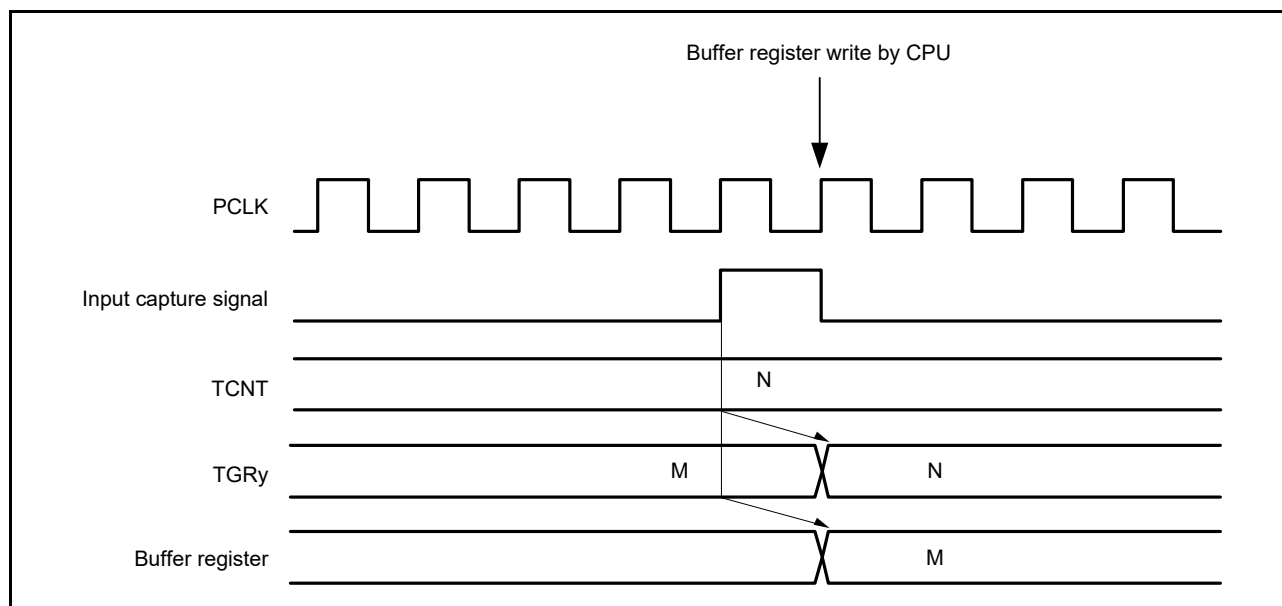


Figure 28.50 Conflict between Buffer Register Write and Input Capture

28.10.11 TCNT Simultaneous Input Capture in Cascade Operation

When TPU1.TCNT and TPU2.TCNT are cascaded for operation as a 32-bit counter, the counter value may not be captured correctly even if the input capture signal is input to the TIOC1A and TIOC2A pins or the TIOC1B and TIOC2B pins at the same time. This is because a difference of up to 1 clock cycle in the timing of the capture signal to input to TPU1.TCNT and TPU2.TCNT may occur due to internal delays.

For example, the counter value is captured at an overflow of TPU2.TCNT as in counting up from 0A31 FFFFh to 0A32 0000h, the value captured may be 0A31 0000h or 0A32 FFFFh.

The same applies to cascaded operation of TPU4.TCNT and TPU5.TCNT.

28.10.12 Conflict between Overflow/Underflow and Counter Clearing

If overflow/underflow and counter clearing*¹ occur simultaneously, TPUm.TCNT is cleared with the generation of the compare match interrupt and an overflow interrupt is generated.

Figure 28.51 shows the operation timing when a TPUm.TGRy compare match is specified as the clearing source and FFFFh is set in TGRy.

Note 1. There are four counter clearing sources:

- Compare match
- Input capture
- Synchronous clearing
- Counter restart operation by an event signal

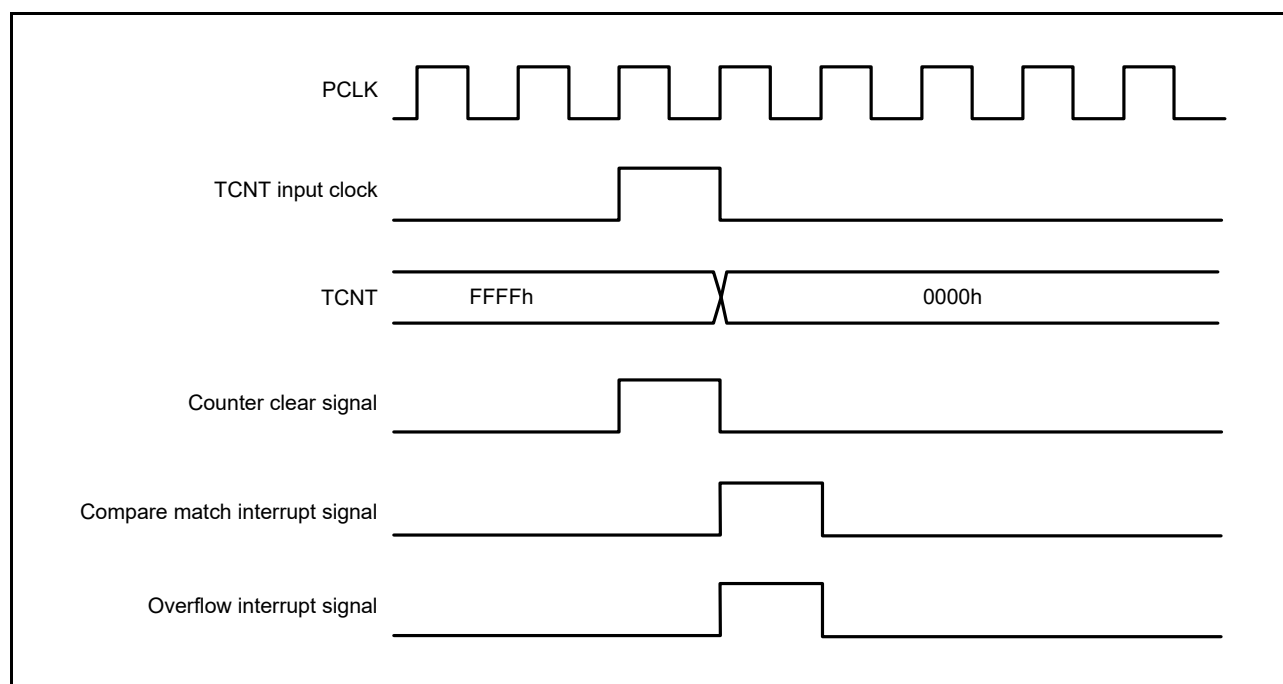


Figure 28.51 Conflict between Overflow and Counter Clearing

28.10.13 Conflict between TPUm.TCNT Write and Overflow/Underflow

If an overflow/underflow occurs due to increment/decrement in a TCNT write cycle, the TCNT write takes precedence. Figure 28.52 shows the operation timing when there is conflict between TCNT write and overflow.

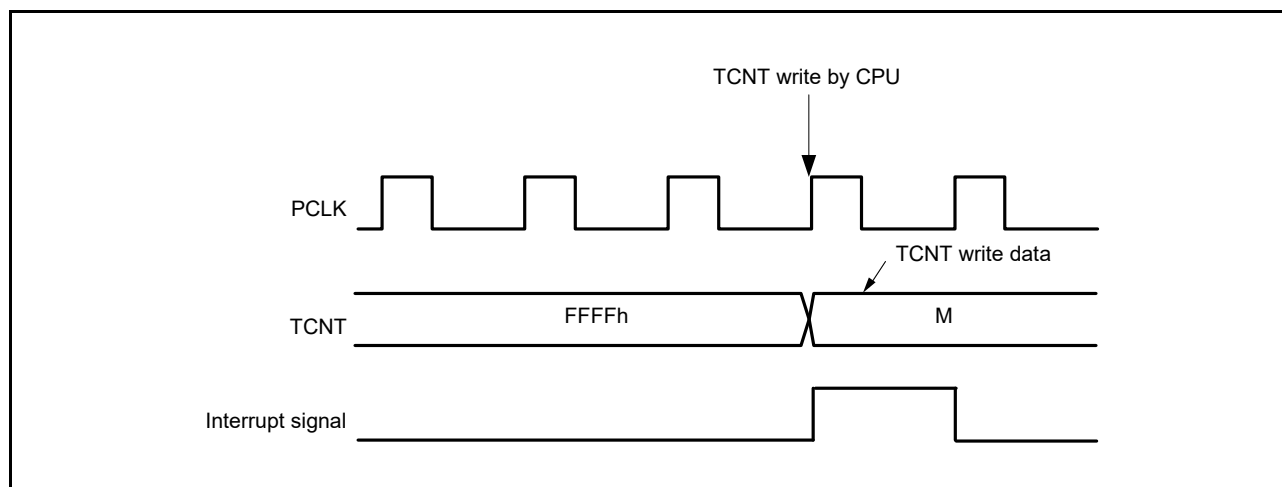


Figure 28.52 Conflict between TPUm.TCNT Write and Overflow

28.10.14 Multiplexing of I/O Pins

In this MCU, the TCLKA input pin is multiplexed with the TIOCB5 I/O pin, the TCLKB input pin with the TIOCB2 I/O pin, the TCLKC input pin with the TIOCB1 I/O pin, the TCLKD input pin with the TIOCB0 I/O pin, the TCLKB input pin with the TIOCD0 I/O pin, the TCLKC input pin with the TIOCC3 I/O pin, and the TCLKD input pin with the TIOCD3 I/O pin. When an external clock is input, compare match output should not be performed from a multiplexed pin.

28.10.15 Continuous Output of Compare-Match Pulse Interrupt Signal

When TGR is set to 0000h, PCLK/1 is set as the count clock, and compare match is set as the counter clear source, the TCNT remains 0000h and is not updated, and a compare-match pulse interrupt signal is output continuously to form a flat signal level.

When a pulse interrupt signal is used, the interrupt controller cannot detect the second and subsequent interrupts.

Figure 28.53 shows an operation timing when the compare-match pulse interrupt signal is continuously output.

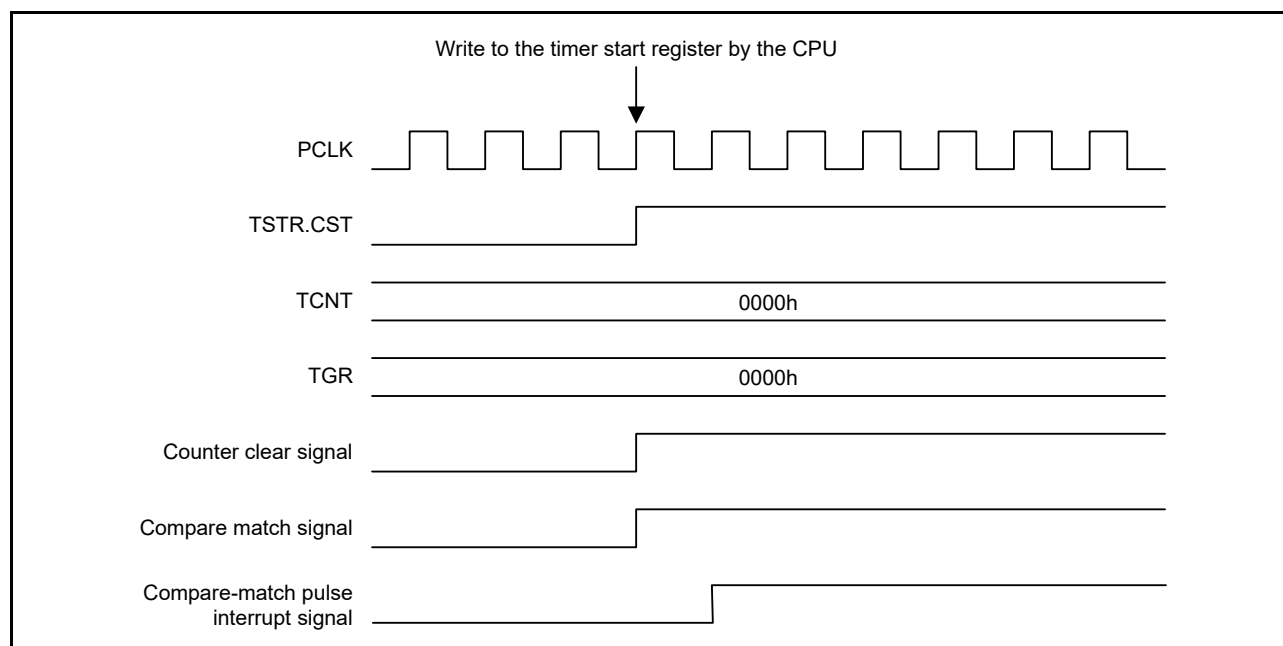


Figure 28.53 Continuous Output of Compare-Match Pulse Interrupt Signal

28.10.16 Continuous Output of Input-Capture Pulse Interrupt Signal

When input-capture signal is set on both edges and when the pulse width of the input-capture input equals to one PCLK cycle detected by internal sampling, input capture is generated continuously on the rising and falling edges. Therefore, an input-capture pulse interrupt signal is output continuously to form a flat signal level.

When a pulse interrupt signal is used, the interrupt controller cannot detect the second and subsequent interrupts.

Figure 28.54 shows an operation timing when the input-capture pulse interrupt signal is output continuously.

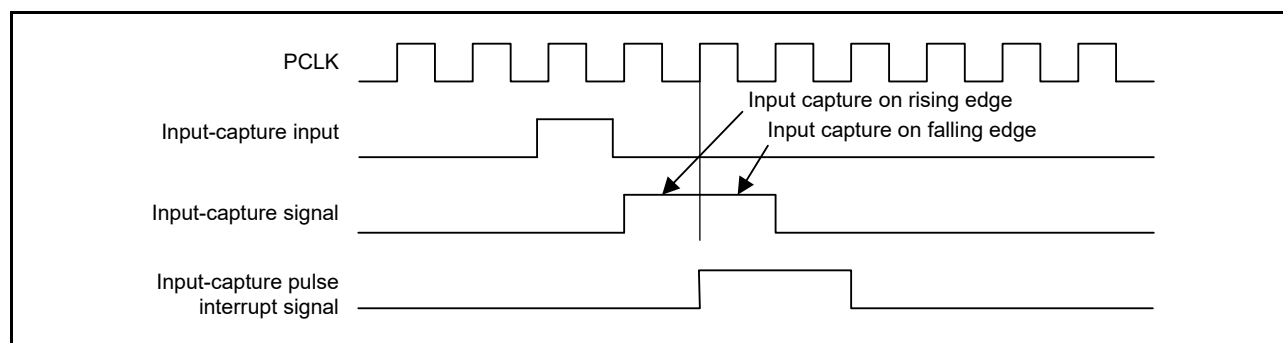


Figure 28.54 Continuous Output of Input-Capture Pulse Interrupt Signal

28.10.17 Continuous Output of Underflow Pulse Interrupt Signal

If two external clock signals' same direction edges to be phase counted are generated within two PCLK cycles in phase counting mode 1, with TGR being 0000h, and compare match set as the counter clear source, the TCNT remains 0000h and is not updated, and a compare-match pulse interrupt signal and an underflow interrupt signal are output continuously to form a flat signal level.

When a pulse interrupt signal is used, the interrupt controller cannot detect the second and subsequent interrupts.

Figure 28.55 shows an operation timing when the underflow pulse interrupt signal is output continuously.

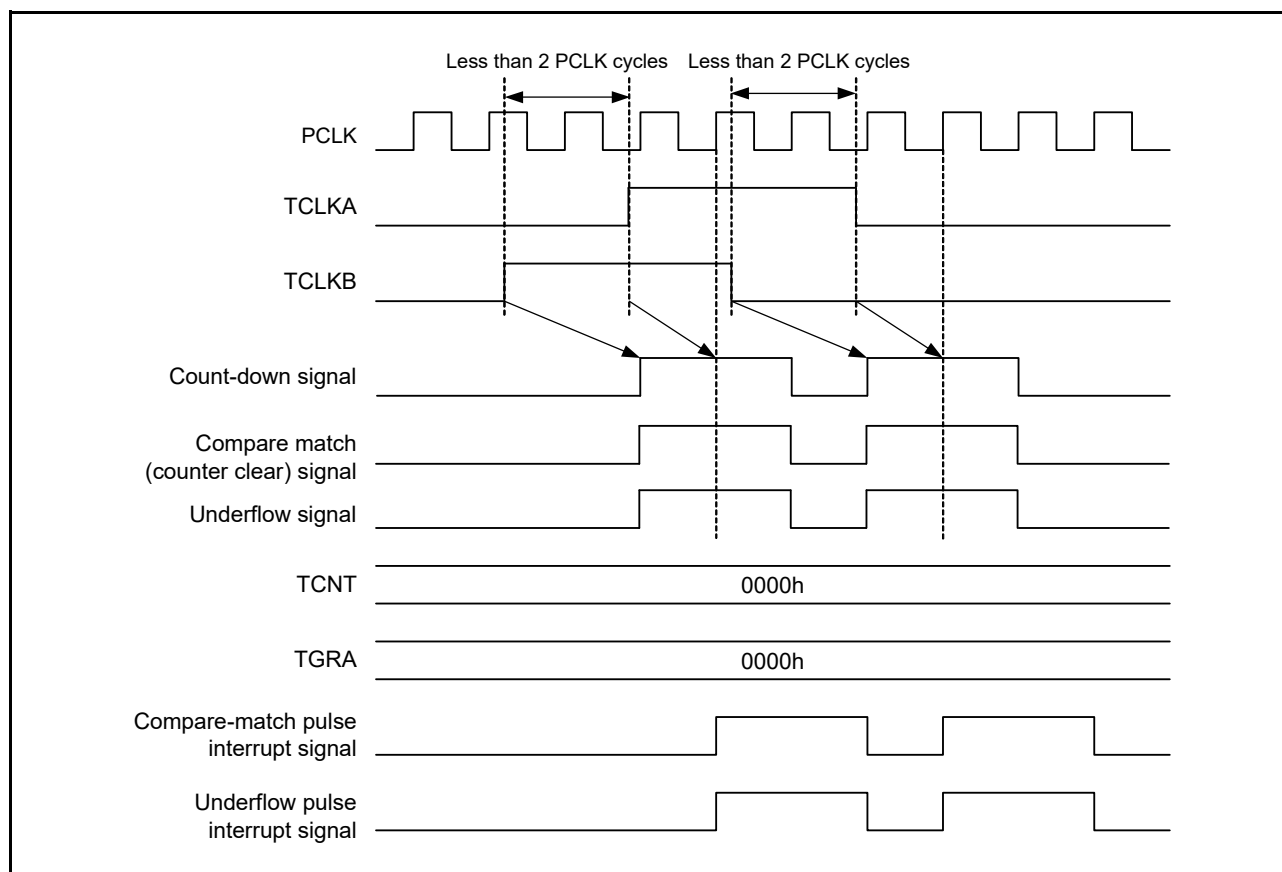


Figure 28.55 Continuous Output of Underflow Pulse Interrupt Signal

28.11 Event Link Operation

28.11.1 Event Signal Output to ELC

The TPU uses the ELC (event link controller) to perform link operation to the previously specified module using the interrupt request signal as the event signal.

The event signal can be output regardless of the setting of the corresponding interrupt request enable bits (TGIEA, TGIEB, TGIEC, or TGIED, and TCIEV, or TCIEU).

28.11.2 Event Signal Input from ELC

The TPU can perform any of the following three operations using the event link setting register of the ELC (event link controller).

(1) Start Counting

When an event signal is input while the TPU count start operation is selected, the CSTn bit in TSTRA register (the timer start register) is set to 1 and counting starts.

However, if this event is generated for the channels when the CSTn bit is set to 1, the event is ignored.

Table 28.30 lists the TSTRA.CSTn bit used for each channel.

Figure 28.56 shows the timing of the count start operation.

For details on the setting procedure to start counting, see section 28.3.1, (1) Counter Operation.

Table 28.30 Correspondence between Channels and TSTRA.CSTn bit

Channel No.	TSTRA.CSTn bit
TPU0	CST0
TPU1	CST1
TPU2	CST2
TPU3	CST3

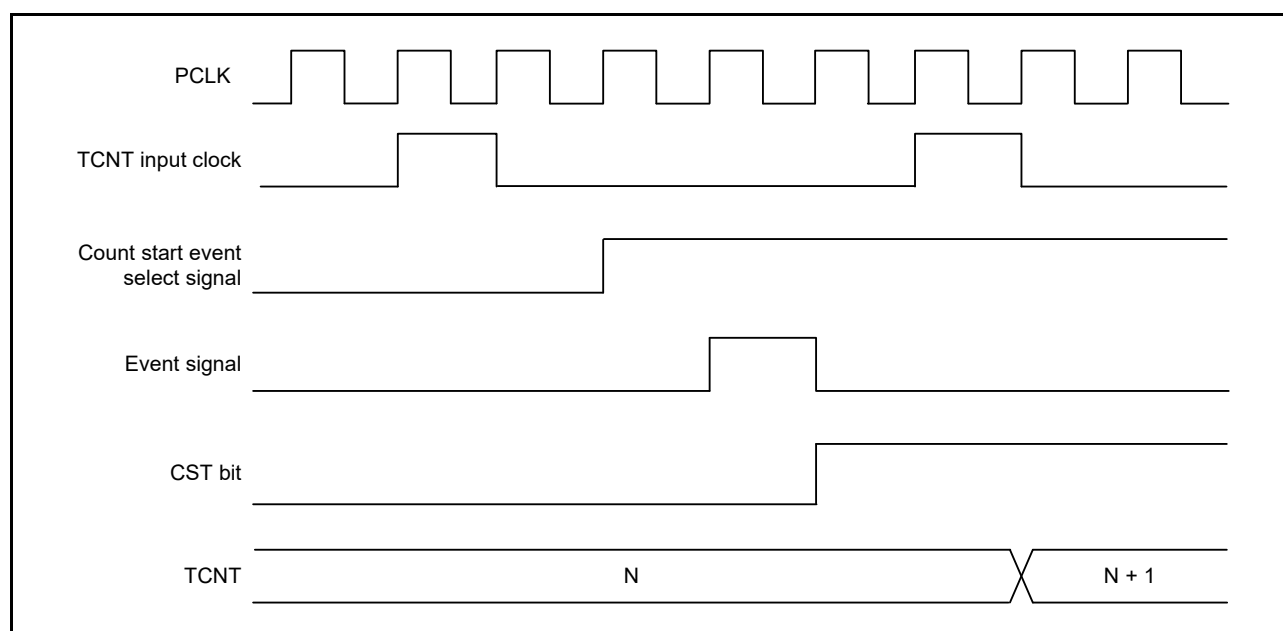


Figure 28.56 Start Counting on Input of the Event Signal

(2) Restart Counting

When an event signal is input while the TPU count restart operation is selected, the value of TCNTn counter (the timer start register) is returned to its initial value (0000h). If the CSTn bit in TSTRA register (the timer start register) is set to 1, however, counting operation will then continue.

Table 28.30 lists the TSTRA.CSTn bit used for each channel.

Figure 28.57 shows the timing of the count restart operation.

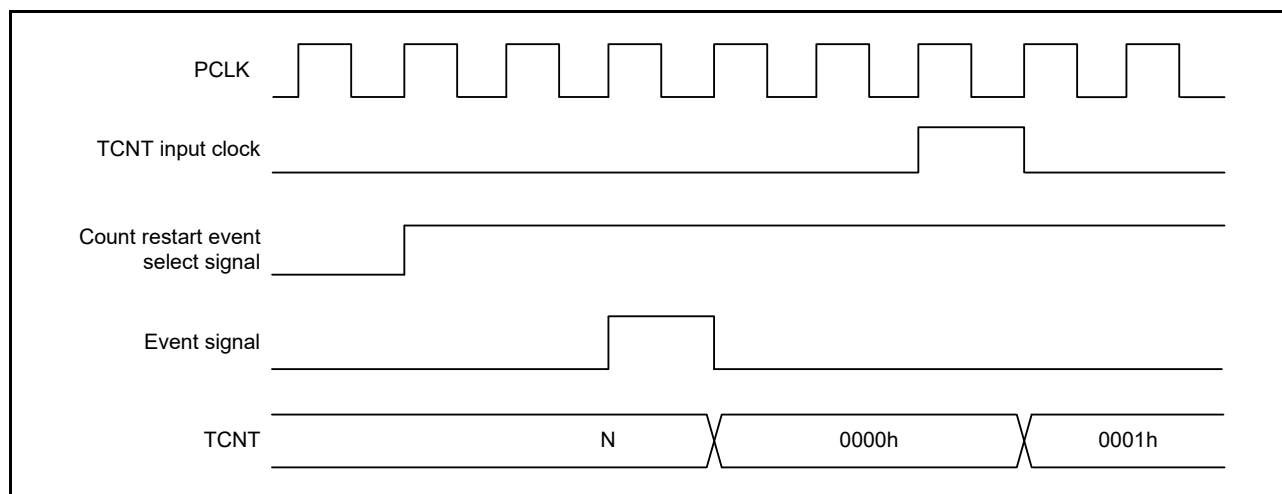


Figure 28.57 Restart Counting on Input of the Event Signal

(3) Input Capture Operation

When an event signal is input while the TPU input capture operation is selected, the value of TCNT counter (the timer count register) for the corresponding channel is captured in TGR register (the timer general register). When using input capture due to the event link, set the bit in TIOR (the timer I/O control register) to specify input capture, and then set the CSTn bit in TSTRA register (the timer start register) to 1 to start counting.

Table 28.31 lists TGR register and TIOR register bits used for each channel. For the TSTRA.CSTn bit used for each channel, see Table 28.30.

Figure 28.58 shows the timing of input capture operation.

When input capture operation by event linking is selected, the setting of TIOR register and the corresponding input capture (the linkage of the TIOCnA pin (input capture pin) input with the specific operation of other channels) are not effective (this also applies when the event select signal is set to 1 at the same time).

For details on the setting procedure for input capture, see section 28.3.1, (3) Input Capture Function.

Table 28.31 TGR and TIOR Used for Input Capture by ELC

Channel No.	Capture Destination Registers	Bits in TIOR
TPU0	TGRA register (channel 0)	IOA[3:0] bits (TIORH0)
TPU1	TGRA register (channel 1)	IOA[3:0] bits (TIOR1)
TPU2	TGRA register (channel 2)	IOA[3:0] bits (TIOR2)
TPU3	TGRA register (channel 3)	IOA[3:0] bits (TIORH3)

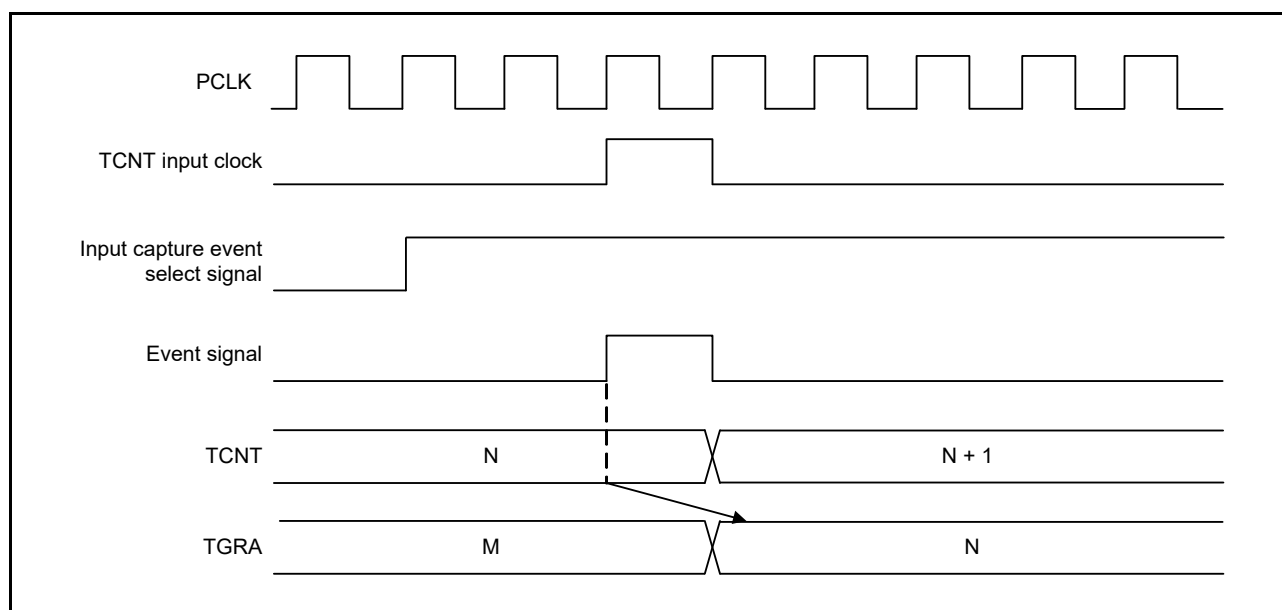


Figure 28.58 Input Capture on Input of the Event Signal

28.11.3 Usage Notes on Operation on Input of the Event Signal

The followings are the notes on using the TPU for event link operations.

(1) Start Counting

When writing to the TSTRA.CSTn bit (the timer start register) and a counting start are in contention, writing to the CSTn bit does not proceed since setting of the CSTn bit to 1 in response to the event takes priority.

Figure 28.59 shows the timing in this case.

Furthermore, even when a counting start due to the event link is selected, CPU writing to the TSTRA.CSTn bit proceeds if the event signal is low.

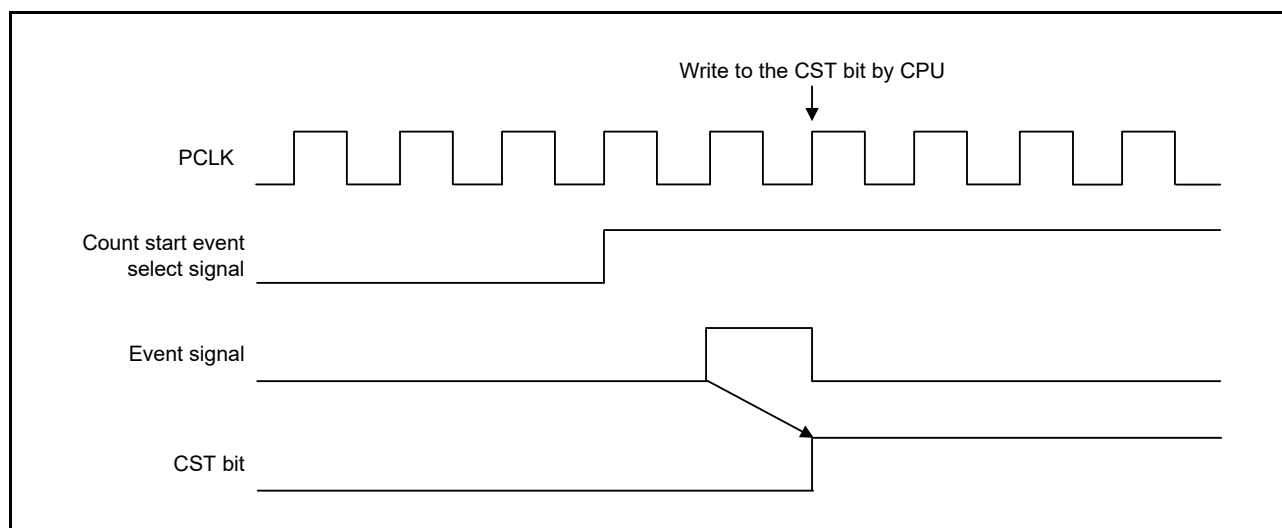


Figure 28.59 Conflict between Writing to the CSTn Bit and Counting Start

(2) Restart Counting

When a TCNTn counter (the timer count register) write cycle and a counting restart are in contention, writing to TCNTn counter does not proceed since the counter value initialization in response to the counting restart takes priority.

Figure 28.60 shows the timing in this case.

Furthermore, even when a counting restart due to the event link is selected, CPU writing to TCNTn counter proceeds if the event signal is low.

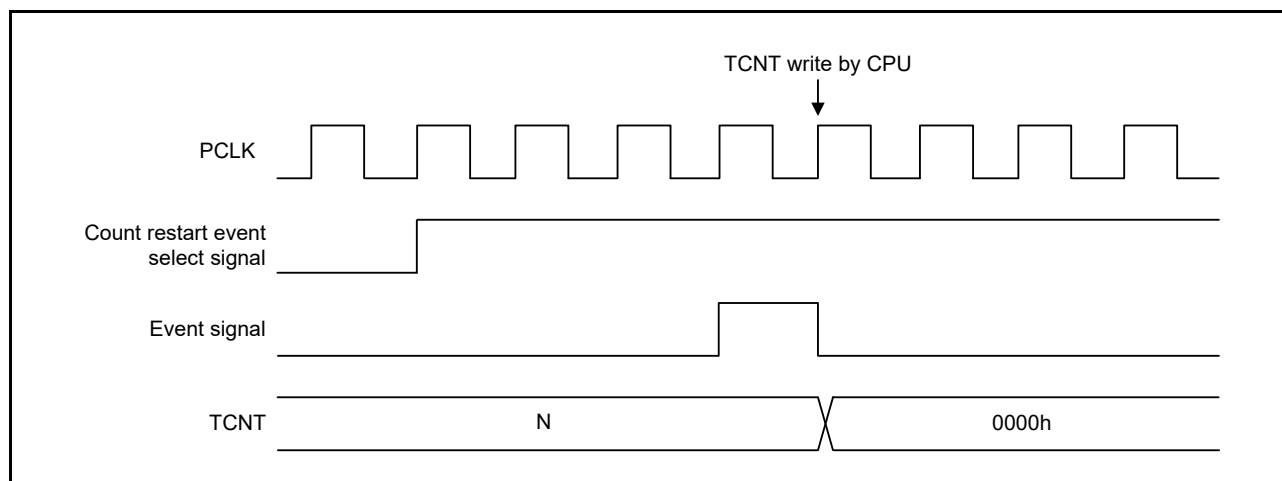


Figure 28.60 Conflict between TCNTn Write Cycle and Counting Restart

(3) Input Capture Operation

If a TGRA register (the timer general register) read/write cycle and input capture operation are in contention, operation proceeds as follows:

(a) Conflict between TGR Read Cycle and Input Capture

The internal data bus reads the data before input capture transfer.

Figure 28.61 shows the timing in this case.

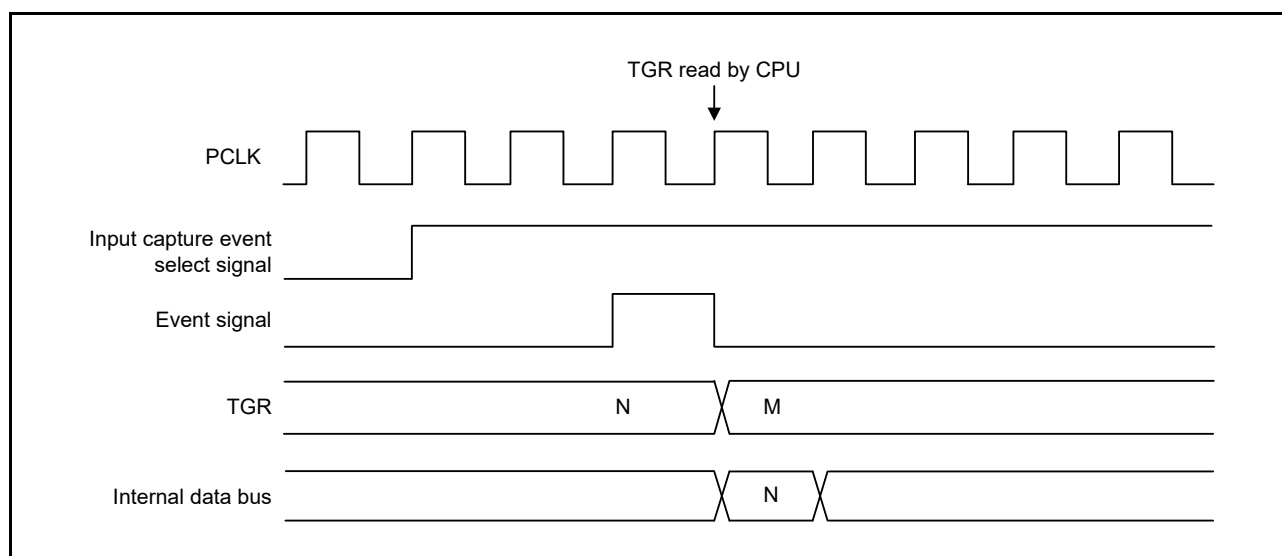


Figure 28.61 Conflict between TGR Read Cycle and Input Capture Operation

(b) Conflict between TGR Write Cycle and Input Capture

Writing to TGRA register does not proceed since input capture takes priority.

Figure 28.62 shows the timing in this case.

Furthermore, even when input capture operation due to the event link is selected, CPU writing to TGR proceeds if the event signal is low.

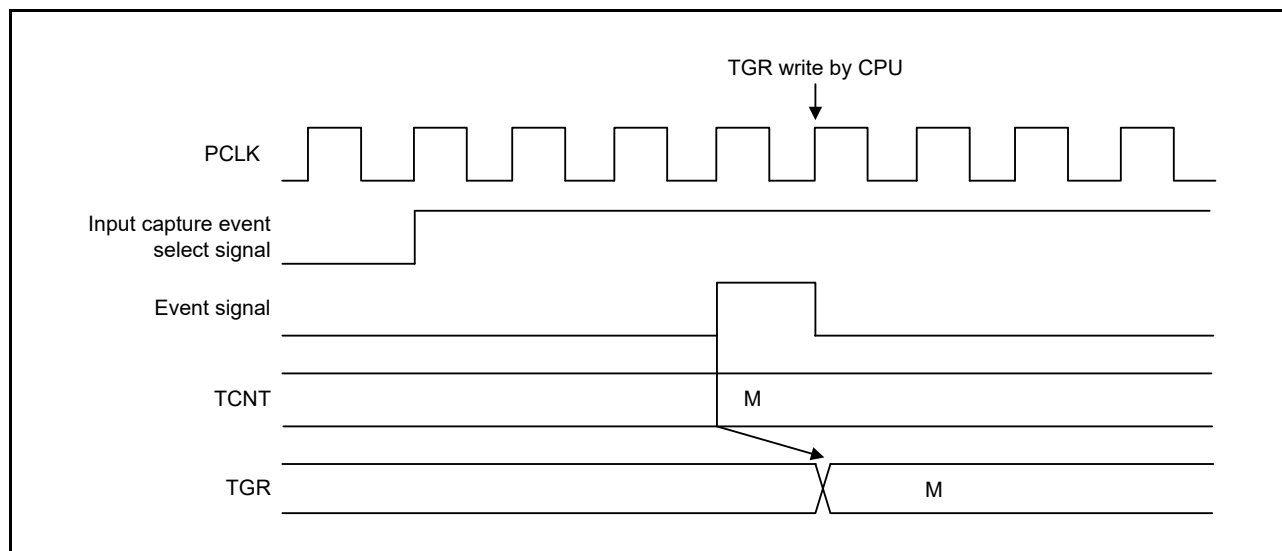


Figure 28.62 Conflict between TGR Write Cycle and Input Capture Operation

28.11.4 Notes on Output of the Event Signal

The followings are the notes on output of the event signal.

(1) Output of the Compare Match Event Signal

When the TGR register is set to 0000h, PCLK/1 is set as the counter clock, and compare match is set as the trigger for clearing of the counter clock (TCRn.TPSC[2:0] = 000b), the value of the TCNT remains 0000h, and the event output signal will be output continuously (i.e. its level will be flat) rather than output over a single cycle.

Figure 28.63 shows the timing for continuous output of the event output signal in response to a compare match.

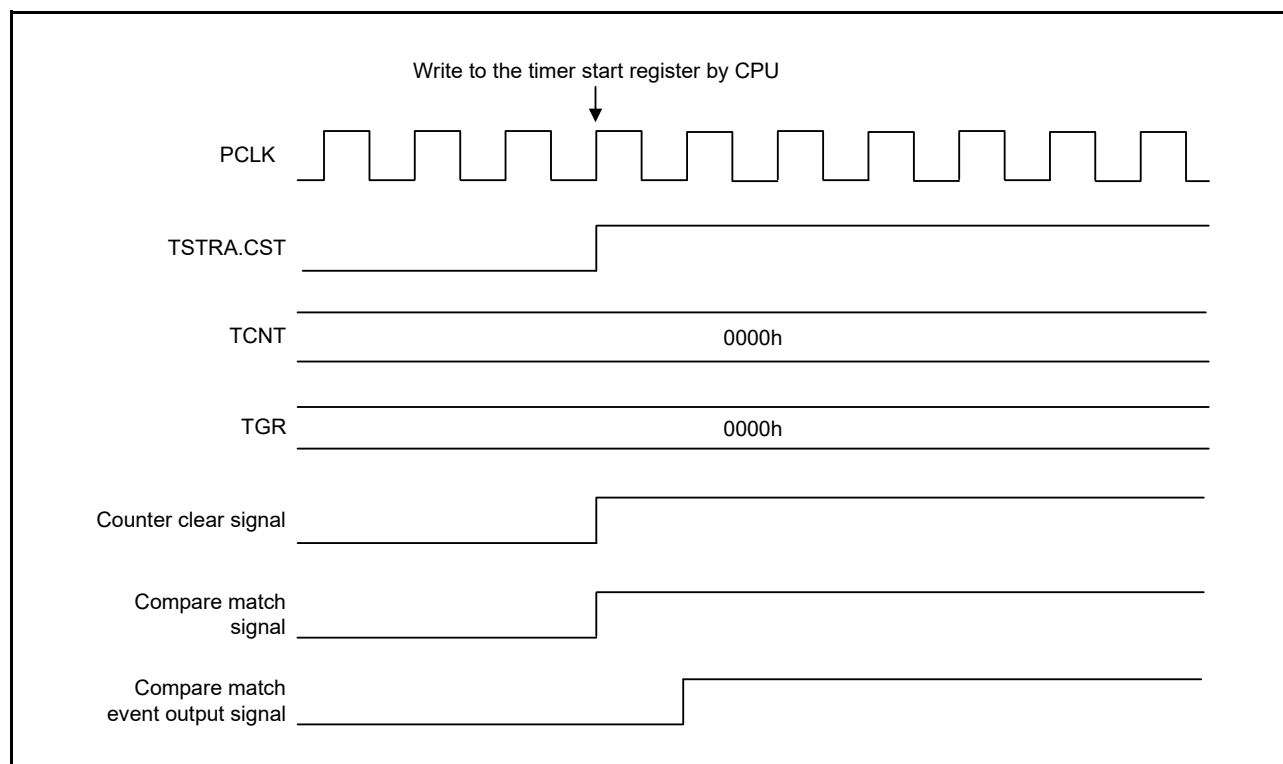


Figure 28.63 Continuous Output of the Compare Match Event Output Signal

(2) Output of the Underflow Event Signal

If two external clock signals' same direction edges to be phase counted are generated within two PCLK cycles in phase counting mode 1, with TGR being 0000h, and compare match set as the counter clear source, the TCNT counter remains 0000h, and a compare-match event signal and an underflow event signal are output continuously to form a flat signal level.

Figure 28.64 shows the timing for continuous output of the event output signal in response to underflow.

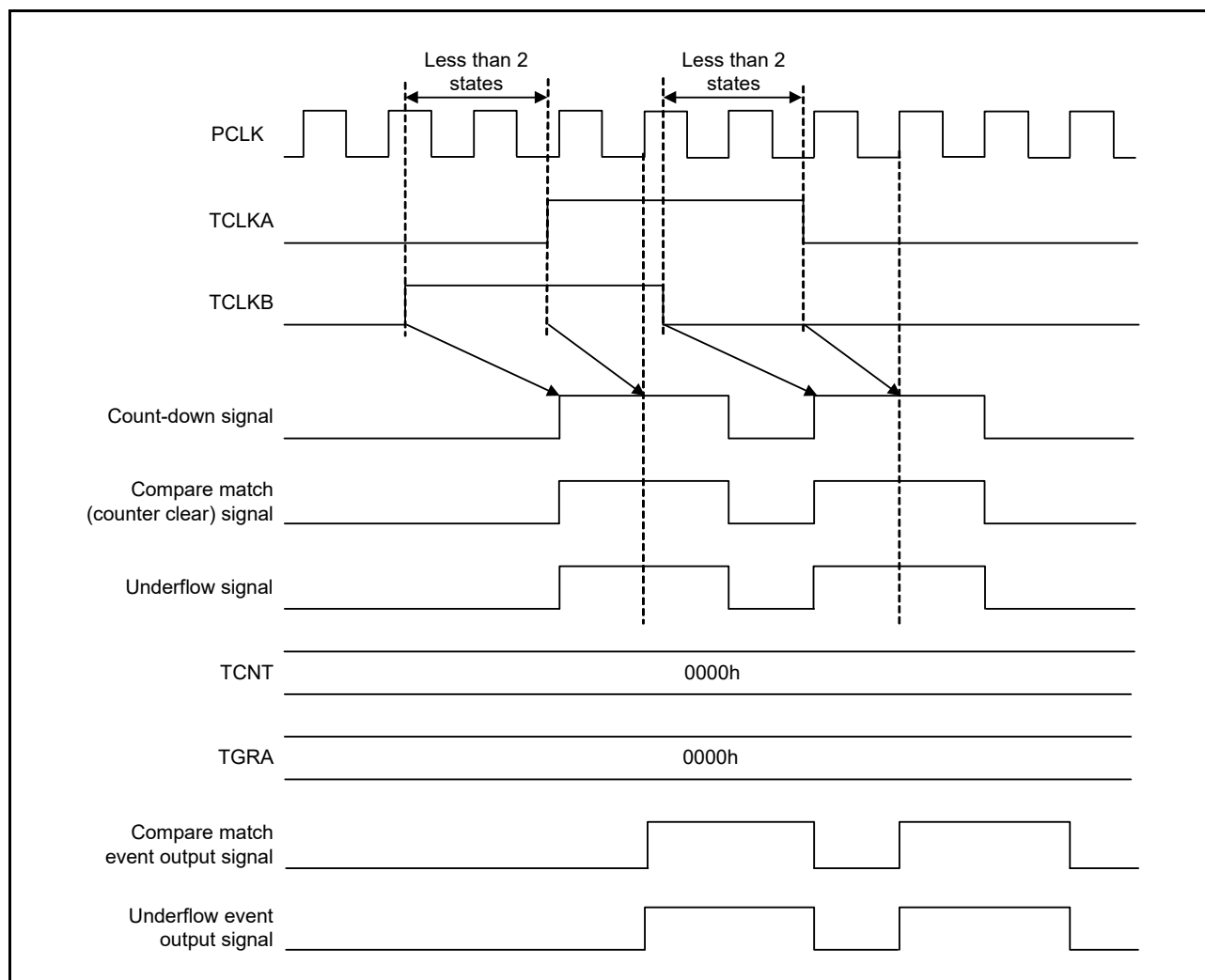


Figure 28.64 Continuous Output of the Underflow Event Output Signal